

Introduction to Algorithms – Study Guide

Study Guide: Introduction to Algorithms (CLRS, 4th Edition)

Generated 2026-06-05. Subject: Computer Science (Algorithms).
Target length: 5000 lines. Coverage: comprehensive – all 35 chapters + 4 appendices.

Chapter-by-Chapter Breakdown

Ch. 1 – The Role of Algorithms in Computing

Named Entities (Terms & Definitions)

- **Algorithm:** well-defined computational procedure taking input $\rightarrow$ output in finite time
- **Computational problem:** specification of desired input/output relationship
- **Problem instance:** specific input needed to compute solution
- **Correct algorithm:** halts with correct output for every instance
- **Data structure:** way to store/organize data to facilitate access and modifications
- **NP-complete:** class of problems with no known efficient algorithm; if one solved efficiently, all are
- **Approximation algorithm:** gives good but not necessarily optimal solution
- **Online algorithm:** receives input over time, must decide without knowing future

Comparisons & Trade-offs

Dimension	Insertion Sort	Merge Sort
Time	$\sim cn^2$	$\sim cn \lg n$
Constant factor	Smaller ($c < c$)	Larger
Winner for large n	Loses	Wins

- **Key result:** Computer A (10B inst/sec, insertion sort $2n^2$) vs Computer B (10M inst/sec, merge sort $50n \lg n$). Sorting 10M numbers: A takes ~ 5.56 hrs, B takes ~ 17 min.

End-of-Chapter Material

- **Ex 1.2-2:** Insertion sort $8n^2$ vs merge sort $64n \lg n$ – insertion beats merge for $n \leq 43$
- **Ex 1.2-3:** $100n^2$ vs 2^n – smallest n where $100n^2$ faster = 15

End-of-Chapter Exercises

- **Ex 1.1-1:** Give real-world example requiring sorting, determining order of matrix multiplication, convex hull
 - Sorting: organizing a list of student GPAs in descending order

- Matrix multiplication order: determining optimal parenthesization for a chain of matrix multiplications in computer graphics
- Convex hull: finding smallest convex polygon containing all points in a robot's obstacle map
- **Ex 1.1-2:** Other measures of efficiency besides speed – memory usage, power consumption, network bandwidth, disk I/O, code maintainability, answer quality (approximation ratio)
- **Ex 1.1-3:** Data structure (doubly linked list) – strengths: $O(1)$ insert/delete at known position; weaknesses: $O(n)$ search, no random access, extra space for pointers
- **Ex 1.1-4:** Similarities between shortest-path and traveling-salesman: both find paths, both involve graphs, both minimize distance. Differences: shortest-path has polynomial solution (Dijkstra), TSP is NP-complete; shortest path finds path between 2 vertices, TSP must visit all vertices.
- **Ex 1.1-5:** Only optimal solution acceptable: medical device control, avionics, cryptographic key generation. Approximate solution acceptable: TV schedule, airline route planning, recommendation systems.
- **Ex 1.1-6:** Problem that can be solved but not in computer-solvable time – subset of NP-complete problems (provided $P \neq NP$). Example: optimal chess move given $10^{40}+$ board positions.

Named Entities (Expanded)

- **Key:** value used to identify a data record (e.g., student ID number)
- **Satellite data:** additional data carried along with a key in a record (e.g., name, address, GPA)
- **Record:** collection of fields containing key + satellite data
- **Problem instance vs. problem:** instance = specific input; problem = class of all valid instances
- **Efficiency:** algorithm that uses fewer computational resources (time, memory) for the same task
- **Correct algorithm:** for every valid input, produces correct output and halts
- **P vs. NP:** Millennium problem: are problems verifiable in polynomial time also solvable in polynomial time?
- **NP-hard:** every NP problem can be reduced to this problem (at least as hard as any NP problem)
- **NP-complete:** both NP and NP-hard – most difficult problems in NP
- **Reduction:** transformation of one problem into another; if A reduces to B and B is easy, then A is easy
- **Traveling-salesman problem (TSP):** given cities and distances, find shortest tour visiting each city exactly once; NP-complete
- **Decision problem:** yes/no output (e.g., “does a tour of length $\leq L$ exist?”)
- **Optimization problem:** minimize/maximize some quantity (e.g., “find shortest tour”)
- **Vertex cover:** set of vertices covering all edges in a graph; NP-complete decision version
- **Vertex cover approximation algorithm:** repeatedly pick uncovered edge, add both endpoints – factor-2 approximation

Classifications & Hierarchies

- **Algorithm classes by scope:** Correct vs. incorrect; Deterministic vs. randomized; Exact vs. approximate; Online vs. offline; Serial vs. parallel
- **Problem difficulty hierarchy:** P NP PSPACE EXPTIME (conjectured $P \neq NP$)

- **NP-completeness proof chain:** Circuit-SAT $\rightarrow$ SAT $\rightarrow$ 3-CNF-SAT $\rightarrow$ CLIQUE $\rightarrow$ VERTEX-COVER $\rightarrow$ HAM-CYCLE $\rightarrow$ TSP

Formulas & Equations

Algorithm speed comparison

A slow computer + fast algorithm > A fast computer + slow algorithm - Formal: $c \cdot n \cdot \lg n$ (merge sort) grows slower than $c \cdot n^2$ (insertion sort) for sufficiently large n

Running time growth

If an algorithm running on n inputs takes $f(n)$ microseconds: | $f(n)$ | 1 second | 1 minute | 1 hour | 1 day | |—|—|—|—|—|—| | $\lg n$ | $2^{(10^6)}$ | $2^{(6 \times 10^7)}$ | $2^{(3.6 \times 10^9)}$ | $2^{(8.64 \times 10^{10})}$ | | n | 10^6 | 6×10^7 | 3.6×10^9 | 8.64×10^{10} | | $n \lg n$ | 62746 | 2.8×10^6 | 1.33×10^8 | 2.76×10^9 | | n^2 | 1000 | 7746 | 60000 | 293939 | | 2^n | 19 | 26 | 32 | 37 |

Edge Cases & Common Pitfalls

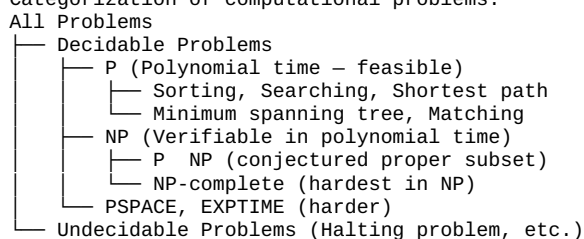
- **Exponential algorithms:** tiny increase in input size causes massive increase in runtime (e.g., 2^n for $n=100$ takes more time than atoms in universe)
- **Confusing problem class vs algorithm class:** P refers to problems solvable in polynomial time, NOT algorithms run in polynomial time
- **Approximation $\neq$ arbitrary:** approximation algorithms have provable worst-case ratio bounds (e.g., factor 2), not heuristics without guarantees
- **NP-completeness is a worst-case property:** some NP-complete instances may be easy; average-case complexity is different

Diagrams & Visuals

Running time growth for algorithms of varying efficiency:

n	$\lg n$	n	$n \lg n$	n^2	2^n
10	3.3	10	33	100	1024
100	6.6	100	664	10	1.27×10^3
1000	10.0	1000	9966	10	10^{31}
10	20.0	10	2×10	10^{12}	—

Categorization of computational problems:



End-of-Chapter Material (Additional)

- **Problem 1-1:** Comparison of running times – fill table showing max n solvable in 1 sec, 1 min, 1 hr, 1 day, 1 month, 1 year, 1 century for functions: $\lg n$, $\sqrt{n}$, n , $n \lg n$, n^2 , n^3 , 2^n , $n!$
 - Solution approach: set $f(n)$ = time bound, solve for n using growth rate inverses
- **Key terms review:** algorithm, computational problem, instance, correctness, efficiency, data structure, NP-complete, NP-hard,

approximation ratio, online algorithm

Cross-Chapter Links

- Referenced throughout: Ch.2 (sorting), Ch.4 (D&C), Ch.14 (DP), Ch.20 (graph), Ch.22 (shortest path), Ch.24 (max flow), Ch.26 (parallel), Ch.27 (online), Ch.29 (LP), Ch.30 (FFT), Ch.31 (RSA), Ch.33 (ML), Ch.34 (NP-complete), Ch.35 (approximation)
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Ch. 2 – Getting Started

Named Entities

- **Insertion sort**: incremental sorting, $\Theta(n^2)$ worst-case
- **Merge sort**: divide-and-conquer sorting, $\Theta(n \lg n)$ worst-case
- **Loop invariant**: property true before each loop iteration (used to prove correctness)
- **RAM model**: Random-Access Machine – instructions execute sequentially, constant time each
- **Divide-and-conquer**: break problem into subproblems, solve recursively, combine solutions
- **Recurrence**: equation describing running time recursively in terms of smaller inputs
- **Sentinel**: special value placed at end of array to simplify boundary checks

RAM Model Details

- **Instructions**: arithmetic (+, -, *, /, mod), data movement (LOAD, STORE), control (branch, subroutine call)
- **Cost model**: each simple instruction = 1 time unit; each memory access = 1 time unit
- **Word size**: fixed word length (e.g., $c \cdot \lg n$ bits for input size n)
- **No fancy operations**: no exponentiation or arbitrary-precision arithmetic in unit time
- **No parallelism**: instructions execute one after another sequentially
- **No hierarchy**: ignores cache, virtual memory, disk (RAM is flat)
- **Purpose**: provides clean framework for asymptotic analysis; real machines deviate but analysis transfers

Processes / Algorithms / Pathways

INSERTION-SORT(A, n)

- **Steps:**

```
INSERTION-SORT( $A, n$ )
for  $i = 2$  to  $n$ :
     $key = A[i]$ 
    // Insert  $A[i]$  into sorted subarray  $A[1:i-1]$ 
     $j = i - 1$ 
    while  $j > 0$  and  $A[j] > key$ :
         $A[j+1] = A[j]$ 
         $j = j - 1$ 
     $A[j+1] = key$ 
```

- **Loop invariant**: At start of each iteration i , $A[1:i-1]$ is sorted and contains original elements $A[1:i-1]$
- **Correctness proof**:

- **Initialization:** $i=2$, $A[1:1]$ has 1 element $\rightarrow$ trivially sorted
- **Maintenance:** inner while shifts elements $>$ key right; inserting key at correct position preserves sorted order
- **Termination:** $i=n+1$, $A[1:n]$ is sorted $\rightarrow$ algorithm correct
- **Complexity:** Best $\Theta(n)$ (already sorted – while never executes), Worst/Average $\Theta(n^2)$, Space $\Theta(1)$ in-place
- **Example:** $A=[5,2,4,6,1,3]$. $i=2$: key=2, shift 5 $\rightarrow$ 2. $i=3$: key=4, shift 5 $\rightarrow$ 3. $i=4$: no shift. $i=5$: key=1, shift all. $i=6$: key=3, shift 6,5,4 $\rightarrow$ 3. Final: $[1,2,3,4,5,6]$

MERGE(A, p, q, r)

- **Steps:**

```

MERGE(A, p, q, r)
n = q - p + 1
n = r - q
let L[1:n+1] and R[1:n+1] be arrays
for i = 1 to n:
    L[i] = A[p + i - 1]
for j = 1 to n:
    R[j] = A[q + j]
L[n+1] =  $\infty$  // sentinel
R[n+1] =  $\infty$  // sentinel
i = 1
j = 1
for k = p to r:
    if L[i]  $\leq$  R[j]:
        A[k] = L[i]
        i = i + 1
    else:
        A[k] = R[j]
        j = j + 1

```

- **Loop invariant:** At start of each k iteration, $A[p:k-1]$ contains $k-p$ smallest elements of LR in sorted order; $L[i]$ and $R[j]$ are smallest remaining elements of their respective arrays
- **Complexity:** $\Theta(n)$ where $n = r-p+1$
- **Space:** $\Theta(n)$ auxiliary for temporary L and R arrays

MERGE-SORT(A, p, r)

- **Steps:**

```

MERGE-SORT(A, p, r)
if p < r:
    q = (p+r)/2 // divide
    MERGE-SORT(A, p, q) // conquer left
    MERGE-SORT(A, q+1, r) // conquer right
    MERGE(A, p, q, r) // combine

```

- **Correctness:** induction on $n = r-p+1$. Base case $n=1$ ($p \geq r$) trivially sorted. Inductive step: assume MERGE-SORT correctly sorts left and right halves; MERGE produces correct merge of sorted lists.
- **Complexity:** $T(n) = 2T(n/2) + \Theta(n) \rightarrow \Theta(n \lg n)$. Space: $\Theta(n)$ auxiliary
- **Example:** $A=[12,3,7,9,14,6,11,2]$. Divide to singletons, merge: $[3,12] \& [7,9] \rightarrow [3,7,9,12]$; $[6,14] \& [2,11] \rightarrow [2,6,11,14]$; final merge $\rightarrow [2,3,6,7,9,11,12,14]$

Formulas & Equations

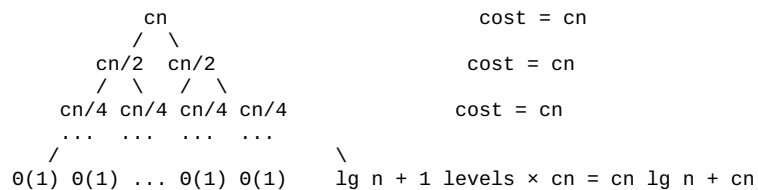
Insertion sort worst-case

$$T(n) = \sum^n (i - 1) = n(n-1)/2 = an^2 + bn + c = \theta(n^2)$$

Merge sort recurrence

$$T(n) = 2T(n/2) + \theta(n) \rightarrow T(n) = \theta(n \lg n)$$

Recursion tree for merge sort



Master theorem proof for merge sort

$$a=2, b=2, f(n)=cn. \log_b a = 1. f(n) = \theta(n^1) \rightarrow \text{Case 2} \rightarrow T(n) = \theta(n^1 \lg n) = \theta(n \lg n)$$

Rules, Laws & Theorems

Loop invariant correctness

- **Initialization:** true before first iteration
- **Maintenance:** true before $\rightarrow$ true after iteration
- **Termination:** loop ends; invariant + !condition $\rightarrow$ correctness

Generic divide-and-conquer recurrence

$T(n) = a \cdot T(n/b) + D(n) + C(n)$ - a = number of subproblems - n/b = size of each subproblem - $D(n)$ = cost to divide - $C(n)$ = cost to combine - If $D(n) + C(n) = \theta(n^d)$, then $T(n) = \theta(n^d)$ if $a < b^d$; $\theta(n^d \lg n)$ if $a = b^d$; $\theta(n^{\log_b a})$ if $a > b^d$

Correctness of MERGE

- **Invariant:** $L[1:n+1]$ and $R[1:n+1]$ are sorted; sentinels ensure no empty array check
- When one array exhausted, sentinel ∞ ensures remaining elements from other array are copied

Data Structures & Types

- **Array (RAM model):** $O(1)$ lookup by index, $\theta(n)$ insert/delete at arbitrary position
- **In-place vs. not:** insertion sort uses $O(1)$ extra space (in-place); merge sort uses $\theta(n)$ extra space (not in-place)

Comparisons & Trade-offs

Aspect	Insertion Sort	Merge Sort
Worst-case	$\theta(n^2)$	$\theta(n \lg n)$
In-place	Yes	No
Small n	Faster (tight constants)	Slower

Diagrams & Visuals

Insertion sort on [5,2,4,6,1,3]:

[5|2,4,6,1,3]→[2,5|4,6,1,3]→[2,4,5|6,1,3]→[2,4,5,6|1,3]→[1,2,4,5,6|3]→[1,2,3,4,5,6]

End-of-Chapter Material

- **Ex 2.2-2:** Selection sort – $\Theta(n^2)$ best and worst
- **Ex 2.3-6:** Binary search: $\Theta(\lg n)$
- **Prob 2-4:** Inversions – pair (i,j) where $i < j$ but $A[i] > A[j]$. Max = $n(n-1)/2$ (reverse sorted)

End-of-Chapter Material (Additional Exercises)

- **Ex 2.1-1:** Trace insertion sort on $A = [31,41,59,26,41,58]$ – both 41s remain in relative order (stable)
- **Ex 2.1-3:** Linear search – write pseudocode, prove loop invariant, $\Theta(n)$ time
- **Ex 2.2-1:** $\Theta(n^3) - n^3/1000 - 100n^2 - 100n + 3 = \Theta(n^3)$, dominant term is $n^3/1000$
- **Ex 2.3-2:** MERGE without sentinels – check bounds explicitly with if statements rather than sentinel values
- **Ex 2.3-7:** $\Theta(n \lg n)$ algorithm to check if set of n integers contains two elements summing to x – sort + two pointers
- **Prob 2-1:** Insertion sort on small arrays in merge sort – merge sort with insertion sort when subarrays $\leq k$: $k = \Theta(\lg n)$ optimizes $\Theta(nk + n \lg(n/k))$
- **Prob 2-3:** Horner's rule for polynomial evaluation – $\Theta(n)$ with n multiplications vs naive $\Theta(n^2)$; loop invariant proves correctness
- **Prob 2-4:** Inversions – pair (i,j) where $i < j$ but $A[i] > A[j]$;
 - Max = $C(n,2) = n(n-1)/2$ (reverse sorted)
 - Insertion sort swaps = number of inversions
 - Lower bound: any algorithm that swaps adjacent elements must perform $\Omega(n^2)$ swaps in worst case

Diagrams & Visuals

Selection sort on [5,2,4,6,1,3]:

[5,2,4,6,1,3] → swap 5↔1 → [1,2,4,6,5,3]

[1,2,4,6,5,3] → no swap → [1,2,4,6,5,3]

[1,2,4,6,5,3] → swap 4↔3 → [1,2,3,6,5,4]

[1,2,3,6,5,4] → swap 6↔4 → [1,2,3,4,5,6]

[1,2,3,4,5,6] → no swap → [1,2,3,4,5,6]

Divide-and-conquer paradigm:

DIVIDE(problem instance)

↓ ↓ ... ↓
subproblem A subproblem B subproblem X

↓ ↓ ↓
solve(A) solve(B) solve(X)

↓ ↓ ↓
COMBINE(sub-solutions) → solution to original problem

Cross-Chapter Links

- Ch.3 (Θ -notation formalized), Ch.4 (master theorem)

Ch. 3 – Characterizing Running Times

Named Entities

- **0-notation:** asymptotic upper bound ($f \leq c \cdot g$)

- **Ω -notation:** asymptotic lower bound ($f \geq c \cdot g$)
- **Θ -notation:** asymptotically tight bound ($cg \leq f \leq cg$)
- **o -notation:** upper bound that is NOT asymptotically tight ($f/g \rightarrow 0$)
- **ω -notation:** lower bound that is NOT asymptotically tight ($f/g \rightarrow \infty$)
- **Asymptotic efficiency:** care about growth rate as $n \rightarrow \infty$; ignore constants and lower-order terms
- **Monotonicity:** f is monotonically increasing if $m \leq n \implies f(m) \leq f(n)$
- **Polynomially bounded:** $f(n) = O(n^k)$ for some constant k
- **Exponential:** $f(n) = O(2^{\epsilon n})$ for some $\epsilon > 0$
- **Polylogarithmic:** $f(n) = O(\lg^k n)$ for some constant k

Formulas & Equations

O -notation (asymptotic upper bound)

$O(g(n)) = \{f(n): c > 0, n > 0 \text{ s.t. } 0 \leq f(n) \leq c \cdot g(n) \text{ } n \geq n\}$ - Examples: $n^2 = O(n^2)$, $2n^2 = O(n^2)$, $1000n^2 + 5n + 1 = O(n^2)$, $n = O(n^2)$ - Non-examples: $n^3 \neq O(n^2)$, $2^n \neq O(n^2)$

Ω -notation (asymptotic lower bound)

$\Omega(g(n)) = \{f(n): c > 0, n > 0 \text{ s.t. } 0 \leq c \cdot g(n) \leq f(n) \text{ } n \geq n\}$ - Examples: $n^2 = \Omega(n^2)$, $n^3 = \Omega(n^2)$, $2^n = \Omega(n^2)$ - Non-examples: $n \neq \Omega(n^2)$, $\lg n \neq \Omega(n)$

Θ -notation (asymptotically tight bound)

$\Theta(g(n)) = \{f(n): c, c > 0, n > 0 \text{ s.t. } 0 \leq c \cdot g(n) \leq f(n) \leq c \cdot g(n) \text{ } n \geq n\}$ - Examples: $n^2/2 + 3n = \Theta(n^2)$, $n \cdot \lg n = \Theta(n \lg n)$ - **Technique:** choose $c \leq \liminf f(n)/g(n)$ and $c \geq \limsup f(n)/g(n)$ - **Polynomial rule:** $a_k n^k + a_{k-1} n^{k-1} + \dots + a = \Theta(n^k)$ if $a_k > 0$

Theorem 3.1 (Relationship between O , Ω , Θ)

$f(n) = \Theta(g(n))$ iff $f(n) = O(g(n))$ and $f(n) = \Omega(g(n))$

o -notation (little-oh, non-tight upper bound)

$o(g(n)) = \{f(n): c > 0, n > 0 \text{ s.t. } 0 \leq f(n) < c \cdot g(n) \text{ } n \geq n\}$ - Equivalent to: $\lim_{n \rightarrow \infty} f(n)/g(n) = 0$ - Examples: $n = o(n^2)$, $n^2 = o(n^3)$, $\lg n = o(n)$ - Non-examples: $n^2 \neq o(n^2)$ (limit is constant, not zero)

ω -notation (little-omega, non-tight lower bound)

$\omega(g(n)) = \{f(n): c > 0, n > 0 \text{ s.t. } 0 \leq c \cdot g(n) \leq f(n) \text{ } n \geq n\}$ - Equivalent to: $\lim_{n \rightarrow \infty} f(n)/g(n) = \infty$ - Examples: $n^2 = \omega(n)$, $n^3 = \omega(n^2)$, $2^n = \omega(n^k)$ for any k - Non-examples: $n^2 \neq \omega(n^2)$ (limit is constant)

Asymptotic notation in equations

- On right side: stands for anonymous function (e.g., $T(n) = 2T(n/2) + \Theta(n)$)
- On left side: equality means "is" (e.g., $2n^2 + 3n + 1 = \Theta(n^2)$ means $2n^2 + 3n + 1 \in \Theta(n^2)$)
- Chain of equalities: e.g., $T(n) = 4T(n/2) + n = 4(2T(n/4) + n/2) + n = 8T(n/4) + 3n = \dots$
- **Asymptotic notation in recurrences:** $T(n) = 2T(n/2) + \Theta(n)$ means there exists $f(n) \in \Theta(n)$ s.t. $T(n) = 2T(n/2) + f(n)$

Growth rates (slowest $\rightarrow$ fastest)

1 $\lg \lg n$ $\lg n$ $(\lg n)^k$ n^ϵ n $n \lg n$ n^2 n^3 2^n $n!$ $2^{\{2^n\}}$ where $\epsilon > 0$, $k > 0$, and “” means “grows asymptotically slower than”

Detailed growth rate comparison table

Function	log scale	n=10	n=100	n=1000	n=10^6
$\lg \lg n$	~3.3 bits	1.7	2.3	3.3	4.3
$\lg n$	~3.3 bits	3.3	6.6	10	20
$\sqrt{n}$	0.5 $\lg n$	3.2	10	31.6	1000
n	1 $\lg n$	10	100	1000	10^6
$n \lg n$	> $\lg n$	33	664	9966	2×10^7
n^2	2 $\lg n$	100	10^4	10^6	10^12
n^3	3 $\lg n$	1000	10^6	10^9	10^18
2^n	$n \lg e$	1024	1.27×10^30	10^301	—
$n!$	$n \lg n$	3.6×10^6	9.3×10^157	—	—

Stirling's approximation

$n! = \sqrt{(2\pi n)} \cdot (n/e)^n \cdot (1 + o(1/n))$ - Useful for: bound on comparison sorting ($n!$ grows faster than 2^n), analyzing permutations - More precise: $n! = \sqrt{(2\pi n)} \cdot (n/e)^n \cdot e^{\{\epsilon_n\}}$ where $1/(12n+1) < \epsilon_n < 1/(12n)$ - Asymptotic: $\lg(n!) = n \lg n - n \lg e + o(\lg n)$

Golden ratio

$\phi = (1 + \sqrt{5}) / 2 \approx 1.61803$ $\phi = (1 - \sqrt{5}) / 2 \approx -0.61803$ - $\phi^2 = \phi + 1$, $\phi\phi = -1$, $\phi + \phi = 1$

Fibonacci numbers

$F = 0$, $F = 1$, $F = F_{i-1} + F_{i-2}$ for $i \geq 2$ - **Closed form (Binet's formula):** $F = (\phi^n - \phi^n) / \sqrt{5}$ - **Asymptotic:** $F = \phi/\sqrt{5} + 1/2$ and $F = o(\phi)$ - Fibonacci grows exponentially: $F \approx \phi/\sqrt{5}$ - Sum: $\sum^n F = F_{n+2} - 1$

Standard logarithms and exponentials

- Change of base: $\log_a n = \log_b n / \log_b a$ (constants matter in θ only via $\lg n$)
- **Natural log:** $\ln n = \log_e n$ where $e \approx 2.71828$
- Exponentiation: $a^{\{\log_b n\}} = n^{\{\log_b a\}}$ (useful for master theorem)
- **L'Hôpital's rule:** $\lim f(n)/g(n) = \lim f'(n)/g'(n)$ when both $\rightarrow \infty$ (compare growth rates)
- **Exponential dominates polynomial:** $n^b = o(c^n)$ for any $b > 0$, $c > 1$
- **Polynomial dominates logarithm:** $\lg^b n = o(n^c)$ for any $b > 0$, $c > 0$
- $(1 - 1/n)^n \rightarrow 1/e \approx 0.3679$ as $n \rightarrow \infty$

Common inequalities

- Arithmetic-geometric mean: $(x + x + \dots + x_n)/n \geq (x \cdot x \cdot \dots \cdot x_n)^{1/n}$
- Bernoulli's: $(1 + x)^n \geq 1 + nx$ for any $n \geq 0$ and $x > -1$
- Union bound: $\Pr[A \vee A \dots A_n] \leq \sum \Pr[A]$
- Markov's: $\Pr[X \geq a] \leq E[X]/a$ for nonnegative X
- Jensen's: $f(E[X]) \leq E[f(X)]$ for convex f ; $f(E[X]) \geq E[f(X)]$ for concave f

Summation formulas

- Arithmetic series: $\sum^n i = n(n+1)/2 = \theta(n^2)$

- Geometric series: $\sum^n r = (r^{n+1} - 1)/(r - 1) = \theta(r^n)$ for $r > 1$; $= \theta(1)$ for $r < 1$
- Harmonic series: $H_n = \sum^n 1/i = \ln n + \gamma + 1/(2n) - \dots$ where $\gamma \approx 0.5772$ (Euler's constant)
- Sum of squares: $\sum^n i^2 = n(n+1)(2n+1)/6 = \theta(n^3)$
- Sum of cubes: $(\sum^n i)^2 = n^2(n+1)^2/4 = \theta(n^4)$

Rules, Laws & Theorems

Theorem 3.1

$f(n) = \theta(g(n))$ iff $f(n) = O(g(n))$ and $f(n) = \Omega(g(n))$

Limit comparison test

- If $\lim_{n \rightarrow \infty} f(n)/g(n) = c$ where $0 < c < \infty$, then $f(n) = \theta(g(n))$
- If $\lim = 0$, then $f(n) = o(g(n))$ (and $g(n) = \omega(f(n))$)
- If $\lim = \infty$, then $f(n) = \omega(g(n))$ (and $g(n) = o(f(n))$)

Properties of asymptotic notation

- **Transitivity:** $f = \theta(g)$ and $g = \theta(h)$ $\implies f = \theta(h)$ (same for O , Ω , o , ω)
- **Reflexivity:** $f = \theta(f)$, $f = O(f)$, $f = \Omega(f)$ (not for o , ω)
- **Symmetry:** $f = \theta(g)$ iff $g = \theta(f)$
- **Transpose symmetry:** $f = O(g)$ iff $g = \Omega(f)$; $f = o(g)$ iff $g = \omega(f)$

Standard notation conventions

- $f(n) = O(g(n))$ means $f(n) \leq O(g(n))$ (one-way abuse of notation)
- $O(f(n)) + O(g(n)) = O(\max(f(n), g(n)))$
- $O(f(n)) \cdot O(g(n)) = O(f(n) \cdot g(n))$
- $\theta(f(n)) + \theta(g(n)) = \theta(\max(f(n), g(n)))$

Classifications & Hierarchies

- **Asymptotic hierarchy (strict):** $o \subset O$; $\omega \subset \Omega$; $\theta = O \cap \Omega$; o , ω are strict subsets of O , Ω respectively
- **Exponential hierarchy:** 2^n grows faster than any polynomial n ; n grows faster than any polylogarithmic $\log n$
- **Standard growth rate ordering:** constant $<$ logarithmic $<$ polylogarithmic $<$ polynomial $<$ exponential $<$ factorial

Data Structures & Types

- **Asymptotic notation is about functions, not algorithms:** any function $f: \rightarrow$ can be classified
- **Polynomial vs exponential classification:** determines tractability in algorithm analysis

Edge Cases & Common Pitfalls

- "At least $O(n^2)$ " is meaningless (O is upper bound) – use Ω instead
- "At most $\Omega(n^2)$ " is meaningless (Ω is lower bound) – use O instead
- Trichotomy fails: some functions are incomparable (e.g., n and $n^{1+\sin n}$)
- **Rate of growth not value:** $f(n) = O(g(n))$ does NOT mean $f(n) \leq g(n)$ at every point – just asymptotically
- **Constants hidden:** $O(100n) = O(n) = O(0.0001n^2)$ but actual runtime differs hugely

- **Multiple variables:** with 2D input sizes, treat each variable separately (e.g., $f(m,n) = O(m+n)$ not $O(\max(m,n))$)
- **Leading coefficient:** $an^2 + bn + c = O(n^2)$ regardless of a, b, c values (as long as $a > 0$)
- **Bases of logarithms:** $O(\log_a n) = O(\log_b n)$ because $\log_a n = \log_b n / \log_b a$ (constant factor)
- **Exponentials with different bases:** $2^n \neq O(1.5^n)$; $2^n = \omega(1.5^n)$ because $(2/1.5)^n \rightarrow \infty$
- **Floor/ceiling in recurrences:** $n/2$ and $\lceil n/2 \rceil$ can usually be ignored in asymptotic analysis
- **o -notation vs O -notation:** o is stricter – $f(n) = o(g(n))$ means $g(n)$ dominates $f(n)$ completely (ratio $\rightarrow 0$)

Standard Notation Conventions for Asymptotics

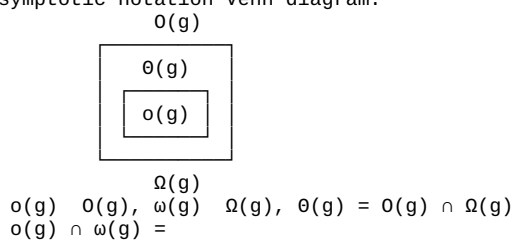
- **One-way equality:** $f(n) = O(g(n))$ means $f \in O(g)$ (abuse of notation)
- **Multiple variables:** $O(m+n)$ grows with both inputs; $O(mn)$ for matrix algorithms
- **Anonymous functions:** $T(n) = 2T(n/2) + o(n)$ means $f(n)$ such that $T(n) = 2T(n/2) + f(n)$
- **Set interpretation:** $O(g(n))$ is a set; $f(n) = O(g(n))$ means membership

Asymptotic Comparison Techniques

Technique	How it works	Example
Limit ratio	$\lim f(n)/g(n)$	n^2 vs $n^2 + n \rightarrow 1 \rightarrow \Theta$
L'Hôpital's rule	Compare derivatives	$\ln n$ vs $n \rightarrow 0 \rightarrow o$
Take logs	Compare log growth	$2^{\{n\}}$ vs $n! \rightarrow \lg(n!) \sim n \lg n$
Stirling	Approximate factorial	$n! \sim 2^n$
Induction	Guess and prove	$T(n) \leq cn \lg n$

Diagrams & Visuals

Asymptotic notation Venn diagram:



Standard growth rates visualized (log scale):

$n!$ ← steepest
 2^n ← steep
 n^3 ← cubic
 n^2 ← quadratic
 $n \lg n$ ← linearithmic
 n ← linear
 $\sqrt{n}$ ← sublinear
 $\lg n$ ← logarithmic
 $\lg \lg n$ ← very slow

Cross-Chapter Links

- Ch.2 (Θ introduced informally), Ch.4 (master theorem), App.A (summations), App.B (sets, relations)

Ch. 4 – Divide-and-Conquer

Named Entities

- **Recurrence:** equation describing function in terms of itself on smaller arguments
- **Substitution method:** guess + induction to prove
- **Recursion-tree method:** tree of recursive costs, sum per level
- **Master method:** cookbook for $T(n)=aT(n/b)+f(n)$
- **Akra-Bazzi method:** general method for unequal subproblem sizes
- **Strassen's algorithm:** $\Theta(n^{\lg 7}) \approx \Theta(n^{2.81})$ matrix multiplication
- **Watershed function:** $n^{\lg_b a}$ – the critical exponent comparing $f(n)$ against
- **Regularity condition:** $a \cdot f(n/b) \leq c \cdot f(n)$ for some $c < 1$ and all sufficiently large n (Master Theorem Case 3)
- **Polynomial growth:** $f(n)$ grows polynomially slower/faster than $n^{\lg_b a}$ for Cases 1/3
- **Matrix multiplication:** standard divide-and-conquer yields 8 multiplications of $n/2 \times n/2$ submatrices

Processes / Algorithms / Pathways

Substitution Method (step-by-step)

- **Goal:** solve recurrence $T(n) = 2T(n/2) + n$
- **Guess:** $T(n) = O(n \lg n)$, i.e., $T(n) \leq cn \lg n$
- **Inductive step:** assume $T(n/2) \leq cn/2 \lg(n/2)$
 - $T(n) \leq 2(cn/2 \lg(n/2)) + n \leq cn \lg(n/2) + n = cn \lg n - cn \lg 2 + n = cn \lg n - cn + n$
 - This holds if $-cn + n \leq 0$, i.e., $c \geq 1$
- **Base case:** $T(1) = \Theta(1) \leq c \cdot 1 \cdot \lg 1 = 0$ – fails, handle separately by choosing n large enough
- **Key technique:** subtract lower-order terms when induction doesn't match exactly
- **Common pitfalls:** need to prove exact form ($cn \lg n$), not just $O(n \lg n)$; base case must hold

Recursion-Tree Method (step-by-step for $T(n)=2T(n/2)+cn$)

Level 0:		cn		$= cn$	
Level 1:	$cn/2$		$cn/2$	$= cn$	
Level 2:	$cn/4$	$cn/4$	$cn/4$	$cn/4$	$= cn$
...					
Level k (leaves):	$\Theta(1)$	...	$\Theta(1)$		$= n \cdot \Theta(1)$

- Number of levels: $\lg n + 1$
- Total: $\sum_{k=0}^{\lg n} cn + n \cdot \Theta(1) = cn \lg n + cn + \Theta(n) = \Theta(n \lg n)$
- **General form** (a subproblems, size n/b): each level cost = $a^k \cdot f(n/b^k)$; leaf cost = $n^{\lg_b a} \cdot \Theta(1)$
- Total = $\sum_{k=0}^{\lg_b n - 1} a^k \cdot f(n/b^k) + \Theta(n^{\lg_b a})$

Strassen's Algorithm

- **Goal:** Multiply $n \times n$ matrices in $\Theta(n^{\lg 7})$
- **Steps:** (1) Partition into $n/2 \times n/2$ submatrices (2) Compute 10 sum/difference matrices (3) Recursively compute 7 products $P \dots P$ (4) Combine via additions/subtractions
- **Products:**
 - $P = A \cdot (B - B)$
 - $P = (A + A) \cdot B$
 - $P = (A + A) \cdot B$
 - $P = A \cdot (B - B)$
 - $P = (A + A) \cdot (B + B)$

- $P = (A - A) \cdot (B + B)$
- $P = (A - A) \cdot (B + B)$
- **Combine:**
 - $C = P + P - P + P$
 - $C = P + P$
 - $C = P + P$
 - $C = P + P - P - P$
- **Complexity:** $T(n) = 7T(n/2) + \Theta(n^2) \rightarrow \Theta(n^{\lg 7}) = \Theta(n^{2.80735})$
- **Practical note:** Strassen's is often slower than naive for $n < 100$ due to overhead; Coppersmith-Winograd $\Theta(n^{2.376})$ is mainly theoretical

Standard D&C recurrence derivation

- $T(n) = aT(n/b) + f(n)$ where:
 - a = number of subproblems
 - b = factor by which input shrinks
 - $f(n)$ = cost of dividing + combining

Formulas & Equations

Master Theorem (Theorem 4.1)

$T(n) = aT(n/b) + f(n)$ where $a > 0$, $b > 1$, f asymptotically positive

- **Case 1:** $f(n) = O(n^{\log_b a - \epsilon})$ for some $\epsilon > 0$ $T(n) = \Theta(n^{\log_b a})$
 - Intuition: leaves dominate total cost (polynomially more work at leaves)
- **Case 2:** $f(n) = \Theta(n^{\log_b a} \cdot \lg^k n)$ for $k \geq 0$ $T(n) = \Theta(n^{\log_b a} \cdot \lg^{k+1} n)$
 - Intuition: roughly equal work at each level; total = (height) $\times$ (cost per level)
 - Special case $k = 0$: $T(n) = \Theta(n^{\log_b a} \cdot \lg n)$
- **Case 3:** $f(n) = \Omega(n^{\log_b a + \epsilon})$ for some $\epsilon > 0$ AND $a \cdot f(n/b) \leq c \cdot f(n)$ for some $c < 1$ (regularity condition)
 - $T(n) = \Theta(f(n))$
 - Intuition: root dominates total cost (polynomially more work at root)

Master method examples – expanded

Recurrence	a	b	$\log_b a$	$f(n)$	Case	Solution
$9T(n/3) + n$	9	3	2	$n = O(n^{\lg 2 - \epsilon})$	1	$\Theta(n^2)$
$T(2n/3) + 1$	1	3/2	0	$1 = \Theta(n^{\lg 0} \lg^0 n)$	2	$\Theta(\lg n)$
$3T(n/4) + n \lg n$	3	4	0.793	$\Omega(n^{\lg 0.793 + \epsilon})$	3	$\Theta(n \lg n)$
$2T(n/2) + n \lg n$	2	2	1	$\Theta(n^{\lg 1} \lg^1 n)$	2	$\Theta(n \lg^2 n)$
$7T(n/2) + \Theta(n^2)$	7	2	$\lg 7 \approx 2.807$	$O(n^{\lg 2.807 - \epsilon})$	1	$\Theta(n^{\lg 7})$
$T(n/2) + 1$	1	2	0	$1 = \Theta(n^{\lg 0} \lg^0 n)$	2	$\Theta(\lg n)$
$4T(n/2) + n$	4	2	2	$n = O(n^{\lg 2 - \epsilon})$	1	$\Theta(n^2)$
$4T(n/2) + n^2$	4	2	2	$n^2 = \Theta(n^{\lg 2} \lg^0 n)$	2	$\Theta(n^2 \lg n)$
$4T(n/2) + n^3$	4	2	2	$n^3 = \Omega(n^{\lg 2 + \epsilon}); 4 \cdot (n/2)^3 = n^3/2 \leq cn^3$ for $c=1/2$	3	$\Theta(n^3)$
$2T(n/2) + 1$	2	2	1	$1 = O(n^{\lg 1 - \epsilon})$	1	$\Theta(n)$
$T(n/2) + n$	1	2	0	$n = \Omega(n^{\lg 0 + \epsilon})$	3	$\Theta(n)$

Master Theorem doesn't apply when:

- $f(n)$ is not polynomial (e.g., $f(n)=n \lg n$ with $a=2$, $b=2$ – actually Case 2 applies)
- $f(n)$ falls between cases (gap between Case 1 and Case 2, or Case 2 and Case 3)
- Regularity condition fails in Case 3 (e.g., $T(n)=2T(n/2)+n \lg n$ – actually this fits Case 2)
- Counterexample: $T(n) = 2T(n/2) + n/\lg n$ – $f(n)$ grows slower than n but not polynomially slower → master theorem fails, need Akra-Bazzi

Akra-Bazzi method

For recurrences with possibly unequal subproblem sizes: $T(n) = \sum a_i T(n/b_i) + f(n)$ - **Step 1:** Find unique real p satisfying $\sum a_i \cdot b_i^{-p} = 1$ - **Step 2:** Solution: $T(n) = \Theta(n^p + n^p \cdot \int^n f(x)/x^{p+1} dx)$ - **Examples:** - $T(n) = 2T(n/2) + n$: $a=2$, $b=2 \rightarrow 2 \cdot (1/2)^p = 1 \rightarrow p = 1 \rightarrow T(n) = \Theta(n + n \cdot \int^n (x/x^2) dx) = \Theta(n + n \cdot \ln n) = \Theta(n \lg n)$ - $T(n) = 2T(n/2) + n/\lg n$: $p = 1 \rightarrow T(n) = \Theta(n + n \cdot \int^n 1/(x \lg x) dx) = \Theta(n + n \cdot (\lg \lg n + C)) = \Theta(n \lg \lg n)$ - $T(n) = T(n/2) + T(n/3) + n$: $a=1$, $b=2$, $a=1$, $b=3 \rightarrow$ find p : $(1/2)^p + (1/3)^p = 1 \rightarrow p \approx 0.7878 \rightarrow T(n) = \Theta(n^p + n^p \cdot \int^n x/x^{p+1} dx) = \Theta(n^p + n^{p \cdot n\{1-p\}}) = \Theta(n)$

Substitution method – detailed proof for merge sort

Show $T(n) = 2T(n/2) + n = O(n \lg n)$ - **Guess:** $T(n) \leq cn \lg n$ for all $n \geq n_0$ (will determine c and n_0) - **Inductive hypothesis:** assume holds for all $m < n$ - **Inductive step** ($n \geq 2$): $T(n) = 2T(n/2) + n \leq 2 \cdot c \cdot n/2 \cdot \lg(n/2) + n \leq cn \cdot \lg(n/2) + n$ (since $n/2 \leq n/2$ and $n/2 \leq n$) $= cn \cdot \lg n - cn \cdot \lg 2 + n = cn \cdot \lg n - cn + n = cn \cdot \lg n + (1 - c)n \leq cn \cdot \lg n$ (if $c \geq 1$) - **Base case:** $T(1) = \Theta(1)$. Choose $n_0 = 2$, and pick c large enough so that $T(n) \leq cn \lg n$. For $n=2$: $T(2) = 2T(1) + 2 \leq c \cdot 2 \cdot \lg 2 = 2c$, works if $c \geq T(1) + 1$. For $n=3$: $T(3) = 2T(1) + 3 \leq 3c \cdot \lg 3$, works for sufficiently large c . - **Subtlety:** base case at $n=1$ fails because $\lg 1 = 0$; avoid by starting induction at $n = 2$

Substitution method – tight lower bound for merge sort

Show $T(n) = 2T(n/2) + n = \Omega(n \lg n)$ - **Guess:** $T(n) \geq cn \lg n$ - **Inductive step:** $T(n) \geq 2 \cdot c \cdot n/2 \cdot \lg(n/2) + n \geq cn \cdot \lg(n/2) + n = cn \cdot \lg n - cn + n \geq cn \cdot \lg n$ (if $c \leq 1$) - Works for any $c \leq 1$ (and sufficiently large n) - Combined: $T(n) = \Theta(n \lg n)$

Comparisons & Trade-offs

Algorithm	Multiplications	Additions/Subtractions	Time
Naive D&C	8	4 (combine C-C)	$\Theta(n^3)$
Strassen	7	18 (10 sum/diff + 8 combine)	$\Theta(n^{\{2.81\}})$
Coppersmith-Winograd	—	—	$\Theta(n^{\{2.376\}})$ (impractical)
Best known (2020+)	—	—	$\sim \Theta(n^{2.37286})$
Method	Best for		When to use
Master Theorem	Standard $aT(n/b)+f(n)$		Check cases quickly

Akra-Bazzi	Unequal subproblem sizes, floor/ceiling gaps	When master fails
Substitution	Any recurrence	When you have a good guess
Recursion tree	Visual intuition	Forming the initial guess

Diagrams & Visuals

Recursion tree for $T(n) = aT(n/b) + f(n)$:

Level 0:	$f(n)$	$\text{cost} = a^0 \cdot f(n/b^0)$
Level 1:	$f(n/b) \quad f(n/b) \quad \dots \quad f(n/b)$	$\text{cost} = a^1 \cdot f(n/b^1)$
Level 2:	$f(n/b^2) \dots f(n/b^2) \dots$	$\text{cost} = a^2 \cdot f(n/b^2)$
...		
Level i:	a^i nodes, each $f(n/b^i)$	$\text{cost} = a^i \cdot f(n/b^i)$
...		
Level L:	$\theta(1) \dots \theta(1)$ (leaves) where $L = \log_b n$	$\text{cost} = a^L \cdot \theta(1) = \theta(n^{\log_b a})$

$$\text{Total} = \sum_{i=0}^{L-1} a^i \cdot f(n/b^i) + \theta(n^{\log_b a})$$

Recursion tree for $T(n) = 2T(n/3) + T(n/4) + n$:

	n	$\text{cost} = n$
	$2 \cdot n/3 \quad 2 \cdot n/3 \quad n/4$	$\text{cost} \approx 1.417n$
	$4 \cdot n/9 \dots \dots \dots$	$\text{cost} \approx 2.007n$
	...	

(Akra-Bazzi needed – unequal sizes)

Substitution method flow:

Guess → Inductive hypothesis → Plug into recurrence →
Simplify → Choose constants → Verify base case → Done!

Edge Cases & Common Pitfalls

- **Master Theorem Case 3 regularity:** must check $a \cdot f(n/b) \leq c \cdot f(n)$ – not automatic for $f(n) = \Omega(n^{\log_b a + \epsilon})$
- **Master theorem gap:** when $f(n)$ is between cases (e.g., $f(n) = n \lg n$ with $a=2, b=2$: Case 2 applies with $k=1$, not a gap)
- **Floor/ceiling in recurrences:** usually ignorable for asymptotic; use induction to prove rigorously
- **$T(n) = 2T(n/2) + n \lg n$:** NOT Case 3 – $f(n) = n \lg n = \theta(n^1 \lg^1 n) =$ Case 2 with $k=1 \rightarrow \theta(n \lg^2 n)$
- **$T(n) = 3T(n/3) + n/2$:** Case 2 with $k=0 \rightarrow \theta(n \lg n)$, NOT Case 1
- **Akra-Bazzi integral divergence:** if $\int$ diverges, $T(n)$ dominated by integral term
- **Initial conditions:** recurrences typically assume $T(1) = \theta(1)$; choosing wrong base changes constants but not asymptotics
- **Negative work:** asymptotic analysis works for positive functions only
- **Recurrence with subtraction:** $T(n) = T(n-1) + T(n-2) + 1$ – exponential; no master theorem applies

End-of-Chapter Material (Additional Exercises & Problems)

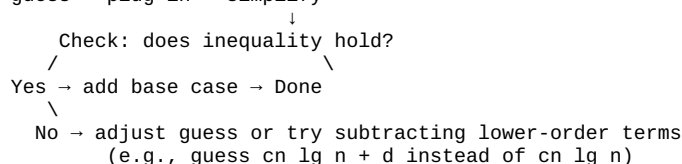
- **Ex 4.1-1:** FIND-MAXIMUM-SUBARRAY when all $A[j] < 0$ – returns smallest magnitude negative element (least loss)
- **Ex 4.1-5:** Linear-time maximum subarray algorithm – Kadane's algorithm: scan left to right, track max ending here = $\max(A[i], \text{max_ending_here} + A[i])$
- **Ex 4.2-1:** Trace Strassen on 2×2 matrices; verify C through C
- **Ex 4.3-1:** Solve $T(n) = T(n-1) + n$ by substitution – guess $\theta(n^2)$, prove cn^2 ; or recognize arithmetic series $\rightarrow \theta(n^2)$

- **Ex 4.4-1:** Recursion tree for $T(n) = 3T(n/2) + n$ – each level cost = $3^i \cdot n/2^i = n \cdot (3/2)^i$; geometric series sum = $\Theta(n^{\lg 3}) \approx \Theta(n^{1.585})$
- **Ex 4.5-1:** Use master theorem to solve:
 - $T(n) = 2T(n/4) + 1$: $a=2, b=4, \log_b a = 0.5, f(n)=1 = \Theta(n^{\{0.5-0.5\}})$ case 2 with $k=0$ $\Theta(\sqrt{n})$
 - $T(n) = 2T(n/4) + \sqrt{n}$: $a=2, b=4, \log_b a = 0.5, f(n)=n^{\{0.5\}}$ case 2 $\Theta(\sqrt{n} \lg n)$
 - $T(n) = 2T(n/4) + n$: $a=2, b=4, \log_b a = 0.5, f(n)=n = \Omega(n^{\{0.5+0.5\}})$ case 3 $\Theta(n)$
- **Ex 4.6-1:** Recurrence where master theorem does not apply – $T(n) = T(n/2) + n(2 - \cos n)$; $f(n)$ oscillates so no case applies
- **Prob 4-1:** Solve recurrences by master theorem (various $a, b, f(n)$ combinations)
- **Prob 4-3:** For $T(n) = T(n-1) + 1/k - \Theta(n)$; $T(n) = T(n-2) + 2/k - \Theta(n)$
- **Prob 4-5:** Chip testing – D&C: pair chips and test, discard pairs that don't agree, keep one from agreeing pairs; $O(n)$ time to find good chip
- **Prob 4-6:** Monge arrays – verify Monge property in $O(n)$ time with D&C

Diagrams & Visuals

Substitution method – finding the right guess:

Recurrence → try guess → plug in → simplify



Master Theorem decision flowchart:

$T(n) = aT(n/b) + f(n)$

↓

Compute $c^* = \log_b a$

↓

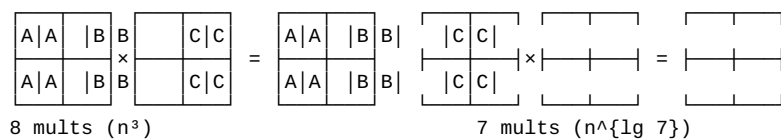
Compare $f(n)$ with $n^{\{c^*\}}$

- └ $f(n) = O(n^{\{c^*-\epsilon\}})$ → Case 1: $T(n) = \Theta(n^{\{c^*\}})$
- └ $f(n) = \Theta(n^{\{c^*\}} \cdot \lg^k n)$ → Case 2: $T(n) = \Theta(n^{\{c^*\}} \cdot \lg^{\{k+1\}} n)$
- └ $f(n) = \Omega(n^{\{c^*+\epsilon\}})$ AND $a \cdot f(n/b) \leq c \cdot f(n)$ → Case 3: $T(n) = \Theta(f(n))$

Divide-and-conquer matrix multiplication:

Standard:

Strassen:



Cross-Chapter Links

- Ch.2 (merge sort D&C intro), Ch.9 (selection recurrence), Ch.30 (FFT), App.A (summations)

Ch. 5 – Probabilistic Analysis and Randomized Algorithms

Named Entities

- **Probabilistic analysis:** use probability to analyze average case assuming input distribution
- **Randomized algorithm:** behavior determined by input + random choices
- **Indicator random variable:** $I\{A\} = 1$ if A occurs, 0 otherwise
- **Linearity of expectation:** $E[\sum X] = \sum E[X]$ (always holds, even with

dependence)

- **Average-case running time:** expected time over distribution of inputs
- **Expected running time:** expectation over algorithm's random choices (not input distribution)
- **Conditional expectation:** $E[X \mid Y = y]$ – expected value of X given $Y = y$
- **Bernoulli trial:** random experiment with two outcomes (success probability p , failure probability $1-p$)
- **Binomial random variable:** $X = \text{\#successes in } n \text{ independent Bernoulli trials}$; $\Pr[X=k] = C(n,k) \cdot p^k \cdot (1-p)^{n-k}$
- **Geometric distribution:** number of trials until first success; $E[X] = 1/p$; $\Pr[X=k] = (1-p)^{k-1} \cdot p$
- **Poisson distribution:** approximation for binomial when n large, p small; $\lambda = np$; $\Pr[X=k] = e^{-\lambda} \cdot \lambda^k / k!$
- **Random permutation:** each of $n!$ orderings equally likely (probability $1/n!$)
- **Priority queue permutation:** assign each element random priority, sort by priority; uniform permutation
- **Permute by sorting:** assign random numbers to elements, sort by those numbers; $O(n \lg n)$ time
- **Random sample:** uniformly random subset of size k from n elements

Indicator Random Variables – Detailed Theory

- **Definition:** $X_A = I\{A\} = \{ 1 \text{ if } A \text{ occurs, } 0 \text{ otherwise} \}$
- **Expectation:** $E[X_A] = 1 \cdot \Pr[A] + 0 \cdot \Pr[\bar{A}] = \Pr[A]$
- **Variance:** $\text{Var}[X_A] = \Pr[A] \cdot (1 - \Pr[A])$ (since $X_A^2 = X_A$)
- **Key insight:** convert event probability calculations into expectation of sum of indicators
- **Standard technique:** express random variable X as sum of indicator variables X_i , then $E[X] = \sum \Pr[\text{event}_i]$
- **Example – hiring problem:** Let $X_i = I\{\text{candidate } i \text{ is hired}\}$. Candidate i is hired iff i is better than all $1..i-1$. Since all orders are equally likely, $\Pr[X_i=1] = 1/i$. So $E[\text{\#hires}] = \sum_{i=1}^n 1/i = H_n = \ln n + O(1)$.

Variance of indicator variable sums

- $\text{Var}[\sum X_i] = \sum \text{Var}[X_i] + 2 \sum_{i < j} \text{Cov}[X_i, X_j]$
- For independent indicators: $\text{Var}[\sum X_i] = \sum p_i (1-p_i)$
- For birthday paradox: Covariance matters because matches are not independent

Processes / Algorithms / Pathways

HIRE-ASSISTANT(n)

- **Steps:**

```
HIRE-ASSISTANT( $n$ )
best = 0 // lowest-qualified dummy candidate
for i = 1 to  $n$ :
    interview candidate  $i$  (cost  $c$ )
    if candidate  $i$  is better than best:
        best =  $i$ 
        hire candidate  $i$  (cost  $c$ )
```

- **Cost:** $O(c \cdot n + c \cdot m)$ where $m = \text{number of hires}$
- **Expected hires:** $E[m] = \ln n + O(1)$ (harmonic series)

- **Expected cost:** $E[\text{total}] = c \cdot n + c \cdot (\ln n + \gamma) + O(1/n)$
- **Worst-case:** $O(c \cdot n + c \cdot n) = O(n)$ if candidates arrive in increasing order – every candidate hired
- **Best-case:** $O(c \cdot n + c) = O(n)$ if first candidate is best – only 1 hire
- **Variance of hires:** $\text{Var}[m] = \sum (1/i)(1 - 1/i) = \ln n + O(1)$ – standard deviation $\approx \sqrt{\ln n}$

RANDOMLY-PERMUTE(A, n) – uniform permutation

- **Steps:**

```

RANDOMLY-PERMUTE(A, n)
for i = 1 to n:
    swap A[i] with A[RANDOM(i, n)]

```

- **Correctness proof:** after i-th iteration, $A[1:i]$ is uniformly random i-permutation of original elements
- **Complexity:** $O(n)$ time, $O(1)$ extra space
- **Alternative – PERMUTE-BY-SORTING:** assign each $A[i]$ random priority $P[i]$, sort A by P; $O(n \lg n)$ time, risks collision if priorities not distinct

Hiring with online hiring (secretary problem variant)

- **Goal:** hire the best candidate when candidates interviewed one at a time, decision immediate
- **Strategy:** interview and reject first k candidates (training phase), then hire first candidate better than all seen
- **Success probability:** $\Pr[\text{success}] = (k/n) \cdot \sum_{i=k+1}^n 1/(i-1) = (k/n) \cdot (H_{n-1} - H_{k-1})$
- **Optimal k:** $k \approx n/e \approx 0.368n$. At optimal k, $\Pr[\text{success}] \rightarrow 1/e \approx 0.3679$ as $n \rightarrow \infty$
- **Intuition:** optimal strategy observes ~37% of candidates, then picks the first best-so-far

Formulas & Equations

Expected hires

$E[X] = \sum^n 1/i = H_n = \ln n + \gamma + 1/(2n) - 1/(12n^2) + O(1/n)$ where $\gamma \approx 0.57721566$ (Euler's constant)

Hiring variance

$\text{Var}[\text{#hires}] = \sum^n (1/i)(1 - 1/i) = H_n - \sum^n 1/i^2$ As $n \rightarrow \infty$, $\sum 1/i^2 \rightarrow \pi^2/6 \approx 1.6449$, so $\text{Var} \rightarrow \ln n + \gamma - \pi^2/6$

Birthday paradox – full derivation

- **Setting:** k people, n = 365 possible birthdays
- **Probability no match:** $\Pr[\text{no match}] = (n/n) \cdot ((n-1)/n) \cdot ((n-2)/n) \cdot \dots \cdot ((n-k+1)/n) = \prod_{i=1}^{k-1} (1 - i/n)$
- **Approximation:** for small x, $1 - x \approx e^{-x}$ $\Pr[\text{no match}] \approx \exp(-\sum_{i=1}^{k-1} i/n) = \exp(-k(k-1)/(2n))$
- **Threshold:** $\Pr[\text{no match}] \leq 1/2$ when $k(k-1)/(2n) \geq \ln 2 \approx 0.693$
 - For n=365: $k \approx 23$ (exact: $k \geq 23$, $\Pr[\text{match}] = 0.507$)

- General formula: $k \approx 1 + \sqrt{(1 + 8n \cdot \ln 2) / 2} \approx \sqrt{(2n \cdot \ln 2)} \approx 1.177\sqrt{n}$
- **Expected number of matches:** $E[\text{\#matches}] = C(k, 2)/n$
 - For $k=23$, $n=365$: $E[\text{\#matches}] = 253/365 \approx 0.693$
- **Expected minimum k for at least one match:** about $\sqrt{(\pi n/2)} \approx \sqrt{(1.571n)}$
- **Your birthday match:** probability someone shares your birthday = $1 - (364/365)^{k-1} \approx (k-1)/365$ for small k
 - For $k=23$: only ~6.1% chance someone shares your birthday specifically

Coupon collector – full analysis

- **Setting:** b distinct coupon types, each box contains one uniform random coupon
- **Let X = #boxes to collect all b types**
- **X partitioned into stages:** X_i = #boxes to get new type after having $i-1$ types
 - X_i follows geometric distribution with success probability $p_i = (b-i+1)/b$
- **Expectation:** $E[X] = \sum E[X_i] = \sum b/(b-i+1) = b \cdot \sum 1/j = b \cdot H_b$
- **Asymptotic:** $E[X] = b \cdot (\ln b + \gamma) + 1/2 + O(1/b)$
- **Variance:** $\text{Var}[X] = \sum (1 - p_i)/p_i^2 \leq (\pi^2/6) \cdot b^2$
- **Tail bound:** $\Pr[X > b \cdot \ln b + c \cdot b] \leq e^{-c}$ (exponentially small probability of needing many more than expectation)
- **Examples:**
 - $b = 6$ (dice faces): $E[X] = 6 \cdot (1 + 1/2 + 1/3 + 1/4 + 1/5 + 1/6) = 6 \cdot (49/20) = 14.7$
 - $b = 365$ (birthday types): $E[X] = 365 \cdot H \approx 365 \cdot 6.183 \approx 2257$ boxes
 - $b = 100$ (Pokémon cards): $E[X] = 100 \cdot H \approx 519$ expected packs

Streaks – longest run of heads

- n independent fair coin flips
- **Expected longest run:** $E[L_n] \approx \log n + O(1)$
- **More precisely:** expected longest run is $\sim \log n + \gamma/\ln 2 - 3/2 + O(\log n / \sqrt{n})$
- **Tail bound:** $\Pr[L_n \geq k] \leq n/2^k$ (union bound over positions)
- **Better bound:** $\Pr[L_n \geq c \cdot \lg n] \leq n^{-(c-1)}$ for $k = c \cdot \lg n$
 - For $c=3$ (streak of $3 \cdot \lg n$): probability $\leq 1/n^2$
- **Balls into bins:** throwing n balls into n bins – max load $\approx \ln n / \ln \ln n$ (with high probability)
- **When $m = \Omega(n \lg n)$ balls:** all bins have $O(m/n)$ balls with high probability

Online hiring (secretary problem) – full analysis

- **Setup:** n candidates in random order; after each interview, must accept/reject immediately; goal: maximize probability of selecting the best candidate
- **Strategy:** reject first k candidates; then select first candidate better than all first k
- **Success probability:**
 - Let M = position of best candidate. Best is selected if $M > k$ AND best among first $M-1$ is among first k
 - $\Pr[\text{success}] = \sum_{m=k+1}^n (1/n) \cdot (k/(m-1))$
 - $= k/n \cdot (1/k + 1/(k+1) + \dots + 1/(n-1))$
 - $= k/n \cdot (H_{n-1} - H_{k-1})$
- **Optimal k :** maximize $f(k) = (k/n) \cdot (H_{n-1} - H_{k-1})$
 - Approximation: treat k as continuous, set $f'(k) = 0 \rightarrow \ln(n/k) \approx 1 \rightarrow k \approx n/e$

- At $k = n/e$: $\Pr[\text{success}] = (1/e) \cdot (\ln e) = 1/e \approx 0.3679$
- **Alternative strategy**: sample ~37% of candidates, then pick first best-so-far; success probability exactly $1/e$ in the limit

Bernoulli trials and binomial distribution

- **Bernoulli**: single trial with success probability p
- **Binomial**(n, p): $X = \sum_{i=1}^n X_i$ where $X_i \sim \text{Bernoulli}(p)$
- $E[X] = np$, $\text{Var}[X] = np(1-p)$
- **Chebyshev**: $\Pr[|X - np| \geq \epsilon \cdot np] \leq (1-p)/(\epsilon^2 \cdot n \cdot p)$
- **Chernoff bound** (upper tail): $\Pr[X \geq (1+\delta)\mu] \leq (e^{\delta/(1+\delta)} \{1+\delta\})^{\mu}$ for $\mu = E[X]$
- **Chernoff bound simplified**: $\Pr[X \geq (1+\delta)\mu] \leq \exp(-\mu \cdot \delta^2/3)$ for $0 \leq \delta \leq 1$

Union bound (Boole's inequality)

$\Pr[A_i] \leq \sum \Pr[A_i]$ - Useful when individual events have small probability but many events - Example: random permutation - probability any element is in its original position $\leq n \cdot 1/n = 1$ (trivial), but more precise is $1 - 1/e$

Rules, Laws & Theorems

Linearity of expectation (always holds)

$E[aX + bY + c] = a \cdot E[X] + b \cdot E[Y] + c$ - no independence required! - **Key consequence**: can decompose complex random variable into sum of simple indicators

Indicator method for expected value

1. Identify events of interest $E_1, E_2, \dots, E_k$
2. Define indicator $X_i = I\{E_i\}$ for each
3. Express target random variable $X = \sum a_i X_i$
4. Take expectation: $E[X] = \sum a_i \Pr[E_i]$

Law of total expectation

$E[X] = E[E[X | Y]]$ - expectation of conditional expectation - Useful for recursive analysis (e.g., RANDOMIZED-SELECT, RANDOMIZED-QUICKSORT)

Birthday paradox result

With $k = \Theta(\sqrt{n})$ people, probability of at least one shared birthday is $\geq 1 - e^{-\Theta(1)}$ - For $n=365$: $k=23$ gives 50.7% probability - General: $k = \sqrt{(2n \cdot \ln 2)}$ suffices for $\geq 50\%$

Coupon collector expectation

$E[\text{\#boxes to collect all } b \text{ types}] = b \cdot H_b = b \cdot \ln b + O(b)$ - Variance: $O(b^2)$ - With $m = b \cdot \ln b + c \cdot b$ boxes: probability of missing at least one type $\leq e^{-c}$

Data Structures & Types

- **Random variable types**: discrete (countable outcomes) vs. continuous (uncountable)
- **Probability distributions for algorithm analysis**:
 - Uniform: each outcome equally likely (random permutation)

- Binomial: number of successes (hiring, hashing chain length)
- Geometric: time until event (coupon collector stages)
- Poisson: approximation for rare events (hashing collisions)
- **Random bits:** model of randomness – each call to RANDOM uses $O(\lg \text{range})$ random bits

Comparisons & Trade-offs

Aspect	Deterministic	Randomized
Input dependency	Behavior fixed per input	Behavior differs per run
Worst-case	Can construct bad input	No single bad input
Correctness	Always correct	May have small error probability
Analysis	Simple worst-case	Expected or high-probability
Performance	Fixed runtime	Runtime is random variable
Permutation method	Time	Uniform?
RANDOMLY-PERMUTE	$O(n)$ Yes (proved)	$O(1)$
PERMUTE-BY-SORTING	$O(n \lg n)$ Yes (distinct priorities)	$O(n)$
Shuffle (naive)	$O(n)$ No (biased)	$O(1)$
Online hiring	Deterministic	Randomized (random-order)
Success prob at $k=n/e$	$1/e$ (0.3679)	Same
Threshold	Fixed k	Same
Adversary	Can foil	Random order neutralizes adversary

Edge Cases & Common Pitfalls

- **Linearity of expectation DOES NOT require independence** – common misconception; $E[X+Y] = E[X] + E[Y]$ always
- **Variance does require independence:** $\text{Var}[X+Y] = \text{Var}[X] + \text{Var}[Y]$ only when X, Y uncorrelated
- **Expected $\neq$ high probability:** expected hires $\approx \ln n$, but with non-negligible probability m could be larger
- **Random permutation $\neq$ random priority:** assigning uniform randoms from finite set can produce collisions; need tie-breaking
- **Birthday paradox scales:** for $n=10^6$ birthdays, need only $k \approx 1178$ for 50% match probability – $O(\sqrt{n})$
- **Coupon collector is expensive:** needing $b \cdot \ln b$ samples to collect all types means linear time in b is insufficient
- **Randomized algorithm $\neq$ average-case analysis:** randomized guarantees hold for any input; average-case assumes input distribution
- **Probabilistic vs worst-case:** a randomized algorithm may have worst-case $\Theta(n^2)$ but expected $\Theta(n)$ against any input
- **Pseudorandomness:** real computers use PRNGs, not true randomness; analysis assumes true randomness

Cross-Chapter Links

- Appendix C (probability – counting, permutations, binomial, expectation, variance, geometric distribution, Poisson)
- Ch.7 (randomized quicksort uses indicator variables for expected comparisons)
- Ch.11 (hashing – expectation over random hash function, birthday paradox bounds chain length)
- Ch.13 (randomly built BST height analysis)
- Ch.30 (FFT randomization)

End-of-Chapter Material (Additional Exercises & Problems)

- **Ex 5.1-1:** Show that procedure HIRE-ASSISTANT always hires the best candidate – by definition, best = max over all seen
- **Ex 5.1-2:** Implement RANDOM(a,b) using RANDOM(0,1) – generate binary representation of range; discard out-of-range (rejection sampling)
- **Ex 5.1-3:** Implement BIASED-RANDOM that returns 1 w.p. p, 0 w.p. 1-p, produce unbiased RANDOM(0,1) – call twice: if 01→0, 10→1, else repeat (expected $1/(2p(1-p))$ calls)
- **Ex 5.2-1:** Probability HIRE-ASSISTANT hires exactly once – only possible if first candidate is best: probability = $1/n$
- **Ex 5.2-2:** Probability HIRE-ASSISTANT hires exactly twice – best candidate not first; first best among first $i-1$ and $i-1 < n$: probability $\approx (\ln n)/n$
- **Ex 5.2-4:** Birthday paradox with $n=365$ – expected number of people with same birthday as you = $(k-1)/n \approx k/365$. For $k=253$, expected ≈ 0.693
- **Ex 5.3-2:** Prof. Kelp's RANDOMLY-PERMUTE with RANDOM(1,n-i) instead of RANDOM(i,n) – does NOT produce uniform permutation; element 1 never moves past position 1
- **Ex 5.3-3:** Permute-by-sorting with duplicate priorities – not uniform permutation; $P(\text{all distinct}) = 1 \cdot (2^{w-1})^2 w \dots = \prod (1 - i/2^w)$ which is ≈ 1 for large w
- **Ex 5.4-1:** Birthday paradox with $n=365$ – how many people needed for 99% probability of a match? Solve $\prod (1 - i/365) \leq 0.01 \rightarrow k \approx 57$ (actually 57 gives 99.0%)
- **Ex 5.4-3:** Birthday paradox – expected number of pairs with same birthday = $C(k,2)/365$
- **Ex 5.4-4:** How many people must you ask so your birthday match probability exceeds 1/2? Solve $1 - (364/365)^{k-1} \geq 0.5 \rightarrow k \geq 254$ (since $(364/365)^{253} \approx 0.499$)
- **Ex 5.4-6:** Coupon collector with $b=6$ (dice) – expected tosses to see all 6 faces = $6 \cdot H = 6 \cdot (49/20) = 14.7$
- **Prob 5-1:** Probabilistic counting – estimate count n with $\lg n$ bits of randomness; $E[Y] = \lg n + O(1)$ where $Y = \max$ zero-run length in binary hashed values
- **Prob 5-2:** Searching unsorted array – random order vs deterministic; random search in expected $(n+1)/2$ comparisons
- **Prob 5-3:** Birthday paradox near matches – expected number of pairs with birthdays within 1 day = $O(k^2/n)$
- **Prob 5-4:** Balls into bins – throwing n balls into n bins: expected max load = $\Theta(\ln n / \ln \ln n)$; probability max load $> 3 \cdot \ln n / \ln \ln n \leq 1/n$

Diagrams & Visuals

Indicator variable method for birthday paradox:

Let $X = I\{\text{people } i \text{ and } j \text{ share birthday}\}$

$E[X] = \sum_{i < j} \Pr[i \text{ and } j \text{ share birthday}] = C(k,2) \cdot 1/n$

Expected matches = $C(k,2)/365 \approx k^2/730$

Variance of matches = $C(k,2)(1/365)(1 - 1/365) \approx k^2/730 \cdot (364/365)$

Coupon collector stages ($b=6$):

Stage 1: need 1st coupon, $p=6/6$, expected 1 box

Stage 2: need new type, $p=5/6$, expected $6/5 = 1.2$ boxes

Stage 3: need new type, $p=4/6$, expected $6/4 = 1.5$ boxes

Stage 4: need new type, $p=3/6$, expected $6/3 = 2$ boxes

Stage 5: need new type, $p=2/6$, expected $6/2 = 3$ boxes

Stage 6: need new type, $p=1/6$, expected $6/1 = 6$ boxes

Total expected = $1 + 1.2 + 1.5 + 2 + 3 + 6 = 14.7$ boxes

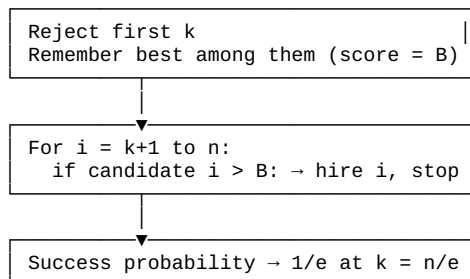
Hiring problem – indicator breakdown:

Candidate 1: hired (first) $X=1$ always

Candidate 2: hired if better than 1 $\Pr=1/2$ $X=I\{\text{candidate 2 best so far}\}$
 Candidate 3: hired if best of 1..3 $\Pr=1/3$ $X=I\{\text{candidate 3 best so far}\}$
 ...
 Candidate n: hired if best of 1..n $\Pr=1/n$ $X_n=I\{\text{candidate n best so far}\}$

$$E[\text{\#hires}] = 1 + 1/2 + 1/3 + \dots + 1/n = H_n = \ln n + \gamma$$

Secretary problem decision tree:
 n candidates, random order



Summary of Key Probability Distributions

Distribution	Parameters	PMF/PDF	E[X]	Var[X]	Used for
Bernoulli	p	$\Pr[1]=p, \Pr[0]=1-p$	p	$p(1-p)$	Single trial
Binomial	n, p	$C(n,k) p^k (1-p)^{n-k}$	np	$np(1-p)$	#successes in n trials
Geometric	p	$(1-p)^{k-1} p$	$1/p$	$(1-p)/p^2$	Wait until success
Poisson	λ	$e^{-\lambda} \lambda^k / k!$	λ	λ	Rare events

Common Probabilistic Bounds

Bound	Formula	Condition	Use case
Markov	$\Pr[X \geq a] \leq E[X]/a$	$X \geq 0$	Weak, needs only mean
Chebyshev	$\Pr[X - \mu \geq k\sigma] \leq 1/k^2$	$X - \mu$	
Chernoff (upper)	$\Pr[X \geq (1+\delta)\mu] \leq (e^{\delta/(1+\delta)})^{\mu}$	$X = \sum \text{indep. Bernoulli}$	Strong tail bound
Chernoff (lower)	$\Pr[X \leq (1-\delta)\mu] \leq e^{-\mu\delta^2/2}$	$0 < \delta < 1$	Lower tail
Union bound	$\Pr[A_i] \leq \sum \Pr[A_i]$	Always	Multiple bad events
Jensen	$f(E[X]) \leq E[f(X)]$ (convex f)	Convex/concave f	Expected value bounds

Randomized Algorithms Design Patterns

- **Foiling the adversary:** randomization ensures no single input causes worst-case behavior (e.g., RANDOMIZED-QUICKSORT)
- **Monte Carlo:** may produce incorrect result with bounded probability (e.g., primality testing)
- **Las Vegas:** always correct, running time is random variable (e.g., RANDOMIZED-SELECT, RANDOMIZED-QUICKSORT)
- **Random sampling:** use random subset to estimate properties of full set (e.g., approximation algorithms)
- **Symmetry breaking:** random choices to assign distinct roles (e.g., distributed algorithms)
- **Derandomization:** convert randomized algorithm to deterministic one

using method of conditional expectations

Ch. 7 – Quicksort

Named Entities

- **Quicksort**: Divide-and-conquer sorting algorithm; expected $\Theta(n \lg n)$, worst-case $\Theta(n^2)$.
- **Pivot**: Element chosen for partitioning.
- **Low side**: Elements $\leq$ pivot; **High side**: Elements $\geq$ pivot.
- **Hoare partition**: Original partitioning by C.A.R. Hoare using two pointers i, j from both ends.
- **Lomuto partition**: The simpler single-direction partition given in Section 7.1 (attributed to N. Lomuto).
- **Tail-recursion elimination**: Technique to convert recursive quicksort into iterative control structure.

Processes / Algorithms / Pathways

QUICKSORT(A, p, r)

- **Type**: Algorithm (divide-and-conquer, sorting, in-place)
- **Goal**: Sort subarray $A[p:r]$
- **Steps**:
 1. if $p < r$:
 2. $q = \text{PARTITION}(A, p, r)$ // divide
 3. QUICKSORT($A, p, q-1$) // conquer low side
 4. QUICKSORT($A, q+1, r$) // conquer high side
- **Combine**: Do nothing (already in place after conquer)

PARTITION(A, p, r)

- **Type**: Algorithm (subroutine)
- **Goal**: Partition $A[p:r]$ around pivot $A[r]$; return pivot's final index
- **Steps**:
 1. $x = A[r]$ // pivot
 2. $i = p - 1$
 3. for $j = p$ to $r-1$:
 4. if $A[j] \leq x$:
 5. $i += 1$; exchange $A[i]$ with $A[j]$
 6. exchange $A[i+1]$ with $A[r]$
 7. return $i+1$
- **Loop invariant**: $A[p:i] \leq x$, $A[i+1:j-1] > x$, $A[j:r-1]$ unknown, $A[r] = x$
- **Complexity**: Time $\Theta(n)$ on n elements
- **Example**: $A = [13, 19, 9, 5, 12, 8, 7, 4, 21, 2, 6, 11]$, $p=1$, $r=12$, pivot $A[12]=11$. $j=1$: $13 > 11 \rightarrow$ no swap. $j=2$: $19 > 11 \rightarrow$ no swap. $j=3$: $9 \leq 11 \rightarrow i=1$, swap $A[1]=13 \leftrightarrow A[3]=9 \rightarrow [9, 19, 13, 5, 12, 8, 7, 4, 21, 2, 6, 11]$. $j=4$: $5 \leq 11 \rightarrow i=2$, swap $19 \leftrightarrow 5 \rightarrow [9, 5, 13, 19, 12, 8, 7, 4, 21, 2, 6, 11]$. $j=5$: $12 > 11$. $j=6$: $8 \leq 11 \rightarrow i=3$, swap $13 \leftrightarrow 8 \rightarrow [9, 5, 8, 19, 12, 13, 7, 4, 21, 2, 6, 11]$. $j=7$: $7 \leq 11 \rightarrow i=4$, swap $19 \leftrightarrow 7 \rightarrow [9, 5, 8, 7, 12, 13, 19, 4, 21, 2, 6, 11]$. $j=8$: $4 \leq 11 \rightarrow i=5$, swap $12 \leftrightarrow 4 \rightarrow [9, 5, 8, 7, 4, 13, 19, 12, 21, 2, 6, 11]$. $j=9$: $21 > 11$. $j=10$: $2 \leq 11 \rightarrow i=6$, swap $13 \leftrightarrow 2 \rightarrow [9, 5, 8, 7, 4, 2, 19, 12, 21, 13, 6, 11]$. $j=11$: $6 \leq 11 \rightarrow i=7$, swap $19 \leftrightarrow 6 \rightarrow [9, 5, 8, 7, 4, 2, 6, 12, 21, 13, 19, 11]$. Final: swap $A[8]=12 \leftrightarrow A[12]=11 \rightarrow [9, 5, 8, 7, 4, 2, 6, 11, 21, 13, 19, 12]$; return $q=8$.

RANDOMIZED-PARTITION(A, p, r)

- **Type:** Algorithm
- **Goal:** Choose random pivot, then partition
- **Steps:**
 1. $i = \text{RANDOM}(p, r)$
 2. exchange $A[r]$ with $A[i]$
 3. return $\text{PARTITION}(A, p, r)$

RANDOMIZED-QUICKSORT(A, p, r)

- **Type:** Algorithm
- **Goal:** Sort with random pivot selection
- **Steps:** Same as QUICKSORT but calls RANDOMIZED-PARTITION

HOARE-PARTITION(A, p, r)

- **Type:** Algorithm
- **Goal:** Original partitioning by Hoare; pivot = $A[p]$; two-pointer from both ends
- **Steps:**
 1. $x = A[p]$; $i = p-1$; $j = r+1$
 2. while TRUE:
 3. repeat $j-$ until $A[j] \leq x$
 4. repeat $i++$ until $A[i] \geq x$
 5. if $i < j$: exchange $A[i]$ with $A[j]$
 6. else: return j
- **Key property:** Returns j where $p \leq j < r$; $A[p:j] \leq A[j+1:r]$

Classifications & Hierarchies

- **Partitioning schemes:** Lomuto (text, simpler) vs Hoare (original, faster on equal elements)
- **Recursion handling:** Standard (two recursive calls) vs Tail-recursion eliminated (iterative for one side)

Comparisons & Trade-offs

Dimension	Quicksort	Merge sort	Heapsort
Worst-case	$\Theta(n^2)$	$\Theta(n \lg n)$	$\Theta(n \lg n)$
Expected	$\Theta(n \lg n)$	$\Theta(n \lg n)$	$\Theta(n \lg n)$
In-place	Yes	No (needs $\Theta(n)$ aux)	Yes
Stable	No	Yes	No
Constant factor	Small	Moderate	Larger
Cache-friendly	Yes	Sequential	Random access
Practical speed	Fastest	average	—

Formulas & Equations

Worst-case recurrence

$$T(n) = T(n-1) + \Theta(n) \rightarrow T(n) = \Theta(n^2)$$

Best-case recurrence

$$T(n) = 2T(n/2) + \Theta(n) \rightarrow T(n) = \Theta(n \lg n)$$

Balanced split (α -to- β where $\alpha+\beta=1$)

$T(n) = T(\alpha n) + T(\beta n) + \Theta(n) \rightarrow T(n) = O(n \lg n)$ for any constant split

Expected comparisons (randomized)

$$\begin{aligned} E[X] &= \sum_{i=1}^{n-1} \sum_{j=i+1}^n \frac{2}{(j-i+1)} \\ &< \sum_{i=1}^{n-1} \sum_{k=1}^{n-i} \frac{2}{k} \\ &< \sum_{k=1}^n \frac{2n}{k} = O(n \lg n) \end{aligned}$$

Rules, Laws & Theorems

Lemma 7.1

- **Statement:** Running time of QUICKSORT on n -element array is $O(n + X)$ where X = number of element comparisons. ##### Lemma 7.2
- **Statement:** Elements z_i and z_j are compared iff one is chosen as pivot before any other element in $Z_{ij} = \{z_i, \dots, z_j\}$. ##### Lemma 7.3
- **Statement:** Probability that z_i and z_j are compared = $2/(j - i + 1)$. ##### Theorem 7.4
- **Statement:** Expected running time of RANDOMIZED-QUICKSORT is $O(n \lg n)$.

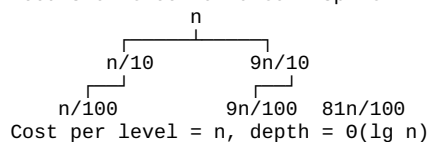
Edge Cases & Common Pitfalls

- **Already sorted array:** Deterministic quicksort with last-element pivot gives worst-case $\Theta(n^2)$ on sorted or reverse-sorted input.
- **All elements equal:** PARTITION returns $q = r$ (all on low side), leading to $\Theta(n^2)$.
- **Hoare vs Lomuto on equal elements:** Hoare partition performs better – Lomuto degrades to $\Theta(n^2)$.
- **Non-distinct elements:** Use ordered pairs $(A[i], i)$ to enforce distinctness at $O(n)$ extra space.
- **Stack depth:** Worst-case recursion depth $\Theta(n)$ can overflow stack; use tail-recursion elimination.
- **Median-of-3:** Choosing median of 3 random elements improves pivot quality; probability of bad split decreases.

Visual Patterns

PARTITION regions:
 $A[p : i] \leq x$ (tan)
 $A[i+1 : j-1] > x$ (blue)
 $A[j : r-1]$ unknown (white)
 $A[r] = x$ (yellow)

Recursion tree for 9-to-1 split:



End-of-Chapter Material

- **Key terms:** quicksort, pivot, partition, divide-and-conquer, randomized algorithm, tail recursion, median-of-3
- **Exercises:** 7.1-1 through 7.4-6 covering partition execution, performance analysis, randomized version, worst-case proof, expected analysis.
- **Problems:** 7-1 (Hoare partition), 7-2 (equal elements), 7-3 (alternative analysis), 7-4 (stooge sort), 7-5 (stack depth), 7-6 (median-of-3), 7-7 (fuzzy sorting).

Cross-Chapter Links

- **Requires:** Ch. 2 (divide-and-conquer, insertion sort), Ch. 4 (master theorem, substitution method), Ch. 5 (randomized algorithms, indicator random variables, harmonic series)
 - **Referenced in:** Ch. 8 (lower bound), Ch. 9 (RANDOMIZED-SELECT)
-

Ch. 8 – Sorting in Linear Time

Named Entities

- **Comparison sort:** Sorting algorithm that only uses comparisons between elements to determine order.
- **Decision tree:** Full binary tree representing comparisons performed by a comparison sort on n elements.
- **Counting sort:** Non-comparison sort for integers in range $[0, k]$; $\Theta(n + k)$ time, stable.
- **Radix sort:** Non-comparison sort sorting by least significant digit first; uses stable digit sorts.
- **Bucket sort:** Non-comparison sort assuming uniform distribution over $[0, 1]$; average $\Theta(n)$.
- **Stable sort:** Sorting algorithm that preserves relative order of equal elements.
- **Θ -1 sorting lemma:** If an oblivious compare-exchange algorithm sorts all Θ -1 inputs, it sorts all inputs.
- **Columnsort:** Oblivious compare-exchange sorting algorithm for rectangular arrays satisfying $r \geq 2s^2$.
- **Fusion tree:** Data structure for sorting n integers in $O(n \lg n / \lg \lg n)$ time (Fredman & Willard).

Processes / Algorithms / Pathways

COUNTING-SORT(A, n, k)

- **Type:** Algorithm (non-comparison sort, stable)
- **Goal:** Sort n integers in range $[0, k]$; output to $B[1:n]$
- **Steps:**
 1. Initialize $C[0:k] = 0$
 2. for $j = 1$ to n : $C[A[j]] += 1$ // count frequencies
 3. for $i = 1$ to k : $C[i] += C[i-1]$ // cumulative counts
 4. for $j = n$ down to 1 :
 5. $B[C[A[j]]] = A[j]$; $C[A[j]] -= 1$ // place in sorted order (stable)
 6. return B
- **Complexity:** Time $\Theta(n + k)$, Space $\Theta(n + k)$
- **Example:** $A = [6, 0, 2, 0, 1, 3, 4, 6, 1, 3, 2]$, $n=11$, $k=6$. After counting: $C = [2, 2, 2, 2, 1, 0, 2]$ (indices $0-6$). After prefix: $C = [2, 4, 6, 8, 9, 9, 11]$. Reverse scan: $j=11$: $A[11]=2 \rightarrow B[6]=2$, $C[2]=5$. $j=10$: $A[10]=3 \rightarrow B[8]=3$, $C[3]=7$. $j=9$: $A[9]=1 \rightarrow B[4]=1$, $C[1]=3$Final $B = [0, 0, 1, 1, 2, 2, 3, 3, 4, 6, 6]$.

RADIX-SORT(A, n, d)

- **Type:** Algorithm (non-comparison sort)
- **Goal:** Sort n d -digit numbers (digit 1 = least significant)
- **Steps:**
 1. for $i = 1$ to d :
 2. use a stable sort to sort A on digit i
- **Complexity:** Time $\Theta(d(n + k))$ with counting sort as stable sort
- **Example:** 3-digit numbers $[329, 457, 657, 839, 436, 720, 355]$ ($d=3$, $k=10$).

Sort on digit 1 (units): [720,355,436,457,657,329,839] Sort on digit 2 (tens): [720,329,436,839,355,457,657] Sort on digit 3 (hundreds): [329,355,436,457,657,720,839,839] → sorted.

BUCKET-SORT(A, n)

- **Type:** Algorithm (non-comparison sort, average-case linear)
- **Goal:** Sort n numbers uniformly distributed in $[0, 1)$
- **Steps:**
 1. Create n empty lists $B[0:n-1]$
 2. for $i = 1$ to n : insert $A[i]$ into $B[n \cdot A[i]]$
 3. for $i = 0$ to $n-1$: sort $B[i]$ with insertion sort
 4. Concatenate $B[0], B[1], \dots, B[n-1]$
- **Complexity:** Average-case $\Theta(n)$, Worst-case $\Theta(n^2)$
- **Example:** $A = [.79, .13, .16, .64, .39, .20, .89, .53, .71, .42]$, $n=10$. Bucket assignment: $.79 \rightarrow 7, .13 \rightarrow 1, .16 \rightarrow 1, .64 \rightarrow 6, .39 \rightarrow 3, .20 \rightarrow 2, .89 \rightarrow 8, .53 \rightarrow 5, .71 \rightarrow 7, .42 \rightarrow 4$. $B[1]=[.13, .16]$, $B[2]=[.20]$, $B[3]=[.39]$, $B[4]=[.42]$, $B[5]=[.53]$, $B[6]=[.64]$, $B[7]=[.71, .79]$, $B[8]=[.89]$. Sort each bucket (insertion sort), concatenate.

Classifications & Hierarchies

- **Sorting by method:** Comparison sorts (insertion, merge, heap, quick) vs Non-comparison sorts (counting, radix, bucket)
- **Non-comparison assumptions:** Counting sort (integers in small range), Radix sort (fixed-digit numbers), Bucket sort (uniform distribution)
- **Oblivious compare-exchange algorithms:** Insertion sort variant, column sort; 0-1 sorting lemma applies

Comparisons & Trade-offs

	Dimension Counting sort	Radix sort	Bucket sort
Time	$\Theta(n + k)$	$\Theta(d(n + k))$	Avg $\Theta(n)$, Wrst $\Theta(n^2)$
Key type	Integers $[0, k]$	d -digit base- k	Uniform real $[0, 1)$
In-place	No	No	No
Stable	Yes	Yes (if stable sub-sort)	Depends on sub-sort
Space	$\Theta(n + k)$	$\Theta(n + k)$	$\Theta(n)$
	Dimension	Comparison sort	Non-comparison sort
Lower bound		$\Omega(n \lg n)$	$O(n)$ possible
Input assumptions		None	Specific (integers, uniform, bounded)
Generality		Any comparable data	Restricted to specific key types

Formulas & Equations

Lower bound for comparison sorting

$h \geq \lg(n!) = \Omega(n \lg n)$ - h = decision tree height (worst-case comparisons) - n = number of elements

Radix sort with b -bit numbers, r bits per digit

$T(n) = \Theta((b/r)(n + 2^r))$ - b = bits per key - r = bits per digit ($r \leq b$) - Optimal $r = \lg n$ when $b \geq \lg n \rightarrow T(n) = \Theta(bn / \lg n)$

Bucket sort expected value analysis

$E[n_i] = 1$ (expected elements per bucket) $E[n_i^2] = \text{Var}[n_i] + E[n_i]^2 = (1-1/n) + 1 = 2 - 1/n$ $E[T(n)] = \Theta(n) + n \cdot O(2 - 1/n) = \Theta(n)$

Rules, Laws & Theorems

Theorem 8.1 (Lower bound for comparison sorts)

- **Statement:** Any comparison sort requires $\Omega(n \lg n)$ comparisons in the worst case.
- **Proof:** Binary decision tree of height h has $\leq 2^h$ leaves; $n!$ permutations require $n! \leq 2^h \rightarrow h \geq \lg(n!) = \Omega(n \lg n)$.
- **Corollary 8.2:** Heapsort and merge sort are asymptotically optimal comparison sorts.

Lemma 8.3 (Radix sort correctness)

- **Statement:** Radix sort correctly sorts n d -digit numbers in $\Theta(d(n+k))$ time if stable sort takes $\Theta(n+k)$.

Lemma 8.4 (Radix sort with bits)

- **Statement:** Given n b -bit numbers and $r \leq b$, radix sort sorts in $\Theta((b/r)(n+2^r))$ time.

0-1 Sorting Lemma

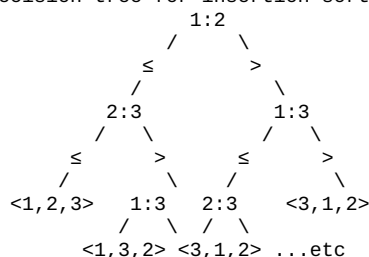
- **Statement:** If an oblivious compare-exchange algorithm correctly sorts all 0-1 inputs, it sorts all inputs.

Edge Cases & Common Pitfalls

- **Counting sort stability:** Must iterate in reverse ($j = n$ down to 1) to maintain stability; forward iteration produces correct output but unstable.
- **Radix sort digit order:** Must sort LSB first; sorting MSD first does not work and creates many intermediate piles.
- **Bucket sort worst case:** All elements fall into one bucket $\rightarrow \Theta(n^2)$. Use insertion sort (or any $O(n^2)$ sort) within buckets.
- **Input distribution:** Bucket sort's linear-time guarantee depends on uniform independent distribution; non-uniform input degrades performance.
- **Counting sort range:** When $k = \omega(n)$, counting sort is worse than comparison sorts.
- **Radix sort trade-off:** Small r increases number of passes; large r increases counting sort array size (2^r).

Visual Patterns

Decision tree for insertion sort on 3 elements:



Counting sort illustration (Fig 8.2):
A: [2,5,3,0,2,3,0,3], $k=5$

C after count: [2,0,2,3,0,1]
C after prefix: [2,2,4,7,7,8]
Reverse fill B: [0,0,2,2,3,3,5]

End-of-Chapter Material

- **Key terms:** comparison sort, decision tree, counting sort, stable sort, radix sort, bucket sort, oblivious compare-exchange, 0-1 sorting lemma, k-sorted
- **Exercises:** 8.1-1 through 8.4-6 covering lower bound, counting sort, radix sort, bucket sort analysis.
- **Problems:** 8-1 (probabilistic lower bounds), 8-2 (sorting in place in linear time), 8-3 (variable-length items), 8-4 (water jugs), 8-5 (average sorting / k-sorted), 8-6 (lower bound on merging), 8-7 (0-1 sorting lemma & columnsort).

Cross-Chapter Links

- **Requires:** Ch. 2 (insertion sort), Ch. 3 (Stirling's approximation, equation 3.28), Ch. 5 (probabilistic analysis), App. C (permutations, binomial distribution, harmonic series), Ch. 6 (heapsort, priority queues)
 - **Referenced in:** Ch. 9 (comparison model discussion)
-

Ch. 9 – Medians and Order Statistics

Named Entities

- **i-th order statistic:** The i-th smallest element of a set.
- **Lower median:** $i = (n+1)/2$; **Upper median:** $i = (n+1)/2$.
- **Selection problem:** Given set A of n distinct numbers and integer i ($1 \leq i \leq n$), find element > exactly i-1 others.
- **RANDOMIZED-SELECT:** Randomized divide-and-conquer selection; expected $\Theta(n)$, worst-case $\Theta(n^2)$.
- **SELECT:** Deterministic selection algorithm; worst-case $\Theta(n)$.
- **Middle half:** Elements excluding smallest $n/4-1$ and largest $n/4-1$.
- **Helpful partitioning:** Partitioning where $|A(j)| \leq (3/4)|A(j-1)|$.
- **Weighted median:** Element x_k satisfying cumulative weight $< 1/2$ on left and $\leq 1/2$ on right.
- **Post-office location problem:** Find point p minimizing $\sum w_i \cdot d(p, p_i)$.

Processes / Algorithms / Pathways

MINIMUM(A, n)

- **Type:** Algorithm
- **Goal:** Find minimum of n elements
- **Steps:**
 1. $\text{min} = A[1]$
 2. for $i = 2$ to n : if $\text{min} > A[i]$: $\text{min} = A[i]$
 3. return min
- **Complexity:** Time $\Theta(n)$ with exactly $n-1$ comparisons (optimal lower bound).

Simultaneous MIN & MAX

- **Type:** Algorithm
- **Goal:** Find both min and max in $\leq 3n/2$ comparisons

- **Steps:**
 1. Process elements in pairs; compare pair members first
 2. Compare smaller to min, larger to max (3 comparisons per 2 elements)
 3. If n odd: initialize min = max = first element; if n even: 1 comparison on first pair
- **Complexity:** Time $\Theta(n)$, $\leq 3n/2$ comparisons

RANDOMIZED-SELECT(A, p, r, i)

- **Type:** Algorithm (randomized, divide-and-conquer)
- **Goal:** Return i-th smallest element of A[p:r]
- **Steps:**
 1. if p == r: return A[p] // base case, i=1
 2. q = RANDOMIZED-PARTITION(A, p, r)
 3. k = q - p + 1 // number of elements in low side + pivot
 4. if i == k: return A[q] // pivot is answer
 5. elif i < k: recurse on A[p:q-1] with i
 6. else: recurse on A[q+1:r] with i-k
- **Complexity:** Expected $\Theta(n)$, worst-case $\Theta(n^2)$
- **Example:** Find 5th smallest in A=[2,3,0,5,7,9,1,8,6,4], n=10. Suppose pivot=4 (partition around 4): low=[2,3,0,1,4], high=[5,7,9,8,6]. k=5. i=5 == k → return 4.

SELECT(A, p, r, i)

- **Type:** Algorithm (deterministic, worst-case linear)
- **Goal:** Return i-th smallest element with guaranteed $\Theta(n)$ worst-case time
- **Steps:**
 1. While (r-p+1) not divisible by 5: find min and reduce (lines 1-10)
 2. g = n/5 groups of 5 elements each
 3. Sort each group of 5 (insertion sort)
 4. Recursively find median x of the g group medians: x = SELECT(A, p+2g, p+3g-1, g/2)
 5. q = PARTITION-AROUND(A, p, r, x) // partition around x
 6. k = q-p+1; if i==k return A[q]; elif i<k recurse on left; else recurse on right with i-k
- **Complexity:** Time $\Theta(n)$ in worst case
- **Recurrence:** $T(n) \leq T(n/5) + T(7n/10) + \Theta(n) \rightarrow T(n) = \Theta(n)$

Classifications & Hierarchies

- **Selection methods:** Naïve (sort → $O(n \lg n)$); Randomized (RANDOMIZED-SELECT → expected $O(n)$); Deterministic (SELECT → $O(n)$ worst-case)
- **Median determination:** Lower median $((n+1)/2)$ vs Upper median $((n+1)/2)$
- **Selector properties:** Comparison-based (unlike linear-time sorting, can beat $\Omega(n \lg n)$ because it does not sort all elements)

Comparisons & Trade-offs

Dimension	RANDOMIZED-SELECT	SELECT	Sort + index
Worst-case	$\Theta(n^2)$	$\Theta(n)$	$O(n \lg n)$
Expected	$\Theta(n)$	—	—
Practical	Yes (small constants)	Mostly theoretical	Simple
Assumptions	Distinct elements	Distinct elements	None

Dimension	Find min only	Find min & max separately	Find min & max together
Comparisons	$n-1$	$2n-2$	$\leq 3n/2$
Method	Single pass	Two passes	Pairwise processing

Formulas & Equations

RANDOMIZED-SELECT recurrence (helpful partitioning)

$$T(n) \leq T(3n/4) + \theta(n) \rightarrow T(n) = \theta(n) \text{ (case 3, master theorem)}$$

SELECT recurrence

$$T(n) \leq T(n/5) + T(7n/10) + \theta(n) \rightarrow T(n) = \theta(n) - n/5: \text{ recursive call to find median of group medians} - 7n/10: \text{ worst-case size of the side of partition containing the answer}$$

Minimum comparisons lower bound

$n - 1$ comparisons needed to find minimum (each non-winner must lose ≥ 1 comparison)

Simultaneous MIN & MAX comparisons

$$\leq 3n/2 \text{ comparisons total}$$

Rules, Laws & Theorems

Lemma 9.1

- **Statement:** A partitioning is helpful (reduces remaining elements by $\geq 1/4$) with probability at least $1/2$.
- **Proof:** Random pivot has probability $\geq 1/2$ of falling in middle half (excluding smallest $n/4-1$ and largest $n/4-1$).

Theorem 9.2

- **Statement:** RANDOMIZED-SELECT has expected running time $\theta(n)$ on distinct elements.
- **Proof:** Break into generations between helpful partitionings; each generation has expected size ≤ 2 ; total comparisons $< n \cdot \sum_{k=0}^{\infty} (3/4)^k \cdot 2 = O(n)$.

Theorem 9.3

- **Statement:** SELECT has worst-case running time $\theta(n)$.
- **Proof:** Groups of 5, median-of-medians pivot guarantees at most $7n/10$ elements on the larger side; recurrence solves to $T(n) = \theta(n)$.

Edge Cases & Common Pitfalls

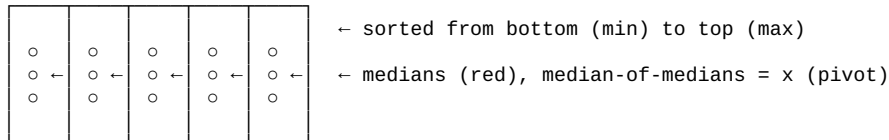
- **RANDOMIZED-SELECT never calls 0-length subarray:** When $i \neq k$, the pivot $A[q]$ is excluded and there are enough elements on correct side.
- **Worst-case sequence:** Always selecting maximum as pivot leads to $\theta(n^2)$ – pick pivots 9,8,7,6,5,... when searching for minimum.
- **SELECT with groups of 3:** Fails to give linear time (gives $O(n \lg n)$) because the $7n/10$ bound becomes $n - \theta(n) \rightarrow$ recurrence does not converge.

- **SELECT with groups of 7:** Still works in linear time (any odd constant ≥ 5 works).
- **Non-distinct elements:** Use ordered pairs $(A[i], i)$ to enforce distinctness.
- **SELECT for small i :** When i is small (e.g., constant), simpler tournament methods use fewer comparisons.

Visual Patterns

SELECT grouping (Fig 9.3):

5 elements per column, sorted bottom-to-top
Columns: 1 2 3 ... g ($g = n/5$)



At least $3g/2$ elements guaranteed $\leq x$ (blue region)
At least $3g/2$ elements guaranteed $\geq x$ (yellow region)
Remaining $\leq 7n/10$ elements could be on either side

RANDOMIZED-SELECT recursion:

```
A[p:r] (n elements)
└─ PARTITION → q (pivot index)
    └─ i == k → return pivot
    └─ i < k → recurse left (A[p:q-1])
    └─ i > k → recurse right (A[q+1:r], i-k)
```

End-of-Chapter Material

- **Key terms:** order statistic, median, selection problem, minimum, maximum, randomized selection, helpful partitioning, median-of-medians, weighted median, post-office location problem, quantiles
- **Exercises:** 9.1-1 through 9.3-10 covering min/max lower bounds, RANDOMIZED-SELECT analysis, SELECT with different group sizes, weighted median, quantiles, pipeline problem.
- **Problems:** 9-1 (largest i numbers), 9-2 (variant of randomized selection), 9-3 (weighted median), 9-4 (small order statistics), 9-5 (alternative analysis), 9-6 (select with groups of 3).

Cross-Chapter Links

- **Requires:** Ch. 2 (sorting by sorting + indexing), Ch. 4 (master theorem), Ch. 5 (indicator random variables, Bernoulli trials, geometric distribution), Ch. 7 (RANDOMIZED-PARTITION, PARTITION), Ch. 8 (comparison model lower bound)
- **Referenced in:** Ch. 13 (order-statistic trees), Problems 9-3 (post-office location)

Chapter-by-Chapter Breakdown

Ch. 10 – Elementary Data Structures

Named Entities (Terms & Definitions)

- **Array:** contiguous sequence of bytes; element i at address $a + b(i - s)$ for 1-origin ($s=1$) or $a + bi$ for 0-origin ($s=0$); $O(1)$ access
- **Matrix:** $m \times n$ grid; stored row-major or column-major in one or more arrays
- **Row-major order:** $M[i, j]$ at index $n(i-1)+j$ (1-origin); rows stored

contiguously

- **Column-major order:** $M[i,j]$ at index $i + m(j-1)$ (1-origin); columns stored contiguously
- **Block representation:** matrix divided into blocks, each stored contiguously
- **Stack:** LIFO dynamic set; INSERT=PUSH, DELETE=POP; attribute S.top
- **Queue:** FIFO dynamic set; INSERT=ENQUEUE, DELETE=DEQUEUE; attributes Q.head, Q.tail; uses circular array
- **Deque:** double-ended queue; insertion/deletion at both ends
- **Linked list:** linear order determined by pointers (not array indices)
- **Doubly linked list:** each node has key, next, prev pointers
- **Singly linked list:** each node has next but no prev pointer
- **Circular list:** tail.next = head, head.prev = tail
- **Sentinel:** dummy node (L.nil) simplifying boundary conditions; replaces NIL references
- **Left-child, right-sibling representation:** each node has x.left-child (leftmost child) and x.right-sibling (next sibling); $O(n)$ space for n-node tree

Processes / Algorithms / Pathways

STACK-EMPTY(S)

- **Type:** Algorithm
- **Steps:** (1) if S.top == 0 return TRUE (2) else return FALSE
- **Complexity:** $O(1)$

PUSH(S, x)

- **Type:** Algorithm
- **Goal:** Insert element x onto stack
- **Steps:** (1) if S.top == S.size → error “overflow” (2) S.top = S.top + 1 (3) S[S.top] = x
- **Complexity:** $O(1)$
- **Example:** Stack S = [1,4,9,16], S.top=4. PUSH(S,25): S.top=5, S[5]=25. Stack now [1,4,9,16,25].

POP(S)

- **Type:** Algorithm
- **Goal:** Remove and return top element
- **Steps:** (1) if STACK-EMPTY(S) → error “underflow” (2) S.top = S.top - 1 (3) return S[S.top + 1]
- **Complexity:** $O(1)$
- **Example:** POP from [1,4,9,16]: returns 16, S.top=3.

ENQUEUE(Q, x)

- **Type:** Algorithm
- **Steps:** (1) Q[Q.tail] = x (2) if Q.tail == Q.size → Q.tail = 1 else Q.tail = Q.tail + 1
- **Complexity:** $O(1)$
- **Example:** Q=[, , , 15, 6, 9, 8, 4], head=4, tail=1. ENQUEUE(Q,17): Q[1]=17, tail=2.

DEQUEUE(Q)

- **Type:** Algorithm
- **Steps:** (1) x = Q[Q.head] (2) if Q.head == Q.size → Q.head = 1 else Q.head = Q.head + 1 (3) return x

- **Complexity:** $O(1)$

LIST-SEARCH(L, k)

- **Steps:** (1) $x = L.head$ (2) while $x \neq NIL$ and $x.key \neq k$: $x = x.next$ (3) return x
- **Complexity:** $O(n)$ worst-case
- **Example:** List $L = [9, 16, 4, 1]$. LIST-SEARCH(L, 4) returns pointer to 3rd element. LIST-SEARCH(L, 7) returns NIL.

LIST-PREPEND(L, x)

- **Steps:** (1) $x.next = L.head$ (2) $x.prev = NIL$ (3) if $L.head \neq NIL$: $L.head.prev = x$ (4) $L.head = x$
- **Complexity:** $O(1)$

LIST-INSERT(x, y)

- **Steps:** Insert x immediately after y : (1) $x.next = y.next$ (2) $x.prev = y$ (3) if $y.next \neq NIL$: $y.next.prev = x$ (4) $y.next = x$
- **Complexity:** $O(1)$

LIST-DELETE(L, x)

- **Steps:** (1) if $x.prev \neq NIL$: $x.prev.next = x.next$ else $L.head = x.next$ (2) if $x.next \neq NIL$: $x.next.prev = x.prev$
- **Complexity:** $O(1)$ (plus $O(n)$ to find the element via LIST-SEARCH)

LIST-SEARCH'(L, k) – sentinel version

- **Steps:** (1) $L.nil.key = k$ (2) $x = L.nil.next$ (3) while $x.key \neq k$: $x = x.next$ (4) if $x == L.nil$ return NIL else return x
- **Complexity:** $O(n)$; eliminates one comparison per iteration

Comparisons & Trade-offs

Dimension	Array	Doubly Linked List
Access k-th element	$O(1)$	$O(k)$
Insert at front	$O(n)$ (must shift)	$O(1)$ (LIST-PREPEND)
Delete first element	$O(n)$	$O(1)$
Search	$O(n)$ (unsorted)	$O(n)$
Space	n elements	n elements + $2n$ pointers
Sentinels benefit	N/A	Simplifies code, small constant speedup

Formulas & Equations

Array element address (1-origin)

address = $a + b(i - 1)$ - a = base memory address [bytes] - b = element size [bytes] - i = 1-based index

Row-major index (1-origin)

index = $n(i - 1) + j$ - n = number of columns - i = row index, j = column

- **Requires:** Ch. 2 (RAM model, Big-O notation), Appendix B (tree math)
- **Referenced in:** Ch. 6 (heaps as arrays), Ch. 11 (chaining uses linked lists), Ch. 12 (BST uses binary tree representation), Ch. 13 (red-black trees use sentinel)

Ch. 11 – Hash Tables

Named Entities (Terms & Definitions)

- **Direct-address table:** array $T[0:m-1]$ where slot k stores element with key k ; $O(1)$ dictionary operations
- **Hash table:** array $T[0:m-1]$ where element with key k stored in slot $h(k)$; size $m \geq |U|$
- **Hash function h :** $U \rightarrow \{0, 1, \dots, m-1\}$; maps keys to slots
- **Collision:** two distinct keys hash to same slot
- **Load factor α :** $\alpha = n/m$ (n elements, m slots); for chaining α can be >1 ; for open addressing $\alpha \leq 1$
- **Chaining:** each slot points to linked list of keys hashing to that slot
- **Independent uniform hashing:** $h(k)$ uniformly random from $\{0, \dots, m-1\}$, independent across keys
- **Universal hashing:** family H where for any distinct k, k' , $\Pr[h(k)=h(k')] \leq 1/m$
- **-universal:** collision probability $\leq 1/m$
- **d-independent:** for any d distinct keys, $\Pr[h(k)=q \text{ for } d \text{ keys}] = 1/m^d$
- **Division method:** $h(k) = k \bmod m$ (m prime, not near power of 2)
- **Multiplication method:** $h(k) = \lfloor m(kA \bmod 1) \rfloor$, $0 < A < 1$
- **Multiply-shift:** $h(k) = (ka \bmod 2^w) \gg (w-l)$; for $m=2^l$, w -bit words
- **Open addressing:** all elements in table; no external storage; $\alpha \leq 1$
- **Probe sequence:** $h(k, 0), h(k, 1), \dots, h(k, m-1)$ – must be permutation of $0, \dots, m-1$
- **Double hashing:** $h(k, i) = (h(k) + i \cdot h'(k)) \bmod m$; requires $h(k)$ relatively prime to m
- **Linear probing:** $h(k, i) = (h(k) + i) \bmod m$; suffers primary clustering
- **Independent uniform permutation hashing:** probe sequence of each key equally likely any permutation
- **Primary clustering:** long runs of occupied slots in linear probing; empty slot preceded by i full slots filled with probability $(i+1)/m$
- **Cryptographic hash function:** SHA-256 produces 256-bit output; usable as hash via $h(k) = \text{SHA-256}(k) \bmod m$
- **Wee hash function:** $f(k) = \text{swap}((2k^2 + ak) \bmod 2)$; then iterate r rounds with CBC-style chaining for variable-length inputs

Processes / Algorithms / Pathways

DIRECT-ADDRESS-SEARCH(T, k)

- **Steps:** return $T[k]$
- **Complexity:** $O(1)$

CHAINED-HASH-INSERT(T, x)

- **Steps:** LIST-PREPEND($T[h(x.key)], x$)
- **Complexity:** $O(1)$ worst-case
- **Example:** T size 9, $h(k)=k \bmod 9$. Insert keys 5, 28, 19, 15, 20, 33, 12, 17, 10. $h(5)=5$, $h(28)=1$, $h(19)=1$ (collision → chain at slot 1: 19→28), $h(15)=6$, $h(20)=2$, $h(33)=6$ (chain: 33→15), $h(12)=3$, $h(17)=8$, $h(10)=1$ (chain: 10→19→28)

CHAINED-HASH-SEARCH(T, k)

- **Steps:** return LIST-SEARCH($T[h(k)], k$)
- **Complexity:** $\Theta(1+\alpha)$ average under independent uniform hashing; $\Theta(n)$

worst-case

CHAINED-HASH-DELETE(T, x)

- **Steps:** LIST-DELETE($T[h(x.key)], x$)
- **Complexity:** $O(1)$ if doubly linked list

HASH-INSERT(T, k) – open addressing

- **Steps:** (1) $i=0$ (2) repeat: $q = h(k, i)$; if $T[q] == \text{NIL}$: $T[q]=k$, return q ; else $i=i+1$ (3) until $i=m$ (4) error “hash table overflow”
- **Complexity:** Expected $O(1/(1-\alpha))$ probes under independent uniform permutation hashing

HASH-SEARCH(T, k) – open addressing

- **Steps:** (1) $i=0$ (2) repeat: $q = h(k, i)$; if $T[q]==k$ return q ; $i=i+1$ (3) until $T[q]==\text{NIL}$ or $i=m$ (4) return NIL

LINEAR-PROBING-HASH-DELETE(T, q)

- **Steps:** (1) $T[q]=\text{NIL}$ (2) search forward; for each key k' found, compute $g(k', q) = (q - h(k')) \bmod m$; if $g(k', q) < g(k', q')$, move k' to vacated slot; continue until empty slot found
- **Complexity:** proportional to distance from $h(k)$ to next empty slot

Classifications & Hierarchies

- **Hash function families by strength:**
 - Uniform: $\Pr[h(k)=q] = 1/m$ for any key k
 - Universal: $\Pr[h(k)=h(k')] \leq 1/m$ for distinct keys
 - -universal: collision prob $\leq$
 - d-independent: d-wise joint distribution uniform
- **Collision resolution:**
 - Chaining (uses external lists)
 - Open addressing (all in table): linear probing, double hashing
- **Hash function construction:**
 - Static: division; multiplication; multiply-shift (fixed h)
 - Random: universal family $H = \{h_b(k) = ((ak+b) \bmod p) \bmod m\}$; multiply-shift with random odd a
 - Cryptographic: SHA-256 + salt; wee hash (RC6-based)

Comparisons & Trade-offs

Dimension	Chaining	Open Addressing
Load factor α	Can exceed 1	≤ 1
Storage	table + pointers	table only (no pointers)
Deletion	Trivial ($O(1)$ doubly linked)	Tricky; needs DELETED marker or linear-probing delete
Search avg ($\alpha=.5$)	$\Theta(1+\alpha) \approx 1.5$ probes	$1/(1-\alpha) \approx 2$ probes (unsuccessful)
Cache behavior	Poor (list traversal)	Better (linear probing sequential)
Worst-case	$\Theta(n)$ (all same slot)	$\Theta(n)$ (table full, must probe all)

Dimension	Linear Probing	Double Hashing
Probe sequences	m distinct	$\Theta(m^2)$ distinct
Primary clustering	Yes	No
Cache locality	Excellent	Poor
Deletion w/o DELETED	Yes (L-P delete)	No

Dimension	Division Method	Multiply-Shift	Universal Family H
Speed	Very fast	3 instructions	Moderate (needs mod p)
m	Prime preferred, not constraint near 2	Power of 2 ($m=2^l$)	Arbitrary
Guarantee	None (static)	None (static)	Universal ($\leq 1/m$ collision)

Formulas & Equations

Load factor

$\alpha = n / m$ - n = number of elements - m = number of slots - For open addressing: $\alpha < 1$ strictly (at least one empty slot)

Expected unsuccessful search – chaining

$\Theta(1 + \alpha)$ - Theorem 11.1: under independent uniform hashing

Expected successful search – chaining

$\Theta(1 + \alpha)$ - Theorem 11.2: under independent uniform hashing

Expected probes unsuccessful search – open addressing

$\leq 1 / (1 - \alpha)$ - Theorem 11.6: under independent uniform permutation hashing, $\alpha < 1$, no deletions

Expected probes successful search – open addressing

$\leq (1/\alpha) \ln(1/(1-\alpha))$ - Theorem 11.8: under independent uniform permutation hashing

Universal hash family H

$h_b(k) = ((ak + b) \bmod p) \bmod m$ - $a \in \{1, \dots, p-1\}$, $b \in \{0, \dots, p-1\}$ - p prime, $p > m$ - $|H| = p(p-1)$

Multiply-shift hash

$h(k) = (ka \bmod 2^w) \gg (w-l)$ - $m = 2^l$, w-bit words - a = odd w-bit constant - 2/m-universal (Theorem 11.5)

Double hashing

$h(k, i) = (h(k) + i \cdot h'(k)) \bmod m$ - $h(k)$ must be relatively prime to m - Common: m prime, $h'(k) = 1 + (k \bmod m')$, m' slightly $< m$

Linear probing hash

$h(k, i) = (h(k) + i) \bmod m$

Edge Cases & Common Pitfalls

- **Worst-case chaining:** all keys hash to same slot $\rightarrow \Theta(n)$ per operation
- **m power of 2 with division:** $h(k) = k \bmod 2^\ell$ uses only low-order ℓ bits – bad if keys are patterned
- **m near power of 2 with division:** bad distribution
- **Open addressing deletion:** setting slot to NIL breaks search; use DELETED marker instead (but then α no longer bounds search time)
- **Double hashing non-relatively-prime:** if $\gcd(h(k), m) = d > 1$, only $1/d$ of table probed
- **Linear probing primary clustering:** empty slot preceded by i full slots filled with probability $(i+1)/m$
- **$\alpha \rightarrow 1$ in open addressing:** probes $\rightarrow \infty$; unsuccessful search at $\alpha=1$ probes all m slots
- **Static hash function:** adversary can choose n keys all same slot

Diagrams & Visuals

Direct-address table:

Keys $U=\{0, \dots, 9\}$ $K=\{2, 3, 5, 8\}$

T: [NIL][NIL][•→e][•→e][NIL][•→e][NIL][NIL][•→e][NIL]
 0 1 2 3 4 5 6 7 8 9

Chaining:

T: [NIL]→[•→k→k]→[•→k→k→k]→[NIL]→...
 0 1 2 3

Open addressing – double hashing:

$h(k)=k \bmod 13$, $h'(k)=1+(k \bmod 11)$

Key 14: $h(14)=1$ (occupied) $\rightarrow$ probe $1+5=6$ (occupied) $\rightarrow$ probe $6+5=11$ (occupied) $\rightarrow$ probe $11+5=16 \bmod 13 = 3$ (occupied) $\rightarrow$ probe $3+5=8$ (occupied) $\rightarrow$ probe $8+5=13 \bmod 13=0$ (occupied) $\rightarrow$ probe $0+5=5$ (occupied) $\rightarrow$ probe $5+5=10$ (occupied) $\rightarrow$ probe $10+5=15 \bmod 13=2$ (occupied) $\rightarrow$ probe $2+5=7$ (occupied) $\rightarrow$ probe $7+5=12$ (occupied) $\rightarrow$ probe $12+5=17 \bmod 13=4$ (occupied) $\rightarrow$ probe $4+5=9$ (empty!) insert at 9

(modified from text: 14 inserted at slot 9 after probing slots 1,5)

Multiply-shift:

k (w bits) $\rightarrow \times a$ (w bits) $\rightarrow [r$ (high w) | r (low w)] $\rightarrow$ take ℓ MSB of r

End-of-Chapter Material

- **Key terms:** direct-address table, hash table, hash function, collision, chaining, load factor, independent uniform hashing, universal hashing, -universal, d-independent, division method, multiplication method, multiply-shift, open addressing, probe sequence, double hashing, linear probing, primary clustering, cryptographic hash, wee hash, random oracle
- **Theorem 11.1–11.2:** chaining search $\Theta(1+\alpha)$ average
- **Theorem 11.4:** H is universal
- **Theorem 11.5:** multiply-shift is $2/m$ -universal
- **Theorem 11.6:** open addressing unsuccessful search $\leq 1/(1-\alpha)$
- **Theorem 11.8:** open addressing successful search $\leq (1/\alpha) \ln(1/(1-\alpha))$
- **Theorem 11.9:** linear probing with 5-independent h and $\alpha \leq 2/3$ takes expected constant time
- **Problem 11-3:** max chain length $E[M] = O(\lg n / \lg \lg n)$ for n keys, n slots

Cross-Chapter Links

- **Requires:** Ch. 10 (linked lists for chaining), Ch. 5 (indicator random variables, expectation), Ch. 31 (modular arithmetic for universal hashing)
 - **Referenced in:** Ch. 17 (amortized analysis of hash tables), Ch. 22 (graph representations)
-

Ch. 12 – Binary Search Trees

Named Entities (Terms & Definitions)

- **Binary search tree (BST):** binary tree with BST property: for any node x , all keys in left subtree $\leq x.key \leq$ all keys in right subtree
- **BST property:** if y in left subtree of x then $y.key \leq x.key$; if y in right subtree then $y.key \geq x.key$
- **Inorder tree walk:** recursively visit left, print root, visit right; prints keys in sorted order in $\Theta(n)$ time
- **Preorder tree walk:** print root, visit left, visit right
- **Postorder tree walk:** visit left, visit right, print root
- **Successor:** next node visited in inorder walk; if right subtree exists $\rightarrow$ minimum of right subtree; else lowest ancestor whose left child is also ancestor
- **Predecessor:** symmetric to successor; last node visited before x in inorder walk
- **Radix tree (trie):** tree on bit strings; go left if bit=0, right if bit=1; sorts strings in $\Theta(\text{total length})$ time
- **Randomly built BST:** insert n keys in random order (all $n!$ permutations equally likely); expected height $O(\lg n)$
- **Catalan number:** $b = (1/(n+1))C(2n,n)$; number of distinct binary trees with n nodes

Processes / Algorithms / Pathways

INORDER-TREE-WALK(x)

- **Type:** Algorithm
- **Goal:** Print keys of BST in sorted order
- **Steps:** (1) if $x \neq \text{NIL}$: (2) INORDER-TREE-WALK($x.\text{left}$) (3) print $x.\text{key}$ (4) INORDER-TREE-WALK($x.\text{right}$)
- **Complexity:** $\Theta(n)$ (Theorem 12.1)
- **Example:** Tree root=6, left child=5 (with left=2, right=5), right child=7 (with right=8). Output: 2,5,5,6,7,8

TREE-SEARCH(x, k)

- **Type:** Algorithm
- **Goal:** Find node with key k in subtree rooted at x
- **Steps:** (1) if $x == \text{NIL}$ or $k == x.\text{key}$: return x (2) if $k < x.\text{key}$: return TREE-SEARCH($x.\text{left}, k$) (3) else: return TREE-SEARCH($x.\text{right}, k$)
- **Complexity:** $O(h)$ where h = tree height
- **Example:** Search key 13 in tree [15,6,7,13, ...]: path 15 $\rightarrow$ 6 $\rightarrow$ 7 $\rightarrow$ 13; returns node with key 13

ITERATIVE-TREE-SEARCH(x, k)

- **Type:** Algorithm
- **Steps:** (1) while $x \neq \text{NIL}$ and $k \neq x.\text{key}$: (2) if $k < x.\text{key}$: $x = x.\text{left}$ (3) else: $x = x.\text{right}$ (4) return x
- **Complexity:** Same, more efficient in practice

TREE-MINIMUM(x)

- **Type:** Algorithm
- **Goal:** Find minimum key in subtree
- **Steps:** (1) while $x.\text{left} \neq \text{NIL}$: $x = x.\text{left}$ (2) return x

- **Complexity:** $O(h)$
- **Example:** From root 15: $15 \rightarrow 6 \rightarrow 3 \rightarrow 2 \rightarrow$ returns 2

TREE-MAXIMUM(x)

- **Type:** Algorithm
- **Goal:** Find maximum key in subtree
- **Steps:** (1) while $x.\text{right} \neq \text{NIL}$: $x = x.\text{right}$ (2) return x
- **Complexity:** $O(h)$

TREE-SUCCESSOR(x)

- **Type:** Algorithm
- **Goal:** Find next node in inorder traversal
- **Steps:** (1) if $x.\text{right} \neq \text{NIL}$: return TREE-MINIMUM($x.\text{right}$) (2) $y = x.p$ (3) while $y \neq \text{NIL}$ and $x == y.\text{right}$: $x = y$; $y = y.p$ (4) return y
- **Complexity:** $O(h)$
- **Example:** Node with key 13 (no right child): go up to parent 7 (13 is right child of 7); go up to parent 6 (7 is right child of 6); go up to 15 (6 is left child of 15) $\rightarrow$ successor is 15

TREE-INSERT(T, z)

- **Type:** Algorithm
- **Goal:** Insert node z into BST T ($z.\text{key}$ set, children NIL)
- **Steps:** (1) $x = T.\text{root}$; $y = \text{NIL}$ (2) while $x \neq \text{NIL}$: $y = x$; if $z.\text{key} < x.\text{key}$: $x = x.\text{left}$ else $x = x.\text{right}$ (3) $z.p = y$ (4) if $y == \text{NIL}$: $T.\text{root} = z$ (5) elseif $z.\text{key} < y.\text{key}$: $y.\text{left} = z$ (6) else: $y.\text{right} = z$
- **Complexity:** $O(h)$
- **Example:** Insert key 13 into tree with root 15, left=6, right=20. Path: $15 \rightarrow 6 \rightarrow 7 \rightarrow \text{NIL}$ (right of 7). Insert as right child of 7.

TRANSPLANT(T, u, v)

- **Type:** Subroutine
- **Goal:** Replace subtree rooted at u with subtree rooted at v
- **Steps:** (1) if $u.p == \text{NIL}$: $T.\text{root} = v$ (2) elseif $u == u.p.\text{left}$: $u.p.\text{left} = v$ (3) else: $u.p.\text{right} = v$ (4) if $v \neq \text{NIL}$: $v.p = u.p$

TREE-DELETE(T, z)

- **Type:** Algorithm
- **Goal:** Delete node z from BST T
- **Steps:** Cases: (a) z no left child $\rightarrow$ TRANSPLANT(z , $z.\text{right}$) (b) z no right child $\rightarrow$ TRANSPLANT(z , $z.\text{left}$) (c) z has two children: $y = \text{TREE-MINIMUM}(z.\text{right})$; if $y \neq z.\text{right}$: TRANSPLANT(y , $y.\text{right}$); $y.\text{right} = z.\text{right}$; TRANSPLANT(z , y); $y.\text{left} = z.\text{left}$
- **Complexity:** $O(h)$
- **Example:** Delete root 15 (has children 6 and 20). $y = \text{TREE-MINIMUM}(z.\text{right}) = 17$. If 17 is not right child of 15: splice 17 out, make 17's right child (say 19) replace 17, set $17.\text{right} = 20$, TRANSPLANT(15,17), set $17.\text{left} = 6$.

Classifications & Hierarchies

- **BST variants:**
 - Basic BST (Ch. 12): $O(h)$ operations, h could be n
 - Red-black tree (Ch. 13): $h \leq 2 \lg(n+1)$
 - AVL tree (Problem 13-3): $|\text{left}.h - \text{right}.h| \leq 1$

- Radix tree / trie (Problem 12-2): bit-based keys
- Optimal BST (Ch. 14): known search frequencies
- **Tree walks:** inorder (sorted), preorder (root first), postorder (root last)

Comparisons & Trade-offs

Dimension	Unsorted Array	Sorted Array	BST (height h)	Red-Black Tree
SEARCH	$\Theta(n)$	$O(\log n)$	$O(h)$	$O(\log n)$
INSERT	$O(1)$	$\Theta(n)$ (shift)	$O(h)$	$O(\log n)$
DELETE	$\Theta(n)$ (search+shift)	$\Theta(n)$	$O(h)$	$O(\log n)$
MIN/MAX	$\Theta(n)$	$O(1)$	$O(h)$	$O(\log n)$
SUCCESSOR	N/A	$O(1)$ if index given	$O(h)$	$O(\log n)$
Sorted output	$\Theta(n \log n)$ sort	$\Theta(n)$ walk	$\Theta(n)$ inorder	$\Theta(n)$ inorder

Formulas & Equations

BST height bound (random)

$E[\text{height}] = O(\log n)$ - Random BST from random insertion order

Number of distinct binary trees (Catalan)

$b = (1/(n+1)) \cdot C(2n, n) = \Omega(4^n / n^{3/2})$ - $b = 1$; recurrence: $b = \sum_{k=0}^{n-1} b \cdot b_{n-1-k}$

Rules, Laws & Theorems

Theorem 12.1

- **Statement:** INORDER-TREE-WALK on n -node subtree takes $\Theta(n)$ time
- **Proof:** $T(n) \leq T(k) + T(n-k-1) + d$; by substitution $T(n) \leq (c+d)n + c$

Theorem 12.2

- **Statement:** SEARCH, MINIMUM, MAXIMUM, SUCCESSOR, PREDECESSOR each run in $O(h)$ on BST of height h

Theorem 12.3

- **Statement:** INSERT and DELETE run in $O(h)$ on BST of height h

Edge Cases & Common Pitfalls

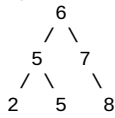
- **Degenerate BST:** inserting sorted keys yields $h = n$ (linked list); operations $\Theta(n)$
- **Equal keys:** need convention about $\leq$ vs $<$ in BST property; Problem 12-1 explores strategies
- **Deleting node with two children:** successor y may be z 's immediate right child (case c) or deeper (case d); two distinct subcases
- **STACK-EMPTY check not needed:** TREE-INSERT accesses NIL child attributes, must stop at NIL
- **TRANSPLANT does not update children of v :** caller's responsibility

(TREE-DELETE handles this)

- **Successor when right subtree empty:** must go up tree until finding a node that is a left child

Diagrams & Visuals

Binary search tree (h=2):



Inorder: 2,5,5,6,7,8

Search path for k=13:

15 → 6 → 7 → 13
(go left at 15, right at 6, right at 7, found)

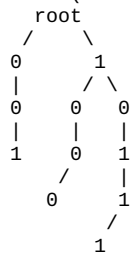
Successor of node 13 (no right child):

13 → parent 7 (13 is right child) → parent 6 (7 is right child) → parent 15 (6 is left child) → successor = 15

Delete cases (z=blue, replacement=orange):

- (a) z no left child: z.right replaces z
- (b) z no right child: z.left replaces z
- (c) z two children, y=successor is z.right: y replaces z
- (d) z two children, y deeper: splice y out first, then y replaces z

Radix tree (trie) for strings {0,011,10,100,1011}:



End-of-Chapter Material

- **Key terms:** binary search tree, BST property, inorder/preorder/postorder tree walk, successor, predecessor, TRANSPLANT, radix tree (trie), Catalan number, randomly built BST
- **Exercises 12.1:** draw BSTs of various heights; compare BST property to min-heap property; nonrecursive inorder walk
- **Exercises 12.2:** search path validity; recursive MIN/MAX; successor has no left child; $\Theta(n)$ inorder via MINIMUM + $n-1$ SUCCESSOR calls
- **Exercises 12.3:** BST sort (worst $\Theta(n^2)$, best $\Theta(n \log n)$); deletion not commutative; successor-only representation
- **Problem 12-3:** average node depth in randomly built BST is $O(\log n)$; links to quicksort
- **Problem 12-4:** Catalan numbers count distinct binary trees

Cross-Chapter Links

- **Requires:** Ch. 10 (binary tree representation, linked lists), Appendix B (tree math)
- **Referenced in:** Ch. 13 (red-black trees), Ch. 14 (optimal BST), Ch. 18 (B-trees), Ch. 19 (Fibonacci heaps)

Ch. 13 – Red-Black Trees

Named Entities (Terms & Definitions)

- **Red-black tree:** BST with 1 extra bit per node (color: RED or BLACK); approximately balanced; height $\leq 2 \lg(n+1)$
- **Red-black properties:** (1) every node red or black (2) root is black (3) every leaf (NIL) is black (4) red node $\rightarrow$ both children black (5) all paths from node to descendant leaves have same #black nodes
- **Black-height $bh(x)$:** number of black nodes on path from x (excluding x) down to a leaf
- **Sentinel $T.nil$:** single object representing all NIL leaves; color BLACK
- **Rotation:** local pointer rearrangement preserving BST property; left rotation and right rotation; $O(1)$ time
- **Left rotation:** $x.right = y$ becomes new root of subtree; x becomes y's left child; y's left child becomes x's right child
- **Right rotation:** symmetric
- **Uncle:** sibling of a node's parent
- **Doubly black / red-and-black:** conceptual extra black on node x after deletion of black y; x is doubly black if $x.color = BLACK$, red-and-black if $x.color = RED$
- **AVL tree:** height-balanced BST; $|left.h - right.h| \leq 1$ for all nodes; height $O(\log n)$
- **Persistent data structure:** maintains past versions; copying only affected path nodes ($O(\log n)$ per op)
- **Join operation (RB-JOIN):** merge two red-black trees S, S and element x where all keys in $S \leq x.key \leq$ all keys in S; $O(\log n)$

Processes / Algorithms / Pathways

LEFT-ROTATE(T, x)

- **Type:** Algorithm
- **Goal:** Perform left rotation on node x ($x.right \neq T.nil$)
- **Steps:** (1) $y = x.right$ (2) $x.right = y.left$ (3) if $y.left \neq T.nil$: $y.left.p = x$ (4) $y.p = x.p$ (5) if $x.p == T.nil$: $T.root = y$ (6) elseif $x == x.p.left$: $x.p.left = y$ (7) else: $x.p.right = y$ (8) $y.left = x$ (9) $x.p = y$
- **Complexity:** $O(1)$
- **Example:** Left rotation on x with right child y: before: $x \rightarrow (\alpha, y \rightarrow (\beta, \gamma))$; after: $y \rightarrow (x \rightarrow (\alpha, \beta), \gamma)$. BST order $\alpha < x < \beta < y < \gamma$ preserved.

RIGHT-ROTATE(T, y)

- **Type:** Algorithm (symmetric to LEFT-ROTATE)
- **Complexity:** $O(1)$

RB-INSERT(T, z)

- **Type:** Algorithm
- **Goal:** Insert node z into red-black tree T maintaining red-black properties
- **Steps:** (1) Same as TREE-INSERT but with $T.nil$ sentinel (2) set $z.left = z.right = T.nil$ (3) $z.color = RED$ (4) call RB-INSERT-FIXUP(T, z)
- **Complexity:** $O(\log n)$; at most 2 rotations
- **Example:** Insert 38 into empty tree $\rightarrow$ root=38, black. Insert 31: root=38(black), left=31(red). Insert 12: 38(black) $\rightarrow$ 31(red) $\rightarrow$ 12(red) $\rightarrow$ violation (two reds). Uncle=NIL(black). Case 2/3: rotate right on 31, recolor $\rightarrow$ 31(black), left=12(red), right=38(red).

RB-INSERT-FIXUP(T, z)

- **Type:** Algorithm
- **Goal:** Restore red-black properties after insertion
- **Invariant:** (a) z is red (b) if z.p is root, z.p is black (c) at most one violation (property 2 or 4, not both)
- **Cases** (z.p is left child of z.p.p):
 - **Case 1** (uncle y red): recolor: z.p=BLACK, y=BLACK, z.p.p=RED; z moves up two levels
 - **Case 2** (uncle black, z is right child): left rotation on z.p; falls through to Case 3
 - **Case 3** (uncle black, z is left child): recolor z.p=BLACK, z.p.p=RED; right rotation on z.p.p
- **Termination:** when z.p is black; line 30 colors root black
- **Complexity:** $O(\log n)$; ≤ 2 rotations

RB-TRANSPLANT(T, u, v)

- **Type:** Subroutine
- **Steps:** (1) if u.p == T.nil: T.root = v (2) elseif u == u.p.left: u.p.left = v (3) else: u.p.right = v (4) v.p = u.p (unconditional)
- **Difference from TRANSPLANT:** uses T.nil, unconditional v.p assignment

RB-DELETE(T, z)

- **Type:** Algorithm
- **Goal:** Delete node z from red-black tree
- **Steps:** (1) y = z; y-original-color = y.color (2) if z has ≤ 1 child: set x = child, RB-TRANSPLANT(z, child) (3) else: y = TREE-MINIMUM(z.right); y-original-color = y.color; x = y.right; splice y out if not z.right; RB-TRANSPLANT(z, y); y.color = z.color (4) if y-original-color == BLACK: call RB-DELETE-FIXUP(T, x)
- **Complexity:** $O(\log n)$; at most 3 rotations

RB-DELETE-FIXUP(T, x)

- **Type:** Algorithm
- **Goal:** Fix red-black properties after deletion of a black node
- **Idea:** x is "doubly black"; push extra black up tree
- **Cases** (x is left child):
 - **Case 1** (sibling w red): w=BLACK, x.p=RED, left-rotate on x.p; new w = x.p.right (black); falls into 2/3/4
 - **Case 2** (w black, both w's children black): remove one black from x and w (w=RED); move extra black up to x.p; x = x.p
 - **Case 3** (w black, w.left red, w.right black): w.left=BLACK, w=RED, right-rotate on w; new w = x.p.right (black with red right child); falls into Case 4
 - **Case 4** (w black, w.right red): w.color = x.p.color; x.p=BLACK; w.right=BLACK; left-rotate on x.p; x = T.root (terminates)
- **Termination:** x = root (extra black vanishes) or x red-and-black (color x black); line 44 colors x black
- **Complexity:** $O(\log n)$; only Case 2 repeats; at most 3 rotations

Classifications & Hierarchies

- **Balanced BSTs:** red-black (Ch. 13), AVL (Problem 13-3), AA-trees, treaps, splay trees, scapegoat trees, weight-balanced trees
- **Red-black variants:** left-leaning red-black trees (Sedgwick); all red nodes are left children

Comparisons & Trade-offs

Dimension	Red-Black Tree	AVL Tree
Height bound	$\leq 2 \lg(n+1)$	$\leq 1.44 \lg n$ (stricter)
Insert rotations	≤ 2	$O(\log n)$
Delete rotations	≤ 3	$O(\log n)$
Lookup speed	$O(\log n)$	Slightly faster (tighter balance)
Implementation	More complex	Simpler (no color; just height)

Dimension	BST (Ch. 12)	Red-Black Tree
Worst-case height	$\Theta(n)$ (degenerate)	$O(\log n)$
SEARCH, INSERT, DELETE	$O(h)$	$O(\log n)$
Space	n nodes + 3 pointers	n nodes + 3 ptrs + 1 color bit
Sentinel	Optional	Required (T.nil)

Formulas & Equations

Height bound

$h \leq 2 \lg(n+1)$ - Lemma 13.1: red-black tree with n internal nodes has height at most $2 \lg(n+1)$ - Proof: (1) subtree at x contains $\geq 2^{bh(x)} - 1$ internal nodes (induction) (2) root black-height $\geq h/2$ (by property 4, $\geq$ half nodes on path are black) (3) $n \geq 2^{(h/2)} - 1 \rightarrow h \leq 2 \lg(n+1)$

Black-height bounds

$h/2 \leq bh(\text{root}) \leq h$ - min black-height = $h/2$ (alternating red-black from root) - max black-height = h (all nodes black)

Rules, Laws & Theorems

Lemma 13.1

- Red-black tree with n internal nodes has height $\leq 2 \lg(n+1)$
- Implies $O(\log n)$ for all BST operations

Property 4 consequence

- A red node cannot have exactly one non-NIL child (Exercise 13.1-8)
- Proof: if red node had one child, that child's black-height would differ from NIL leaf's black-height

Edge Cases & Common Pitfalls

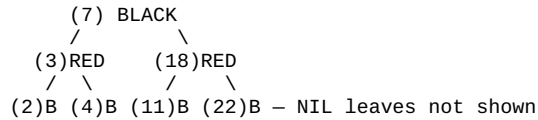
- **Inserting black instead of red:** would violate property 5 (extra black on paths through new node)
- **Deleting red node:** no problem – properties preserved
- **Deleting black root with red child:** new root (x) is red – property 2 violated; fix by coloring root black (line 44)
- **T.nil sentinel:** must never be colored RED; RB-INSERT-FIXUP never sets $T.\text{nil}.color = \text{RED}$ (Exercise 13.3-4)
- **Case 2 entering from Case 1:** new x ($x.p$) will be red-and-black $\rightarrow$ loop terminates next iteration
- **Doubly black:** not a real color attribute; conceptual; $x.color$ attribute stays RED or BLACK
- **$x.p$ must be set in RB-DELETE** even when $x = T.\text{nil}$ (line 16), needed by

RB-DELETE-FIXUP

- **RB-INSERT vs TREE-INSERT:** uses T.nil instead of NIL; sets children to T.nil; colors RED; calls FIXUP

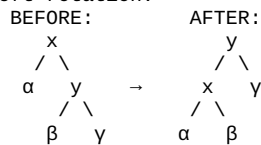
Diagrams & Visuals

Red-black tree structure:



Black-heights: nil=0, 2=bh1, 4=bh1, 3=bh1, 11=bh1, 22=bh1, 18=bh1, 7=bh2

Left rotation:



BST order preserved: $\alpha < x < \beta < \gamma < \gamma$

RB-INSERT-FIXUP cases:

- Case 1: z.p and uncle both red → recolor, move z up 2
- Case 2: z.p red, uncle black, z right child → left rotate → Case 3
- Case 3: z.p red, uncle black, z left child → recolor + right rotate → done

RB-DELETE-FIXUP cases (x is left child):

- Case 1: w red → recolor + left rotate → Case 2/3/4
- Case 2: w black, both children black → push black up
- Case 3: w black, w.left red, w.right black → right rotate → Case 4
- Case 4: w black, w.right red → recolor + left rotate → done

End-of-Chapter Material

- **Key terms:** red-black tree, red-black properties, black-height, sentinel T.nil, rotation (left/right), RB-INSERT, RB-INSERT-FIXUP, RB-DELETE, RB-DELETE-FIXUP, doubly black, red-and-black, uncle, AVL tree, persistent set, join
- **Lemma 13.1:** $h \leq 2 \lg(n+1)$
- **Exercises 13.1:** draw RB tree; red node cannot have exactly one child; largest/smallest internal nodes for given black-height
- **Exercises 13.2:** any BST can be transformed to any other using $O(n)$ rotations; right-conversion $O(n^2)$; exactly $n-1$ possible rotations in n -node BST
- **Exercises 13.3:** tracing insertions 41,38,31,12,19,8 into empty RB tree; black-height labeling
- **Exercises 13.4:** tracing deletions from previous tree; sentinel interactions
- **Problem 13-1:** persistent BST – copy $O(h)$ nodes per insert; parent pointers make it $\Omega(n)$
- **Problem 13-2:** RB-JOIN – $O(\log n)$ by grafting at nodes of matching black-height
- **Problem 13-3:** AVL trees – height $O(\log n)$, at least F_h nodes; BALANCE procedure using rotations; AVL-INSERT $O(\log n)$ with $O(\log n)$ rotations

Cross-Chapter Links

- **Requires:** Ch. 12 (BST operations, TREE-INSERT, TREE-DELETE), Ch. 10 (binary tree representation, sentinel)
- **Referenced in:** Ch. 14 (optimal BST), Ch. 17 (amortized analysis – constant rotations per op matters), Ch. 18 (B-trees), Ch. 19

(mergeable heaps)

Ch. 14 – Dynamic Programming

Named Entities (Terms & Definitions)

- **Dynamic Programming (DP)**: method for solving problems by combining solutions to overlapping subproblems; solves each subproblem once and stores result in a table; applies to optimization problems with optimal substructure and overlapping subproblems.
- **Memoization**: top-down DP technique that saves results of subproblems in an array/table upon first computation; subsequent calls look up the saved value.
- **Optimal substructure**: property where an optimal solution to a problem contains optimal solutions to its subproblems.
- **Overlapping subproblems**: property where a recursive algorithm revisits the same subproblem repeatedly; total distinct subproblems is polynomial.
- **Subproblem graph**: directed graph with one vertex per distinct subproblem; edge (x,y) if solving x requires solving y.
- **Rod-cutting problem**: given rod length n and price table p_i for i=1..n, maximize revenue by cutting rod into integer-length pieces.
- **Matrix-chain multiplication**: given chain of n matrices with dimensions p₀×p₁, p₁×p₂, ..., p_{n-1}×p_n, fully parenthesize product to minimize scalar multiplications.
- **Longest common subsequence (LCS)**: given two sequences X, Y, find maximum-length subsequence common to both.
- **Optimal binary search tree (OBST)**: given keys k₁..k_n with probabilities p_i and dummy keys d₀..d_n with probabilities q_i, build BST minimizing expected search cost.

Processes / Algorithms / Pathways

Four-Step DP Method

- **Type**: Design paradigm
- **Steps**: (1) Characterize structure of optimal solution (optimal substructure). (2) Recursively define value of optimal solution. (3) Compute value bottom-up (or top-down with memoization). (4) Construct optimal solution from computed info.

CUT-ROD (naive recursive)

- **Goal**: compute max revenue for rod length n
- **Steps**: (1) if n=0 return 0. (2) q = -∞. (3) for i=1..n: q = max(q, p[i] + CUT-ROD(p, n-i)). (4) return q.
- **Complexity**: T(n) = 2ⁿ (exponential); recurrence: T(n) = 1 + Σ_{j=0}ⁿ⁻¹ T(j), T(0)=1

MEMOIZED-CUT-ROD

- **Goal**: compute max revenue with memoization
- **Steps**: (1) initialize r[0..n] = -∞. (2) call MEMOIZED-CUT-ROD-AUX(p,n,r).
- **Complexity**: θ(n²)

BOTTOM-UP-CUT-ROD

- **Goal:** compute max revenue bottom-up
- **Steps:** (1) $r[0] = 0$. (2) for $j=1..n$: $q = -\infty$; for $i=1..j$: $q = \max(q, p[i] + r[j-i])$; $r[j] = q$. (3) return $r[n]$.
- **Complexity:** Time $\Theta(n^2)$, Space $\Theta(n)$
- **Example:** For $n=4$, prices $p=[1,5,8,9]$, $r[1]=1$, $r[2]=5$, $r[3]=8$, $r[4]=10$ (cut as $2+2$)

MATRIX-CHAIN-ORDER

- **Goal:** compute min scalar multiplications for matrix chain $A_1..A_n$
- **Steps:** (1) for $i=1..n$: $m[i,i]=0$. (2) for $l=2..n$ (chain length): for $i=1..n-l+1$: $j=i+l-1$; $m[i,j]=\infty$; for $k=i..j-1$: $q = m[i,k]+m[k+1,j]+p_{i-1} \cdot p_k \cdot p_j$; if $q < m[i,j]$: $m[i,j]=q$, $s[i,j]=k$.
- **Complexity:** Time $\Theta(n^3)$, Space $\Theta(n^2)$
- **Example:** Dimensions: $A_1(30 \times 35)$, $A_2(35 \times 15)$, $A_3(15 \times 5)$, $A_4(5 \times 10)$, $A_5(10 \times 20)$, $A_6(20 \times 25)$. $m[1,6]=15,125$. Optimal parenthesization: $((A_1(A_2A_3))((A_4A_5)A_6))$

LCS-LENGTH

- **Goal:** compute length of LCS of $X[1..m]$, $Y[1..n]$
- **Steps:** (1) initialize $c[i,0]=0$, $c[0,j]=0$. (2) for $i=1..m$: for $j=1..n$: if $x_i=y_j$: $c[i,j]=c[i-1,j-1]+1$, $b[i,j]=""$; else if $c[i-1,j] \geq c[i,j-1]$: $c[i,j]=c[i-1,j]$, $b[i,j]="\uparrow"$; else: $c[i,j]=c[i,j-1]$, $b[i,j]="\leftarrow"$.
- **Complexity:** Time $\Theta(mn)$, Space $\Theta(mn)$
- **Example:** $X=<A,B,C,B,D,A,B>$, $Y=<B,D,C,A,B,A>$. LCS length=4, LCS = $<B,C,B,A>$

OPTIMAL-BST

- **Goal:** compute min expected search cost for BST with keys $k_1..k_n$
- **Steps:** (1) for $i=1..n+1$: $e[i,i-1]=q_{i-1}$, $w[i,i-1]=q_{i-1}$. (2) for $l=1..n$: for $i=1..n-l+1$: $j=i+l-1$; $w[i,j]=w[i,j-1]+p_j+q_j$; $e[i,j]=\infty$; for $r=i..j$: $t=e[i,r-1]+e[r+1,j]+w[i,j]$; if $t < e[i,j]$: $e[i,j]=t$, $root[i,j]=r$.
- **Complexity:** Time $\Theta(n^3)$; can be improved to $\Theta(n^2)$ using Knuth's inequality ($root[i,j-1] \leq root[i,j] \leq root[i+1,j]$)
- **Example:** 5 keys with given probabilities; $e[1,5]=2.75$; root k_2

Classifications & Hierarchies

- **DP implementation approaches:**
 - Top-down with memoization: recursive + table; solves only needed subproblems; higher constant factors
 - Bottom-up: iterative, smallest subproblems first; lower constant factors; solves all subproblems
- **DP problem classification (td/ed):**
 - 2D/1D: matrix-chain ($\Theta(n^3)$ time, $\Theta(n^2)$ subproblems, each depends on $O(n)$ others)
 - 2D/0D: LCS ($\Theta(mn)$ time, each entry depends on $O(1)$ others)

Comparisons & Trade-offs

Dimension	Top-down (Memoization)	Bottom-up
Strategy	Recursive + save results	Iterative, smallest first
Solves all subproblems?	Only those needed	All
Overhead	Recursion + table	Lower constant factors

	maintenance	
When preferred	Not all subproblems needed	All subproblems must be solved

Dimension	DP	Divide-and-Conquer
Subproblems Overlapping		Disjoint
Efficiency	Solves each once	May recompute
Example	Rod cutting, LCS	Merge sort

Formulas & Equations

Rod-cutting recurrence

$$r_n = \max(p_n, \max_{1 \leq i \leq n-1} (r_i + r_{n-i}))$$
 Simpler:
$$r_n = \max_{1 \leq i \leq n} (p_i + r_{n-i}), \quad r_0 = 0$$

Matrix-chain recurrence

$$m[i, j] = \min_{i \leq k < j} (m[i, k] + m[k+1, j] + p_{i-1} \cdot p_k \cdot p_j)$$
 Base: $m[i, i] = 0$

LCS recurrence

$$c[i, j] = 0 \text{ if } i=0 \text{ or } j=0$$

$$c[i, j] = c[i-1, j-1] + 1 \text{ if } x_i = y_j$$

$$c[i, j] = \max(c[i-1, j], c[i, j-1]) \text{ if } x_i \neq y_j$$

OBST recurrence

$$e[i, j] = q_{i-1} \text{ if } j=i-1$$

$$e[i, j] = \min_{i \leq r \leq j} (e[i, r-1] + e[r+1, j] + w[i, j]) \text{ if } j \geq i$$
 where $w[i, j] = w[i, j-1] + p_j + q_j$

Rules, Laws & Theorems

Theorem 14.1 (Optimal substructure of LCS)

- **Statement:** Let $X = \langle x_1..x_m \rangle$, $Y = \langle y_1..y_n \rangle$, Z any LCS. (1) If $x_m = y_n$ then $z_k = x_m = y_n$ and $Z_{\{k-1\}}$ is LCS of $X_{\{m-1\}}$ and $Y_{\{n-1\}}$. (2) If $x_m \neq y_n$ and $z_k \neq x_m$ then Z is LCS of $X_{\{m-1\}}$ and Y . (3) If $x_m \neq y_n$ and $z_k \neq y_n$ then Z is LCS of X and $Y_{\{n-1\}}$.

Edge Cases & Common Pitfalls

- **Longest simple path lacks optimal substructure** (subproblems not independent – sharing vertices makes combination non-simple); while shortest path has it
- **OBST is not necessarily the tree with smallest height**; highest-probability key may not be root
- **Cut-and-paste proof:** must show subproblems are independent (no shared resources)
- **Greedy $\neq$ DP:** rod cutting greedy by density fails (Exercise 14.1-2)

Case Studies & Examples

Rod Cutting ($n=4$)

- Prices: $p_1=\$1$, $p_2=\$5$, $p_3=\$8$, $p_4=\$9$
- Options: $1+1+1+1=\$4$, $2+2=\$10$ (optimal), $3+1=\$9$, $4=\$9$

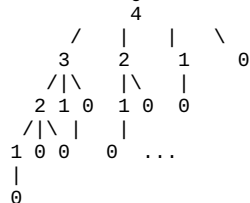
- Optimal: cut into two 2-inch pieces, revenue \$10

Matrix Chain (n=3)

- $A_1(10 \times 100)$, $A_2(100 \times 5)$, $A_3(5 \times 50)$
- $((A_1 A_2) A_3)$: $10 \cdot 100 \cdot 5 + 10 \cdot 5 \cdot 50 = 5000 + 2500 = 7500$
- $(A_1 (A_2 A_3))$: $100 \cdot 5 \cdot 50 + 10 \cdot 100 \cdot 50 = 25000 + 50000 = 75000$
- Ratio: 10x difference

Diagrams & Visuals

Rod-cutting recursion tree (n=4):



Subproblem graph: collapses repeated nodes

Vertices: $\{0, 1, 2, 3, 4\}$

Edges: $i \rightarrow j$ for $j < i$

End-of-Chapter Material

- **Key terms:** optimal substructure, overlapping subproblems, memoization, bottom-up, subproblem graph, Catalan numbers, prefix, dummy key
- **Exercises:** 14.1-1 ($T(n)$ recurrence), 14.1-2 (greedy counterexample), 14.2-1 (optimal parenthesization for $\langle 5, 10, 3, 12, 5, 50, 6 \rangle$), 14.4-1 (LCS of binary sequences), 14.5-1 (construct OBST from root table)

Cross-Chapter Links

- **Requires knowledge of:** Ch 2 (divide-and-conquer), Ch 4 (recurrences), Ch 12 (BSTs)
- **Referenced in later chapters:** Ch 15 (greedy vs DP), Ch 34 (NP-completeness)

Ch. 15 – Greedy Algorithms

Named Entities (Terms & Definitions)

- **Greedy algorithm:** makes locally optimal choice at each step, hoping for globally optimal solution.
- **Greedy-choice property:** globally optimal solution can be assembled by making locally optimal (greedy) choices.
- **Activity-selection problem:** given n activities with start/finish times, select maximum-size set of mutually compatible activities.
- **Compatible activities:** activities whose intervals $[s_i, f_i)$ do not overlap.
- **Huffman code:** optimal prefix-free code constructed by repeatedly merging two least-frequent characters.
- **Prefix-free code:** no codeword is a prefix of any other codeword; decoding is unambiguous.
- **Full binary tree:** every nonleaf node has two children; optimal prefix-free codes correspond to full trees.
- **0-1 knapsack problem:** thief can take or leave each item (binary

choice); requires DP.

- **Fractional knapsack problem:** thief can take fractions of items; solvable by greedy.
- **Offline caching:** know entire request sequence in advance; optimal strategy is furthest-in-future.
- **Furthest-in-future:** evict the block in cache whose next access is furthest in the future.

Processes / Algorithms / Pathways

Greedy Algorithm Design Steps

- **Type:** Design paradigm
- **Steps:** (1) Cast problem as making a choice, leaving one subproblem. (2) Prove greedy choice is always safe. (3) Demonstrate optimal substructure: greedy choice + optimal solution to subproblem = optimal solution to original.

RECURSIVE-ACTIVITY-SELECTOR(s, f, k, n)

- **Goal:** select max-size set of compatible activities from S_k
- **Steps:** (1) $m = k+1$. (2) while $m \leq n$ and $s[m] < f[k]$: $m++$. (3) if $m \leq n$: return { a_m } RECURSIVE-ACTIVITY-SELECTOR(s, f, m, n). (4) else return .
- **Complexity:** $\Theta(n)$ (each activity examined once across all recursive calls)

GREEDY-ACTIVITY-SELECTOR(s, f, n)

- **Goal:** iterative greedy activity selection
- **Steps:** (1) $A = \{a_1\}$, $k=1$. (2) for $m=2..n$: if $s[m] \geq f[k]$: $A = A \cup \{a_m\}$, $k=m$. (3) return A.
- **Complexity:** $\Theta(n)$ (after sorting by finish time $O(n \lg n)$)
- **Example:** Activities (sorted by finish): $a_1(1,4)$, $a_2(3,5)$, $a_3(0,6)$, $a_4(5,7)$, $a_5(3,9)$, $a_6(5,9)$, $a_7(6,10)$, $a_8(8,11)$, $a_9(8,12)$, $a_{10}(2,14)$, $a_{11}(12,16)$. Selected: { a_1 , a_4 , a_8 , a_{11} }

HUFFMAN(C)

- **Goal:** construct optimal prefix-free code for alphabet C with frequencies
- **Steps:** (1) $Q = C$ (min-priority queue keyed by freq). (2) for $i=1$ to $|C|-1$: allocate new node z ; $x = \text{EXTRACT-MIN}(Q)$; $y = \text{EXTRACT-MIN}(Q)$; $z.\text{left}=x$; $z.\text{right}=y$; $z.\text{freq}=x.\text{freq}+y.\text{freq}$; $\text{INSERT}(Q, z)$. (3) return $\text{EXTRACT-MIN}(Q)$ (root).
- **Complexity:** $O(n \lg n)$ with binary heap
- **Example:** Frequencies: a:45, b:13, c:12, d:16, e:9, f:5 ($\times 1000$). Code: a=0 (1 bit), b=101 (3), c=100 (3), d=111 (3), e=1101 (4), f=1100 (4). Total bits: 224,000 vs 300,000 fixed-length.

Classifications & Hierarchies

- **Greedy vs DP:** both require optimal substructure; greedy adds greedy-choice property; greedy makes choice before solving subproblems; DP solves subproblems before making choice.
- **Knapsack variants:**
 - 0-1 knapsack: each item taken or left (DP, $O(nW)$)
 - Fractional knapsack: take fractions (greedy by value/weight, $O(n \lg n)$)

Comparisons & Trade-offs

Dimension	Greedy	Dynamic Programming
Choice timing	Before solving subproblems	After solving subproblems
Subproblems	One remains after choice	Multiple considered
Direction	Top-down	Usually bottom-up
Efficiency	Typically faster	More general
When applies	Greedy-choice + optimal substructure	Optimal substructure + overlapping subproblems

Dimension	0-1 Knapsack	Fractional Knapsack
Items	Take or leave	Take fractions
Solution	DP ($O(nW)$)	Greedy by value/weight
Greedy works?	No	Yes

Formulas & Equations

Huffman tree cost

$$B(T) = \sum_{c \in C} c.\text{freq} \cdot dT(c)$$

where $dT(c)$ = depth of leaf for character c (also codeword length)

Huffman cost = sum of merge costs

$B(T) = \sum_{\text{internal nodes}} (\text{freq}(\text{left child}) + \text{freq}(\text{right child}))$ - Each merge adds combined frequency to total cost

Rules, Laws & Theorems

Theorem 15.1 (Greedy choice – activity selection)

- **Statement:** For any nonempty subproblem S_k , let a_m be activity with earliest finish time. Then a_m is included in some maximum-size subset of mutually compatible activities of S_k .
- **Proof:** Take optimal solution A_k ; let a_j be earliest-finish activity in A_k . If $a_j \neq a_m$, substitute a_m for a_j ; compatibility preserved, same size.

Lemma 15.2 (Greedy choice – Huffman)

- **Statement:** Let x, y be two characters with lowest frequencies. There exists an optimal prefix-free code where codewords for x, y have same length and differ only in last bit.
- **Proof:** Swap deepest siblings a, b with x, y ; cost does not increase.

Lemma 15.3 (Optimal substructure – Huffman)

- **Statement:** If T' is optimal for $C' = (C - \{x, y\}) \cup \{z\}$ with $z.\text{freq} = x.\text{freq} + y.\text{freq}$, then replacing leaf z with internal node having children x, y gives optimal tree T for C .

Theorem 15.4

- **Statement:** HUFFMAN produces an optimal prefix-free code.

Theorem 15.5 (Greedy choice – offline caching)

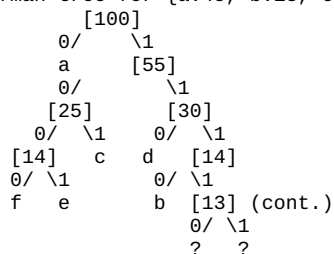
- **Statement:** When cache is full and miss occurs, evicting the block whose next access is furthest in the future is included in some optimal solution.

Edge Cases & Common Pitfalls

- **Not all greedy strategies work:** activity selection by shortest duration, fewest overlaps, or earliest start time all fail (Exercise 15.1-3)
- **0-1 knapsack greedy fails:** item1=10lbs/\$60 (\$6/lb), item2=20lbs/\$100 (\$5/lb), item3=30lbs/\$120 (\$4/lb); greedy picks item1 then can't fill 50lb → suboptimal; optimal is items2+3=\$220
- **Huffman with nearly-uniform frequencies:** no better than fixed-length code (Exercise 15.3-7)
- **Compression cannot guarantee compression for all inputs:** pigeonhole principle (Exercise 15.3-8)

Diagrams & Visuals

Huffman tree for {a:45, b:13, c:12, d:16, e:9, f:5}:

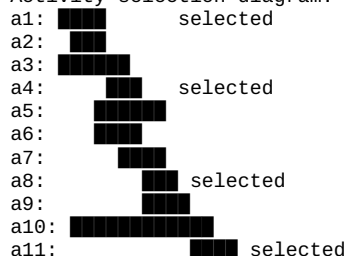


Fixed-length vs Huffman:

Fixed: a=000,b=001,c=010,d=011,e=100,f=101 → 300,000 bits

Huffman: a=0,b=101,c=100,d=111,e=1101,f=1100 → 224,000 bits (25% savings)

Activity selection diagram:



End-of-Chapter Material

- **Key terms:** greedy algorithm, greedy-choice property, optimal substructure, activity selection, compatible, prefix-free code, Huffman code, full binary tree, offline caching, furthest-in-future
- **Exercises:** 15.1-1 (DP for activity selection), 15.1-3 (counterexamples for wrong greedy strategies), 15.2-1 (prove greedy-choice for fractional knapsack), 15.2-2 (DP for 0-1 knapsack $O(nW)$), 15.3-1 (Huffman proof case), 15.3-3 (Huffman for Fibonacci frequencies), 15.4-2 (LRU not optimal example)

Cross-Chapter Links

- **Requires knowledge of:** Ch 14 (DP, optimal substructure), Ch 6 (heaps for Huffman)
- **Referenced in later chapters:** Ch 21 (MST – Kruskal, Prim), Ch 22 (Dijkstra), Ch 35 (greedy set-covering heuristic)

Ch. 16 – Amortized Analysis

Named Entities (Terms & Definitions)

- **Amortized analysis:** averaging time over a sequence of operations; guarantees average per-operation cost in the worst case.
- **Aggregate analysis:** total cost $T(n)$ for n operations; amortized cost = $T(n)/n$ (same for all operations).
- **Accounting method:** assign differing amortized costs; overcharge early operations, store credit on specific objects, use credit to pay for undercharged later operations.
- **Potential method:** assign potential $\Phi(D_i)$ to data structure; amortized cost $\hat{c}_i = c_i + \Phi(D_i) - \Phi(D_{i-1})$; prepaid work as “potential energy” of whole structure.
- **Load factor $\alpha(T)$:** $\text{num_items} / \text{size}$ (number of slots).
- **Table expansion:** when table fills, allocate new table of double size, copy items.
- **Table contraction:** when load factor drops below $1/4$, halve table size.
- **MULTIPOP(S, k):** pops up to k objects from stack.
- **INCREMENT:** increments k -bit binary counter; flips bits from rightmost 0.

Processes / Algorithms / Pathways

MULTIPOP(S, k)

- **Goal:** pop $\min(s, k)$ objects from stack
- **Steps:** while not STACK-EMPTY(S) and $k > 0$: POP(S); $k--$.
- **Cost:** $\min(s, k)$ where s = stack size

INCREMENT(A, k)

- **Goal:** add 1 to k -bit binary counter
- **Steps:** $i=0$; while $i < k$ and $A[i]==1$: $A[i]=0$; $i++$; if $i < k$: $A[i]=1$.
- **Cost:** number of bits flipped ($t_i + 1$, where t_i = # of trailing 1s)

TABLE-INSERT(T, x)

- **Goal:** insert item into dynamic table, expanding if full
- **Steps:** (1) if $T.\text{size}==0$: allocate with 1 slot. (2) if $T.\text{num}==T.\text{size}$: allocate new table with $2 \cdot T.\text{size}$ slots; copy all items; free old; $T.\text{table}=\text{new}$; $T.\text{size}=2 \cdot T.\text{size}$. (3) insert x ; $T.\text{num}++$.
- **Cost:** 1 (no expansion) or i (expansion at i -th insertion)

Classifications & Hierarchies

- **Three amortized analysis methods:**
 - Aggregate: $T(n)/n$, same for all operations, simplest
 - Accounting: per-operation amortized costs, credit on objects, flexible
 - Potential: Φ function on data structure, most general

Comparisons & Trade-offs

Dimension	Aggregate	Accounting	Potential
Amortized cost	Same for all ops	May differ per op type	May differ per op type

Credit storage	None	On specific objects	type Data structure as whole
Difficulty	Simplest	Medium	Most flexible
Key concept	Total cost / n	Overcharge → credit	Φ function

Formulas & Equations

Potential method definition

$$\hat{c}_i = c_i + \Phi(D_i) - \Phi(D_{i-1})$$

$$\text{Total: } \sum \hat{c}_i = \sum c_i + \Phi(D_n) - \Phi(D_0)$$

If $\Phi(D_i) \geq \Phi(D_0)$ for all i , total amortized cost $\geq$ total actual cost.

Accounting method invariant

$$\sum_{i=1}^n \hat{c}_i \geq \sum_{i=1}^n c_i \text{ for all } n \text{ (credit never negative)}$$

Stack – amortized costs (accounting/potential)

Actual: PUSH=1, POP=1, MULTIPOP=min(s,k)

Amortized: PUSH=2, POP=0, MULTIPOP=0

Potential Φ = number of objects in stack

Binary counter – amortized cost (aggregate)

Total flips in n INCREMENT ops: $\sum_{i=0}^{k-1} n/2^i < 2n$

Amortized: $O(1)$ per operation

Binary counter – amortized (accounting)

Setting $0 \rightarrow 1$ costs \$2 (\$1 to set, \$1 stored as credit).

Resetting $1 \rightarrow 0$ costs \$0 (paid by stored credit).

Each INCREMENT sets at most one $0 \rightarrow 1 \rightarrow$ amortized cost $\leq$ \$2.

Binary counter – potential method

Φ = number of 1-bits in counter

If operation resets t_i bits: $c_i \leq t_i + 1$, $\Phi(D_i) - \Phi(D_{i-1}) \leq 1 - t_i$

$$\hat{c}_i \leq (t_i + 1) + (1 - t_i) = 2$$

Dynamic table – aggregate

Cost of i -th insertion: $c_i = i$ if $i-1$ is power of 2, else $c_i = 1$

Total: $\sum c_i \leq n + \sum_{j=0}^{\lg n} 2^j = n + (2n-1) < 3n$

Amortized: < 3 per insertion

Dynamic table – potential

$\Phi = 2 \cdot \text{num} - \text{size}$ for $\alpha \geq 1/2$

$\Phi = \text{size}/2 - \text{num}$ for $\alpha < 1/2$

Amortized cost: insert=3 (with/without expansion), delete=1 (with contraction) or 2 (without contraction crossing $\alpha=1/2$) or 0 (insert crossing up to $\alpha=1/2$)

Dynamic table – expansion/contraction strategy

- Expand: double size when full ($\alpha=1$)
- Contract: halve size when $\alpha < 1/4$ (NOT $1/2$ – avoids thrashing)

Rules, Laws & Theorems

Key invariant for aggregate analysis (stack)

- **Statement:** Number of POP operations (including within MULTIPOP) $\leq$ number of PUSH operations $\leq n$.
- **Implication:** Total cost of n PUSH, POP, MULTIPOP operations = $O(n)$.

Key insight for binary counter

- **Statement:** Bit $A[i]$ flips $n/2^i$ times in sequence of n INCREMENT operations from 0.
- **Implication:** Total flips $< 2n$, so amortized $O(1)$.

Edge Cases & Common Pitfalls

- **Naive bound $O(n^2)$ for stack is not tight:** aggregate analysis shows $O(n)$.
- **Naive bound $O(nk)$ for binary counter is not tight:** aggregate analysis shows $O(n)$.
- **Thrashing:** contracting at load factor $1/2$ causes $\Theta(n^2)$ total cost for alternating insert/delete pattern. Solution: contract at $\alpha < 1/4$, not $\alpha < 1/2$.
- **Potential function choice:** different Φ give different amortized costs; must ensure $\Phi(D_i) \geq \Phi(D_0)$.
- **DECREMENT in binary counter breaks amortized $O(1)$:** n ops can cost $\Theta(nk)$ (Exercise 16.1-2).

Case Studies & Examples

Stack aggregate analysis

Sequence of n operations: PUSH, PUSH, MULTIPOP($S,2$), POP, PUSH, MULTIPOP($S,3$)

Naive worst-case: each MULTIPOP $O(n) \rightarrow O(n^2)$

Actual: each pop corresponds to a previous push $\rightarrow$ at most n pops total $\rightarrow O(n)$

Binary counter flips ($n=16, k=8$)

Value Bits (A7..A0) Flips

0	00000000	—
1	00000001	1
2	00000010	2
3	00000011	1
4	00000100	3
...	...	...
16	00010000	1

Total flips = $31 < 2 \cdot 16 = 32$

Dynamic table expansion sequence

Insert ops 1..16: costs = 1,2,1,4,1,1,1,8,1,1,1,1,1,1,1,16

Total = $1+2+1+4+1+1+1+8+1+1+1+1+1+1+1+16 = 1+2+4+8+16 = 31 < 3 \cdot 16 = 48$

Amortized cost per op = $31/16 < 2$

Diagrams & Visuals

Stack with MULTIPOP:
Initial: [A,B,C,D] (A=top)
MULTIPOP(3): pops A,B,C $\rightarrow$ [D]
Cost: 3 pops = 3 units

Binary counter flips pattern:
Bit A0: flips every time (n times)
Bit A1: flips every 2nd time (n/2)
Bit A2: flips every 4th time (n/4)
...

Dynamic table potential:
 $\Phi = 2 \cdot \text{num} - \text{size}$ (when $\alpha \geq \frac{1}{2}$)
 $\Phi = \text{size}/2 - \text{num}$ (when $\alpha < \frac{1}{2}$)

After expansion ($\alpha=\frac{1}{2}$): $\Phi=0$
As insertions fill table: Φ increases linearly
At full ($\alpha=1$): $\Phi = \text{size}$ (pays for next expansion)

End-of-Chapter Material

- **Key terms:** amortized analysis, aggregate analysis, accounting method, potential method, credit, potential function, load factor, table expansion, contraction, MULTIPOP
- **Exercises:** 16.1-1 (MULTIPUSH impact), 16.1-2 (DECREMENT breaks amortized), 16.1-3 (powers of 2 cost sequence), 16.2-1 (stack with backups), 16.3-1 (re-zeroing potential), 16.3-5 (queue with two stacks), 16.4-1 (first insertion potential), 16.4-3 (accounting for both insert/delete)

Cross-Chapter Links

- **Requires knowledge of:** Ch 10 (stacks), Ch 6 (binary heaps), Ch 11 (hash tables, load factor)
- **Referenced in later chapters:** Ch 17 (augmenting data structures), Ch 19 (disjoint-set forests), Ch 13 (red-black tree amortized analysis)

Ch. 17 – Augmenting Data Structures

Named Entities

- **Order-statistic tree:** red-black tree augmented with `size` attribute (subtree internal node count)
- **Interval tree:** red-black tree keyed by low endpoint, augmented with `max` attribute
- **Interval trichotomy:** any two intervals satisfy exactly one: overlap, left-of, right-of

Processes / Algorithms / Pathways

Four-Step Augmentation Process

1. Choose underlying data structure (e.g., red-black tree)
2. Determine additional information to maintain (e.g., subtree sizes)
3. Verify attribute can be maintained by existing operations (insert/delete)
4. Develop new operations using the augmented info (e.g., OS-SELECT)

OS-SELECT(x, i)

- **Goal:** Return pointer to i -th smallest key in subtree
- **Steps:** $r = x.\text{left.size} + 1$; if $i == r$ return x ; if $i < r$ recurse left; else recurse right with $i - r$
- **Example:** Root has key 26, $\text{left.size} = 7$, $\text{right.size} = 15$. OS-SELECT(root, 17): $r = 8$, $i > r$ so recurse right with $i = 17 - 8 = 9$. Right child key 41 has $\text{left.size} = 5$, $r = 6$, $i > r$ so recurse right with $9 - 6 = 3$. Right child key 47 has $\text{left.size} = 2$ but left is leaf, so recurse right ($3 - 1 = 2$)... tracing continues until matching rank found.
- **Complexity:** $O(\lg n)$

OS-RANK(T, x)

- **Goal:** Return rank of x in inorder walk
- **Steps:** $r = x.\text{left.size} + 1$; while $y \neq \text{root}$: if y is right child, add $\text{left.size} + 1$; $y = y.p$
- **Complexity:** $O(\lg n)$

INTERVAL-SEARCH(T, i)

- **Steps:** $x = \text{root}$; while $x \neq \text{nil}$ and no overlap: if $x.\text{left.max} \geq i.\text{low}$ go left else go right; return x
- **Correctness:** The key invariant is that if an overlapping interval exists, the algorithm's path choice will never miss it. Proof: If $x.\text{left.max} < i.\text{low}$ then no interval in left subtree can overlap i (all end before i starts). Symmetrically, when going left, if any interval overlaps i , it must be in left subtree. This is guaranteed by the max attribute – the maximum right endpoint in the left subtree.
- **Complexity:** $O(\lg n)$

Maintaining size During Rotations

- Left/right rotations change subtree structure but size can be recomputed in $O(1)$:
 - $y.\text{size} = x.\text{size}$
 - $x.\text{size} = x.\text{left.size} + x.\text{right.size} + 1$
- This ensures RB-INSERT-FIXUP and RB-DELETE-FIXUP maintain the size attribute correctly without changing the $O(\lg n)$ time bound.

Formulas & Equations

Subtree size

$$x.\text{size} = x.\text{left.size} + x.\text{right.size} + 1$$

Node max (interval tree)

$$x.\text{max} = \max(x.\text{int.high}, x.\text{left.max}, x.\text{right.max})$$

Rules, Laws & Theorems

- **Theorem 17.1 (Augmenting RB Trees):** Let f be an attribute that depends only on x , $x.\text{left}$, and $x.\text{right}$. If f can be computed from these in $O(1)$ time, then RB-INSERT and RB-DELETE can maintain f without increasing their asymptotic complexity.
- **Theorem 17.2 (INTERVAL-SEARCH Correctness):** INTERVAL-SEARCH returns a node whose interval overlaps i , or nil if no such node exists. The algorithm is correct because the max attribute guarantees safe pruning of subtrees that cannot contain an overlapping interval.

- **Edge case:** When $x.\text{left} = \text{nil}$, treat $x.\text{left}.\text{max}$ as $-\infty$ so the branch always goes right.
- **All-Overlapping Interval Tree:** If all intervals overlap i , INTERVAL-SEARCH returns the first node examined (the root if it overlaps), not necessarily the one with earliest start time. To find all overlapping intervals, call INTERVAL-SEARCH repeatedly, removing found intervals.

Edge Cases & Common Pitfalls

- **Duplicate intervals:** OS-SELECT returns the first encountered node with the target rank; if keys are not unique, the rank ordering may be ambiguous
- **nil children:** Always guard $x.\text{left}.\text{size}$ and $x.\text{right}.\text{size}$ with nil checks; treat $\text{nil}.\text{size} = 0$, $\text{nil}.\text{max} = -\infty$
- **Rotation during maintenance:** After a left or right rotation, the size attribute for the rotated nodes must be recomputed before returning from the fixup procedure
- **Interval tree deletion:** When deleting a node, the max attributes along the path to root must be recomputed ($O(\lg n)$)

Cross-Chapter Links

- Ch.13 (red-black trees), Ch.9 (order statistics)

Ch. 18 – B-Trees

Named Entities

- **B-tree:** balanced search tree with high branching factor, designed for disk
- **Minimum degree t :** every non-root node has $t-1$ to $2t-1$ keys; root has 1 to $2t-1$
- **Full node:** node with exactly $2t-1$ keys
- **2-3-4 tree:** B-tree with $t=2$ (each internal node has 2, 3, or 4 children)

Processes / Algorithms / Pathways

B-TREE-SEARCH(x, k)

- **Linear scan:** Walk through keys left-to-right; if found return; if leaf return nil; DISK-READ appropriate child, recurse
- **Binary search variant:** Within a node, use binary search instead of linear scan to reduce CPU cost to $O(\lg t \cdot \log_t n)$ at the expense of slightly more complex code
- **Complexity:** Disk $O(\log_t n)$, CPU $O(t \log_t n)$ linear, or $O(\log_t n \cdot \lg t)$ with binary variant

B-TREE-INSERT(T, k)

- **Steps:** If root is full, B-TREE-SPLIT-ROOT (creates new root with one key, height increases by 1). Then B-TREE-INSERT-NONFULL: if leaf, shift keys right and insert; else find child to descend into, split if that child is full, then recurse
- **Split-root case:** When the root has $2t-1$ keys, we split it into two nodes around its median key. The median moves up to a new root node (with one key and two children). This is the only way the tree grows in height.

- **Example (t=3):** Insert into root [1,2,3,4,5] (full). Split at median 3: new root [3] with children [1,2] and [4,5]. Then insert k normally.
- **Complexity:** Disk $O(h)$, CPU $O(t \cdot h)$

B-TREE-DELETE(T, k)

- **Goal:** Delete key k while ensuring every node has $\geq t-1$ keys (except root). The algorithm descends once, restructuring before descending so the child being entered always has $\geq t$ keys.
- **Case 1 (leaf):** If k is in a leaf node, simply remove it.
- **Case 2 (internal node, has left/right children):**
 - **Case 2a (predecessor):** If child preceding k has $\geq t$ keys, find predecessor, swap, recursively delete predecessor from that child.
 - **Case 2b (successor):** Symmetric to 2a using successor child.
 - **Case 2c (merge):** If both adjacent children have exactly $t-1$ keys, merge k and both children into one node ($2t-1$ keys total), then recursively delete k from the merged node.
- **Case 3 (internal node, ensuring precondition):**
 - **Case 3a (sibling has $\geq t$ keys):** While descending, if target child has $t-1$ keys but an adjacent sibling has $\geq t$ keys, rotate a key from parent through sibling into the child.
 - **Case 3b (sibling also $t-1$):** If both child and adjacent sibling have $t-1$ keys, merge child with sibling, pulling one key from parent down.
- **Example (t=3):** Delete 4 from internal node [3,4,7] where left child [1,2] and right child [5,6] both have $t-1=2$ keys. Merge: new node [3,4,5,6,7] replaces children, then delete 4 from the merged internal node.
- **Complexity:** Disk $O(h)$, CPU $O(t \cdot h)$

Formulas & Equations

Height bound proof

- Root has ≥ 1 key. All other nodes have $\geq t-1$ keys.
- At depth 1: ≥ 2 nodes (each $\geq t-1$ keys) $\rightarrow \geq 2(t-1)$ keys
- At depth 2: $\geq 2t$ nodes $\rightarrow \geq 2t(t-1)$ keys
- At depth h: $\geq 2t^{h-1}$ nodes $\rightarrow \geq 2t^{h-1}(t-1)$ keys
- Total: $n \geq 1 + (t-1) \cdot 2 \cdot (t^{h-1}-1)/(t-1) = 2t^h - 1$
- Solving: $h \leq \log_t ((n+1)/2)$
- $n=1M$, $t=100 \rightarrow h \approx 3$. Binary tree: $h \approx 20$

Comparisons & Trade-offs

Dimension	B-tree	Red-Black Tree
Branching factor t to 2t (large)	2	2
Height	$O(\log_t n)$	$O(\lg n)$
Disk accesses	$O(\log_t n)$	$O(\lg n)$

Relationship to Red-Black Trees

- A red-black tree is equivalent to a B-tree with $t=2$ (2-3-4 tree) when you interpret black nodes as tree nodes and red nodes as "extra keys" within a B-tree node.
- Specifically: merge each black node with its red children to form a B-tree node with 1-3 keys.
- This correspondence is used in the red-black deletion algorithm.

Edge Cases & Common Pitfalls

- **Root merge in delete:** If after a merge the root has zero keys, delete the old root; the merged child becomes the new root (height decreases)
- **Delete with $t=2$:** Minimum degree $t=2$ (2-3-4 tree) is the most complex case because $t-1 = 1$, so many nodes are exactly at the minimum
- **Insert into B-tree with preexisting duplicate keys:** B-trees typically assume distinct keys; insertion with duplicates requires the convention (insert left, insert right, or reject)
- **DISK-READ/DISK-WRITE costs dominate:** In practice the CPU cost within a node (linear scan of up to $2t-1$ keys) is negligible compared to disk I/O

B-Tree Variants

- **B+ tree:** Internal nodes store only keys (routing); all data stored in leaves linked in a linked list. Used in database indexes (MySQL, PostgreSQL).
- **B* tree:** Nodes are kept $2/3$ full by redistributing keys to siblings before splitting.
- **Pre-splitting:** During insertion, split full nodes proactively on the way down (as done in B-TREE-INSERT-NONFULL) to avoid a second pass.

End-of-Chapter Material

- **Key terms:** minimum degree, full node, split, merge, 2-3-4 tree, B-TREE-DELETE case analysis, B+ tree, B* tree
- **Exercises:** 18.1-1 (min/max keys), 18.2-1 (insert into $t=3$ B-tree)

Cross-Chapter Links

- Ch.13 (red-black trees generalize to B-trees), Ch.12 (BST search)

Ch. 19 – Data Structures for Disjoint Sets

Named Entities

- **Disjoint-set (union-find):** maintains collection of disjoint dynamic sets
- **Union by rank:** attach shorter tree under taller; $\text{rank} \approx \lg(\text{size})$
- **Path compression:** during FIND-SET, make every node on find path point to root
- **Ackermann function $A_k(j)$:** defined recursively: $A_1(j) = 2^{j+1}$ for $j \geq 1$; $A_k(1) = A_{k-1}(2)$ for $k \geq 2$; $A_k(j) = A_{k-1}(A_k(j-1))$ for $k \geq 2, j \geq 2$. $A(1) = A(2) \approx 2^{2048} \gg 10$.
- **$\alpha(n)$:** inverse Ackermann; $\alpha(n) = \min\{k : A_k(1) \geq n\}$. $\alpha(n) \leq 4$ for all practical n (since $A_4(1)$ is astronomically large).

Processes / Algorithms / Pathways

MAKE-SET(x)

- **Steps:** $x.p = x$, $x.\text{rank} = 0$; **Complexity:** $O(1)$

FIND-SET(x)

- **Steps:** if $x \neq x.p$: $x.p = \text{FIND-SET}(x.p)$; return $x.p$

- **Complexity:** Amortized $O(\alpha(n))$

UNION(x, y)

- **Steps:** LINK(FIND-SET(x), FIND-SET(y))
- **Complexity:** $O(\alpha(n))$ amortized

LINK(x, y)

- **Steps:** if $x.\text{rank} > y.\text{rank}$: $y.p = x$ else $x.p = y$; if equal: $y.\text{rank}++$
- **Complexity:** $O(1)$

CONNECTED-COMPONENTS Application

- Process a graph's edges with disjoint-set:
 1. For each vertex v : MAKE-SET(v)
 2. For each edge (u, v) : if FIND-SET(u) $\neq$ FIND-SET(v), UNION(u, v)
- **Example:** Graph with vertices $\{a, b, c, d, e\}$ and edges (a, b) , (c, d) , (b, c) , (d, e) :
 - After (a, b) : $\{a, b\}$, $\{c\}$, $\{d\}$, $\{e\}$
 - After (c, d) : $\{a, b\}$, $\{c, d\}$, $\{e\}$
 - After (b, c) : UNION($\{a, b\}$, $\{c, d\}$) $\rightarrow \{a, b, c, d\}$, $\{e\}$
 - After (d, e) : UNION($\{a, b, c, d\}$, $\{e\}$) $\rightarrow \{a, b, c, d, e\}$
 - Result: single connected component

Linked-List Representation

- **Simple variant:** Each set = linked list with head pointing to first element; each element points to set representative (head). UNION concatenates lists and updates all tail elements' pointers $\rightarrow O(n)$.
- **Weighted variant:** Maintain length of each list; always append shorter list to longer. Each element's representative pointer changes at most $\lg n$ times because each change doubles the size of its set.

Potential Function Analysis (Forest)

- Define potential $\Phi = \sum \{x\} \text{rank}(x)$ after path compression adjustments
- The potential decreases by at least 1 per FIND-SET (amortized cost = actual cost - $\Delta\Phi$)
- Combined with union by rank, the amortized cost per operation is $O(\alpha(n))$
- **Key insight:** Path compression reduces rank on the path, lowering future potential. The money saved from potential reduction pays for the traversal.

Rank Properties

- $x.\text{rank} \leq x.p.\text{rank}$ (rank is non-decreasing along any path to root)
- If $x.\text{rank} = k$, the subtree rooted at x has at least 2^k nodes (proved by induction)
- Number of nodes of rank k is at most $n/2^k$
- Ranks are integers, starting at 0, and $\text{rank}(x) < \lg n$

Path Compression Analysis Sketch

1. Partition nodes by rank into groups where group g contains ranks $[A(g-1)+1, A(g)]$ where A is the Ackermann function
2. For each FIND-SET, charge $O(1)$ cost plus the number of nodes whose parent changes

3. Nodes whose parent changes move to a higher rank group; each node's parent changes at most $\alpha(n) + 1$ times within its group
4. Summing over all operations gives $O(m \alpha(n))$

Formulas & Equations

- **Weighted-union (linked-list):** each object's pointer updated $\leq \lg n$ times, so m operations take $O(m + n \lg n)$
- **Theorem 19.1 (Weighted-Union Heuristic):** With weighted union on linked lists, a sequence of m MAKE-SET, UNION, FIND-SET operations takes $O(m + n \lg n)$ time.
- **Theorem 19.14 (Forest):** Forest with union by rank + path compression: $O(m \alpha(n))$ for m ops
- **Ackermann values:** $A_0(1)=2$, $A_1(1)=3$, $A_2(1)=7$, $A_3(1)=2047$, $A_4(1) \approx 2^{2048}$

Edge Cases & Common Pitfalls

- **Union without FIND-SET:** Calling UNION on the raw nodes (not roots) produces incorrect results; always call UNION(FIND-SET(x), FIND-SET(y))
- **Rank after path compression:** The rank of a node is an upper bound on its height; path compression can make the actual height much less than the rank, but the rank is not updated
- **Recursive FIND-SET:** The recursive implementation of FIND-SET can overflow the stack for very deep trees (before path compression flattens them); iterative two-pass or one-pass variants exist

Cross-Chapter Links

- Ch.16 (potential method for analysis), Ch.20 (connected components), Ch.21 (Kruskal's algorithm)

Ch. 21 – Minimum Spanning Trees

Named Entities (Terms & Definitions)

- **Minimum spanning tree (MST):** acyclic subset TE connecting all vertices with minimum total weight.
- **Cut $(S, V-S)$:** partition of V .
- **Edge crosses cut:** one endpoint in S , other in $V-S$.
- **Cut respects A :** no edge in A crosses the cut.
- **Light edge:** edge crossing a cut with minimum weight.
- **Safe edge:** edge (u,v) such that $A \cup \{(u,v)\}$ some MST.
- **Generic-MST:** grows MST one safe edge at a time.
- **Kruskal's algorithm:** sort edges by weight; add edge if its endpoints are in different components (union-find).
- **Prim's algorithm:** grow single tree; add lightest edge connecting tree to non-tree vertex (priority queue).

Processes / Algorithms / Pathways

Generic-MST(G, w)

- **Type:** Algorithm
- **Steps:** (1) $A = \emptyset$. (2) While A not spanning tree: find safe edge (u,v) , $A = A \cup \{(u,v)\}$. (3) Return A .
- **Loop invariant:** A some MST.

Kruskal(G, w)

- **Type:** Algorithm
- **Steps:** (1) $A = \emptyset$. For each $v \in V$: MAKE-SET(v). (2) Sort edges by weight. (3) For each (u, v) in order: if FIND-SET(u) $\neq$ FIND-SET(v): $A = A \cup \{(u, v)\}$, UNION(u, v). (4) Return A .
- **Complexity:** $O(E \lg V)$ – dominated by sorting; disjoint-set ops $O(E \alpha(V))$.
- **Example:** Graph $V = \{a, b, c\}$, edges: $(a, b) = 1$, $(b, c) = 2$, $(a, c) = 3$. Sorted: $(a, b) \rightarrow \text{add}$; $(b, c) \rightarrow \text{add}$ (different sets); $(a, c) \rightarrow \text{skip}$ (same set). MST weight = 3.

Prim(G, w, r)

- **Type:** Algorithm
- **Steps:** (1) For each u : $u.\text{key} = \infty$, $u.\pi = \text{NIL}$. $r.\text{key} = 0$. $Q = V$. (2) While $Q \neq \emptyset$: $u = \text{EXTRACT-MIN}(Q)$. For each $v \in \text{Adj}[u]$: if $v \in Q$ and $w(u, v) < v.\text{key}$: $v.\pi = u$, $v.\text{key} = w(u, v)$, DECREASE-KEY($Q, v, v.\text{key}$). (3) Return $A = \{(v, v.\pi) : v \in V - \{r\}\}$.
- **Complexity:** $O(E \lg V)$ with binary heap; $O(E + V \lg V)$ with Fibonacci heap.

Classifications & Hierarchies

- **MST algorithm family:**
 - Generic: safe-edge framework
 - Kruskal: forest-growing, global view
 - Prim: tree-growing, local view
 - Borůvka: parallel edge contraction

Comparisons & Trade-offs

Dimension	Kruskal	Prim
Data structure	Union-find (disjoint sets)	Min-priority queue
Edge processing	Sorted globally	Incremental per vertex
Graph type	Sparse (E small)	Dense (E large)
Complexity	$O(E \lg V)$	$O(E \lg V)$ binary, $O(E + V \lg V)$ Fibonacci

Formulas & Equations

Cut property (Theorem 21.1)

If $(S, V - S)$ respects A and (u, v) is a light edge crossing, then (u, v) is safe for A . ##### Cycle property Let e be a max-weight edge on some cycle. Then e is not in any MST.

Rules, Laws & Theorems

Theorem 21.1 (Safe-edge recognition)

Given AMST, cut $(S, V - S)$ that respects A , light edge (u, v) crossing the cut $\rightarrow (u, v)$ is safe for A . ##### Corollary 21.2 If (u, v) is a light edge connecting component C to another component in $G - A$, then (u, v) is safe for A .

Edge Cases & Common Pitfalls

- **Ties in edge weights:** MST may not be unique; any light edge works.
- **Disconnected graph:** Kruskal/Prim assume connected graph; otherwise produce minimum spanning forest.
- **Nonpositive weights:** MST still valid, but minimum-weight connected subgraph might not be a tree (e.g., negative edges).

Diagrams & Visuals

Kruskal's algorithm:

Forest of $|V|$ trees $\rightarrow$ merge with lightest connecting edges
Sorted edges: $(a,b)=1^*$, $(b,c)=2^*$, $(a,c)=3$ (skip)

Prim's algorithm:

Single tree grows from root r
Queue Q holds non-tree vertices, keyed by min edge to tree

End-of-Chapter Material

- **Key terms:** MST, cut, light edge, safe edge, cut property, cycle property, Kruskal, Prim, union-find
- **Exercises:** 21.1-1 (min-weight edge in some MST), 21.2-1 (tie-breaking in Kruskal), 21.2-2 ($O(V^2)$ Prim with adjacency matrix)

Cross-Chapter Links

- **Referenced in:** Ch.22 (Prim $\leftrightarrow$ Dijkstra similarity), Ch.23 (reweighting in Johnson's algorithm), Ch.24 (cut definitions for flow)
- **Requires:** Ch.19 (disjoint-set forest), Ch.15 (greedy algorithms), Ch.6 (binary heaps)

Ch. 22 – Single-Source Shortest Paths

Named Entities (Terms & Definitions)

- **Shortest-path weight $\delta(s,v)$:** min total weight of any path from s to v (∞ unreachable, $-\infty$ if negative-weight cycle reachable).
- **Relaxation:** test whether $u.d + w(u,v) < v.d$; if so, update $v.d$ and $v.\pi$.
- **Initialize-Single-Source(G,s):** set $v.d=\infty$, $v.\pi=\text{NIL}$, $s.d=0$.
- **Triangle inequality:** $\delta(s,v) \leq \delta(s,u) + w(u,v)$ for all edges (u,v) .
- **Upper-bound property:** $v.d \geq \delta(s,v)$ always; once $v.d=\delta(s,v)$, it never changes.
- **Convergence property:** if $u.d=\delta(s,u)$ before relaxing (u,v) on a shortest path, then $v.d=\delta(s,v)$ thereafter.
- **Path-relaxation property:** relaxing edges of a shortest path in order yields final shortest-path weights.
- **Predecessor-subgraph property:** when all $v.d=\delta(s,v)$, G_π is a shortest-paths tree.
- **Bellman-Ford algorithm:** handles negative edges; $O(VE)$; detects negative-weight cycles.
- **DAG-Shortest-Paths:** topological sort + relax once per edge; $O(V+E)$.
- **Dijkstra's algorithm:** nonnegative weights only; $O(V^2)$ array, $O(E \lg V)$ binary heap, $O(V \lg V + E)$ Fibonacci heap.
- **Difference constraints:** system $x_j - x_i \leq b_k$; solved via constraint graph + Bellman-Ford.

Processes / Algorithms / Pathways

Bellman-Ford(G, w, s)

- **Type:** Algorithm
- **Steps:** (1) INIT-SINGLE-SOURCE(G, s). (2) For $i=1$ to $|V|-1$: for each edge (u, v) : RELAX(u, v, w). (3) For each edge (u, v) : if $v.d > u.d + w(u, v)$: return FALSE (negative cycle). (4) Return TRUE.
- **Complexity:** $O(VE)$ time, $O(V)$ space
- **Example:** 5-vertex graph, 4 passes over edges. After each pass, distances improve; final d values = $\delta(s, v)$. If any edge still relaxable after $|V|-1$ passes $\rightarrow$ negative cycle.

DAG-Shortest-Paths(G, w, s)

- **Type:** Algorithm
- **Steps:** (1) Topological sort G . (2) INIT-SINGLE-SOURCE(G, s). (3) For each u in topological order: for each $v \in \text{Adj}[u]$: RELAX(u, v, w).
- **Complexity:** $O(V+E)$ time
- **Application:** critical path in PERT charts (negate weights to find longest path).

Dijkstra(G, w, s)

- **Type:** Algorithm
- **Steps:** (1) INIT-SINGLE-SOURCE(G, s). (2) $S = \emptyset$, $Q = V$ (keyed by d). (3) While $Q \neq \emptyset$: $u = \text{EXTRACT-MIN}(Q)$, $S = S \cup \{u\}$. For each $v \in \text{Adj}[u]$: RELAX(u, v, w); if $v.d$ decreased, DECREASE-KEY($Q, v, v.d$).
- **Complexity:** $O(V^2)$ with array, $O(E \lg V)$ with binary heap, $O(V \lg V + E)$ with Fibonacci heap
- **Requires:** $w(u, v) \geq 0$
- **Example:** Graph with $s \rightarrow a$ (1), $s \rightarrow b$ (4), $a \rightarrow b$ (2). Extract s ($d=0$), relax $a \rightarrow 1$, $b \rightarrow 4$. Extract a ($d=1$), relax $b \rightarrow 1+2=3$ (update). Extract b ($d=3$).

Comparisons & Trade-offs

Dimension	Bellman-Ford	DAG-Shortest	Dijkstra
Negative edges allowed	Yes	Yes (no cycles)	No
Negative cycle detection	Yes	N/A (acyclic)	N/A
Complexity	$O(VE)$	$O(V+E)$	$O(E \lg V)$
Edge relaxations	$V-1$ passes	1 pass	1 per vertex

Formulas & Equations

Relaxation

```
RELAX( $u, v, w$ ):  
  if  $v.d > u.d + w(u, v)$ :  
     $v.d = u.d + w(u, v)$   
     $v.\pi = u$ 
```

System of difference constraints

$x_j - x_i \leq b_k \rightarrow$ constraint graph: edge (v_i, v_j) with weight b_k ; add source v_0 with 0-weight edges to all v_i . Feasible solution: $x_i = \delta(v_0, v_i)$.

Rules, Laws & Theorems

Lemma 22.1 (Optimal substructure)

Subpaths of shortest paths are shortest paths. ##### Lemma 22.10 (Triangle inequality) $\delta(s,v) \leq \delta(s,u) + w(u,v)$ for all edges (u,v) . ##### Upper-bound property (Lemma 22.11) $v.d \geq \delta(s,v)$ always; once equal, never changes. ##### Convergence property (Lemma 22.14) If $u.d = \delta(s,u)$ before relaxing (u,v) on shortest path $su \rightarrow v$, then $v.d = \delta(s,v)$ thereafter. ##### Path-relaxation property (Lemma 22.15) Relax edges of shortest path p in order $\rightarrow v_k.d = \delta(s, v_k)$. ##### Theorem 22.4 (Bellman-Ford correctness) Returns TRUE and correct $\delta(s,v)$ iff no negative-weight cycle reachable from s . ##### Theorem 22.6 (Dijkstra correctness) With nonnegative weights, Dijkstra terminates with $u.d = \delta(s,u)$ for all u .

Edge Cases & Common Pitfalls

- **Negative-weight cycle reachable from s :** $\delta(s,v) = -\infty$; Bellman-Ford returns FALSE.
- **Zero-weight cycles:** can be removed without affecting weight.
- **Dijkstra on negative edges:** may produce incorrect result (selects wrong vertex as settled).
- **Disconnected vertices:** $v.d$ remains ∞ (no-path property).

Diagrams & Visuals

Relaxation:

Before: $u.d = 5, v.d = 9, w(u,v) = 2$
 After RELAX: $v.d = \min(9, 5+2) = 7, v.\pi = u$

Shortest-paths tree:

Root $s \rightarrow \dots \rightarrow v$ (unique simple path in G_π is shortest path)

End-of-Chapter Material

- **Key terms:** relaxation, shortest-path estimate, triangle inequality, convergence, Bellman-Ford, DAG-Shortest-Paths, Dijkstra, difference constraints
- **Exercises:** 22.1-1 (Bellman-Ford on Figure 22.4), 22.2-1 (DAG-Shortest-Paths), 22.3-1 (Dijkstra on Figure 22.2), 22.4-1 (difference constraints feasible?)

Cross-Chapter Links

- **Requires:** Ch.20 (BFS as unweighted SSSP, topological sort)
- **Referenced in:** Ch.23 (Johnson's algorithm uses Bellman-Ford + Dijkstra), Ch.24 (Edmonds-Karp uses BFS)

Ch. 23 – All-Pairs Shortest Paths

Named Entities (Terms & Definitions)

- **All-pairs shortest paths (APSP):** find $\delta(i,j)$ for all $i,j \in V$.
- **Predecessor matrix Π :** π_{ij} = predecessor of j on shortest path from i .
- **EXTEND-Shortest-Paths:** min,+ matrix multiplication analogue; $L^{\{r\}} = L^{\{r-1\}} \cdot W$ yields paths with $\leq r$ edges.
- **SLOW-APSP:** compute $L^{\{1\}}$ through $L^{\{n-1\}}$ by repeated extension; $\Theta(n^3)$.
- **FASTER-APSP:** repeated squaring; $\Theta(n^3 \lg n)$.
- **Floyd-Warshall algorithm:** DP over intermediate vertices; $\Theta(V^3)$; uses $D^{\{k\}}$ where intermediate vertices $\{1..k\}$.
- **Transitive closure:** $T^{\{k\}}$ matrix; boolean; $T^{\{\emptyset\}} = I$ adjacency,

$$T^{\{(k)\}} = T^{\{(k-1)\}} \quad (T^{\{(k-1)\}} \quad T^{\{(k-1)\}}).$$

- **Johnson's algorithm:** reweighting + Dijkstra from each vertex; $O(V^2 \lg V + VE)$ for sparse graphs.

Processes / Algorithms / Pathways

SLOW-APSP(W, $L^{\{(0)\}}$, n)

- **Steps:** $L = L^{\{(0)\}}$. For $r=1$ to $n-1$: $M=\infty$; EXTEND(L, W, M, n); $L=M$. Return L .
- **Complexity:** $\Theta(n)$

FASTER-APSP(W, n)

- **Steps:** $L=W$, $r=1$. While $r < n-1$: $M=\infty$; EXTEND(L, L, M, n); $r=2r$; $L=M$. Return L .
- **Complexity:** $\Theta(n^3 \lg n)$

Floyd-Warshall(W, n)

- **Type:** Algorithm (DP)
- **Recurrence:** $d_{\{ij\}}^{\{(k)\}} = \min(d_{\{ij\}}^{\{(k-1)\}}, d_{\{ik\}}^{\{(k-1)\}} + d_{\{kj\}}^{\{(k-1)\}})$
- **Steps:** $D^{\{(0)\}} = W$. For $k=1..n$: for $i=1..n$: for $j=1..n$: $d_{\{ij\}}^{\{(k)\}} = \min(d_{\{ij\}}^{\{(k-1)\}}, d_{\{ik\}}^{\{(k-1)\}} + d_{\{kj\}}^{\{(k-1)\}})$. Return $D^{\{(n)\}}$.
- **Complexity:** $\Theta(V^3)$ time, $\Theta(V^2)$ space (in-place)
- **Example:** 3-vertex graph. $D^{\{(0)\}}$ = edge weights. After $k=1$: paths via 1. After $k=2$: paths via $\{1,2\}$. After $k=3$: complete.

Johnson(G, w)

- **Type:** Algorithm
- **Steps:** (1) Add vertex s with 0-weight edges to all v . (2) Run Bellman-Ford from $s \rightarrow h(v)=\delta(s,v)$. If negative cycle $\rightarrow$ report. (3) Reweight: $\hat{w}(u,v)=w(u,v)+h(u)-h(v) \geq 0$. (4) For each uV : Dijkstra($G, \hat{w}, u$) $\rightarrow d_{\{uv\}}$. (5) Convert: $d_{\{uv\}}=d_{\{uv\}}+h(v)-h(u)$. (6) Return D .
- **Complexity:** $O(V^2 \lg V + VE)$ with Fibonacci heap

Comparisons & Trade-offs

Dimension	Floyd-Warshall	Johnson	FASTER-APSP
Approach	DP over vertices	Reweighting + Dijkstra	Repeated squaring
Complexity	$\Theta(V^3)$	$O(V^2 \lg V + VE)$	$\Theta(V^3 \lg V)$
Best for	Dense graphs	Sparse graphs	Theoretical interest
Negative edges allowed	Yes (no neg cycles)	Yes (no neg cycles)	Yes (no neg cycles)

Formulas & Equations

Floyd-Warshall recurrence

$$d_{\{ij\}}^{\{(k)\}} = \min(d_{\{ij\}}^{\{(k-1)\}}, d_{\{ik\}}^{\{(k-1)\}} + d_{\{kj\}}^{\{(k-1)\}}) - d_{\{ij\}}^{\{(0)\}} = w_{\{ij\}} \quad (0 \text{ if } i=j, \infty \text{ if no edge}) - \text{Base: } d_{\{ii\}} = 0$$

Reweighting (Johnson)

$\hat{w}(u,v) = w(u,v) + h(u) - h(v)$ - $\hat{w}(u,v) = \delta(s,v)$ from Bellman-Ford on G' with added source s - Preserves shortest paths: $\hat{w}(p) = w(p) + h(s) - h(t)$ - Ensures $\hat{w}(u,v) \geq 0$

Rules, Laws & Theorems

Lemma 23.1 (Reweighting preserves shortest paths)

$\hat{w}(p) = w(p) + h(s) - h(t)$; thus shortest paths under w are shortest under $\hat{w}$.

Edge Cases & Common Pitfalls

- **Negative-weight cycles:** Floyd-Warshall detects if $d_{ii} < 0$ after completion; Johnson detects via Bellman-Ford.
- **Space:** in-place Floyd-Warshall only needs $\Theta(V^2)$ (one matrix D , update in place).
- **Transitive closure vs. shortest paths:** if only reachability needed, use boolean version ($T^{\wedge}\{(k)\}$) for efficiency.

Diagrams & Visuals

Floyd-Warshall DP structure:

$d_{ij}^k = \text{shortest path } i \rightarrow j \text{ using intermediates } \{1..k\}$

$k=0$: direct edges only

$k=1$: include vertex 1 as intermediate

...

$k=n$: all vertices allowed $\rightarrow$ final answer

Johnson's reweighting:

$G' = G \cup \{s\}$ with edges $(s,v):0$

$h(v) = \text{shortest distance from } s \text{ to } v$

$\hat{w}(u,v) = w(u,v) + h(u) - h(v) \geq 0$

End-of-Chapter Material

- **Key terms:** EXTEND-Shortest-Paths, repeated squaring, Floyd-Warshall, transitive closure, Johnson's algorithm, reweighting
- **Exercises:** 23.1-1 (SLOW-APSP and FASTER-APSP on Figure 23.2), 23.2-1 (Floyd-Warshall on Figure 23.2), 23.3-1 (Johnson on Figure 23.2)

Cross-Chapter Links

- **Requires:** Ch.22 (Bellman-Ford, Dijkstra), Ch.14 (DP), Ch.4 (matrix multiplication)
- **Referenced in:** Ch.24 (max-flow/min-cut theorem)

Ch. 24 – Maximum Flow

Named Entities (Terms & Definitions)

- **Flow network:** directed graph $G=(V,E)$ with capacity $c(u,v) \geq 0$, source s , sink t ; no antiparallel edges.
- **Flow f :** function $V \times V \rightarrow \mathbb{R}$ satisfying: (1) capacity constraint: $0 \leq f(u,v) \leq c(u,v)$; (2) flow conservation: $\sum_v f(u,v) = \sum_v f(v,u)$.
- **Flow value $|f|$:** net flow out of source: $\sum_v f(s,v) - \sum_v f(v,s)$.
- **Residual network G_f :** edges with residual capacity $c_f(u,v) = c(u,v) - f(u,v)$

- $f(u,v)$ if $(u,v) \in E$, or $c_f(u,v) = f(v,u)$ if $(v,u) \in E$; else 0.
- **Augmenting path**: simple $s \rightarrow t$ path in G_f .
- **Residual capacity of path p** : $c_f(p) = \min\{c_f(u,v) : (u,v) \in p\}$.
- **Cut (S,T)** : partition with $s \in S$, $t \in T$. Capacity $c(S,T) = \sum_{u \in S, v \in T} c(u,v)$. Net flow $f(S,T) = \sum_{u \in S, v \in T} f(u,v) - \sum_{v \in T, u \in S} f(v,u)$.
- **Max-flow min-cut theorem**: $|f| = c(S,T)$ for some cut f is maximum G_f has no augmenting path.
- **Ford-Fulkerson method**: repeatedly find augmenting path, augment flow.
- **Edmonds-Karp**: Ford-Fulkerson using BFS for augmenting path; $O(VE^2)$.
- **Max bipartite matching**: maximum cardinality matching in bipartite graph via max flow.
- **Integrality theorem**: if capacities are integer, Ford-Fulkerson produces integer flow.

Processes / Algorithms / Pathways

Ford-Fulkerson(G, s, t)

- **Type**: Algorithm
- **Steps**: (1) For each edge (u,v) : $f(u,v) = 0$. (2) While path p from s to t in G_f : $c_f(p) = \min\{c_f(u,v) : (u,v) \in p\}$. For each $(u,v) \in p$: if $(u,v) \in E$: $f(u,v) += c_f(p)$; else $f(v,u) -= c_f(p)$. (3) Return f .
- **Complexity**: $O(E |f|)$ where $|f|$ is max flow value (integer capacities).
- **Example**: Network with $V = \{s, u, v, t\}$, edges $s \rightarrow u(10)$, $s \rightarrow v(10)$, $u \rightarrow v(5)$, $u \rightarrow t(10)$, $v \rightarrow t(10)$. Augmenting paths: $s \rightarrow u \rightarrow t(10)$, $s \rightarrow v \rightarrow t(10)$, $s \rightarrow u \rightarrow v \rightarrow t(5) \rightarrow$ total $|f^*| = 25$. But if bad ordering ($s \rightarrow u \rightarrow v \rightarrow t$, $s \rightarrow v \rightarrow u \rightarrow t$ alternating), may need many iterations.

Edmonds-Karp(G, s, t)

- **Type**: Algorithm
- **Steps**: Same as Ford-Fulkerson but augmenting path found via BFS (shortest path in G_f).
- **Complexity**: $O(VE^2)$ – each of $O(VE)$ augmentations takes $O(E)$.
- **Proof**: Each edge becomes critical at most $O(V/2)$ times; $O(VE)$ critical edges total.

Maximum bipartite matching via flow

- **Construction**: Source $s \rightarrow$ each L (capacity 1) $\rightarrow$ each R (capacity 1) $\rightarrow$ sink t .
- **Steps**: Run Ford-Fulkerson on constructed flow network; matching $M = \{(u,v) : f(u,v) = 1\}$.
- **Complexity**: $O(VE)$ since $|f^*| = O(V)$.

Comparisons & Trade-offs

Dimension	Ford-Fulkerson (unrestricted)	Edmonds-Karp (BFS)
Path selection	Any augmenting path	Shortest (BFS)
Complexity	$O(E)$	f^*
Polynomial?	Pseudopolynomial	Polynomial
Works on irrational?	May not terminate	May not terminate

Formulas & Equations

Residual capacity

$$\begin{aligned} c_f(u,v) &= c(u,v) - f(u,v) && \text{if } (u,v) \in E \\ c_f(u,v) &= f(v,u) && \text{if } (v,u) \in E \\ c_f(u,v) &= 0 && \text{otherwise} \end{aligned}$$

Flow augmentation

$(f \uparrow f')(u,v) = f(u,v) + f'(u,v) - f'(v,u)$ ##### Value of augmented flow $|f \uparrow f'| = |f| + |f'|$

Rules, Laws & Theorems

Theorem 24.6 (Max-flow min-cut theorem)

For any flow f in G , these are equivalent: (1) f is max flow; (2) G_f has no augmenting path; (3) $|f| = c(S,T)$ for some cut (S,T) . ##### Lemma 24.4 (Net flow = $|f|$) For any cut (S,T) , $f(S,T) = |f|$. ##### Corollary 24.5 $|f| \leq c(S,T)$ for any cut (S,T) . ##### Integrality theorem (24.10) If capacities are integer, Ford-Fulkerson produces an integer-valued max flow. ##### Lemma 24.9 (Flow $\leftrightarrow$ Matching) In bipartite graph, matching M corresponds to 0-1 flow f with $|f|=|M|$.

Edge Cases & Common Pitfalls

- **Antiparallel edges:** not allowed in flow network definition; split by adding intermediate vertex.
- **Multiple sources/sinks:** add supersource/supersink with ∞ capacity edges.
- **Irrational capacities:** Ford-Fulkerson may never terminate.
- **Bad augmenting path choice:** can cause $O(|f^*|)$ iterations (e.g., Figure 24.7 with alternating paths).
- **Vertex capacities:** split vertex v into $v_{in} \rightarrow v_{out}$ with capacity = vertex capacity.

Diagrams & Visuals

Ford-Fulkerson flow network:
 $s \rightarrow [\text{intermediate vertices}] \rightarrow t$

Residual network G_f :
 Forward edges: $c_f(u,v)$ = capacity remaining
 Reverse edges: $c_f(v,u)$ = flow that can be "undone"
 Augmenting path: any $s \rightarrow t$ path in G_f

Cut $(\{s,v,v\}, \{v,v,t\})$:
 $c(S,T) = c(v,v) + c(v,v)$
 $f(S,T) = f(v,v) + f(v,v) - f(v,v) = |f|$

Bipartite matching as flow:
 $s \rightarrow L(1) \dots L(1) \rightarrow R(1) \dots R(1) \rightarrow t$
 Matching edges: paths with flow = 1

End-of-Chapter Material

- **Key terms:** flow network, capacity, flow conservation, residual network, augmenting path, cut, min cut, max-flow min-cut, Ford-Fulkerson, Edmonds-Karp, bipartite matching, integrality theorem
- **Exercises:** 24.1-1 (splitting edge equivalence), 24.2-3 (Edmonds-Karp on Figure 24.1), 24.3-1 (Ford-Fulkerson on bipartite matching), 24.3-2 (prove integrality theorem)

Cross-Chapter Links

- **Requires:** Ch.20 (BFS for Edmonds-Karp), Ch.22 (shortest-path

concepts)

- **Referenced in:** Ch.25 (Hopcroft-Karp for bipartite matching), Ch.26 (multicommodity flow)

Chapter-by-Chapter Breakdown

Ch. 25 – Matchings in Bipartite Graphs

Named Entities (Terms & Definitions)

- **Matching:** subset $M \subseteq E$ s.t. each vertex has ≤ 1 incident edge in M
- **Maximum matching:** matching of maximum cardinality
- **M-augmenting path:** alternating path starting & ending with edges in $E-M$, containing one more edge in $E-M$ than M
- **M-alternating path:** simple path whose edges alternate between M and $E-M$
- **Stable matching:** matching with no blocking pair
- **Blocking pair:** unmatched pair who each prefer the other over current partner
- **Assignment problem:** find perfect matching maximizing total weight
- **Feasible vertex labeling:** $l.h + r.h \geq w(l,r)$ for all $l \in L, r \in R$
- **Equality subgraph:** $G_h = (V, E_h)$ where $E_h = \{(l,r) : l.h + r.h = w(l,r)\}$

Processes / Algorithms / Pathways

Hopcroft-Karp (Maximum Bipartite Matching)

- **Type:** Algorithm
- **Goal:** Find maximum matching in $O(\sqrt{V} \cdot E)$ time
- **Steps:** (1) Start with $M = \emptyset$. (2) Find maximal set of vertex-disjoint shortest M -augmenting paths. (3) $M = M \oplus \{\text{paths}\}$. (4) Repeat until no augmenting paths exist.
- **Complexity:** $O(\sqrt{V} \cdot E)$ time
- **Key Phases per iteration:** (a) Direct edges to form G_M ($L \rightarrow R$ for $E-M$, $R \rightarrow L$ for M). (b) BFS to build DAG H of shortest paths. (c) DFS on transpose H^T to find maximal vertex-disjoint paths.
- **Example:** Graph $L=\{l1..l7\}$, $R=\{r1..r8\}$, matching M of size 4. Found two augmenting paths: $(r1, l3, l3, r3, r3, l1)$ and $(r4, l5, l5, r7, r7, l6)$.

Gale-Shapley (Stable Marriage)

- **Type:** Algorithm
- **Goal:** Find a stable matching in complete bipartite graph with rankings
- **Steps:** (1) All free. (2) While free woman w : w proposes to next man m on her list. (3) If m free $\rightarrow$ engage. (4) If m prefers w to current fiancée $\rightarrow$ break engagement, engage with w . (5) Else $\rightarrow$ reject. (6) Return engaged pairs.
- **Complexity:** $O(n^2)$ time
- **Theorem:** Women-optimal, men-pessimal. Always terminates with stable matching.
- **Example:** 4 women (Wanda, Emma, Lacey, Karen) and 4 men (Oscar, Davis, Brent, Hank) $\rightarrow$ stable matching: (Lacey, Brent), (Wanda, Hank), (Karen, Davis), (Emma, Oscar)

Hungarian Algorithm (Assignment Problem)

- **Type:** Algorithm
- **Goal:** Maximum-weight perfect matching in complete bipartite graph
- **Complexity:** $O(n)$ (can be improved to $O(n^3)$)
- **Steps:** (1) Initialize feasible vertex labeling h . (2) Build equality subgraph G_h . (3) Find max matching in G_h . (4) If not perfect, update labels and repeat. (5) Perfect matching in G_h is optimal.

Classifications & Hierarchies

- Matching types: maximal maximum perfect
- Stable matching variants: woman-oriented vs man-oriented Gale-Shapley

Comparisons & Trade-offs

Dimension	Hopcroft-Karp	Hungarian
Problem	Unweighted max matching	Max-weight perfect matching
Graph	Bipartite (general)	Complete bipartite
Time	$O(\sqrt{V} \cdot E)$	$O(n)$
Technique	Augmenting paths	Vertex labeling

Formulas & Equations

Symmetric difference

$$M' = M \oplus P = (M - P) \cup (P - M) \quad |M'| = |M| + 1 \text{ if } P \text{ is } M\text{-augmenting}$$

Maximum matching bound

$$|M^*| \leq |M| + |V|/(q+1) \quad q = \text{length of shortest } M\text{-augmenting path}$$

Rules, Laws & Theorems

Hall's Theorem

- **Statement:** Bipartite graph $G=(L,R,E)$ has perfect matching iff $|A| \leq |N(A)|$ for all $A \subseteq L$
- **Conditions:** $|L| = |R|$

Lemma 25.1

- $M \oplus P$ where P is M -augmenting gives $|M'| = |M|+1$

Corollary 25.4

- M is maximum iff no M -augmenting path exists

Theorem 25.14

- Perfect matching in equality subgraph = optimal solution to assignment problem

Edge Cases & Common Pitfalls

- Stable-roommates problem (non-bipartite) may have no stable matching
- Maximal $\neq$ maximum (e.g., maximal set of augmenting paths may not be maximum)
- Hungarian: zero-weight edges may need careful tie-breaking

Diagrams & Visuals

Hopcroft-Karp BFS layers:
Layer 0: L unmatched vertices
Layer 1: R neighbors via E-M
Layer 2: L neighbors via M
... alternating until R unmatched found

End-of-Chapter Material

- **Key terms:** matching, M-augmenting path, stable matching, blocking pair, equality subgraph, feasible labeling
- **Exercises:** 25.1-1 (Hopcroft-Karp on Fig 25.1), 25.2-1 ($O(n^2)$ Gale-Shapley impl), 25.3-2 (greedy matching $\geq \frac{1}{2}$ max)

Cross-Chapter Links

- Ch 24 (max flow $\rightarrow$ max matching via flow network)
- Ch 34 (NP-completeness of general matching variants)
- Ch 27 (online algorithms, competitive analysis)

Ch. 26 – Parallel Algorithms

Named Entities (Terms & Definitions)

- **Work (T):** total time to execute on 1 processor (sum of strand times)
- **Span (T_∞):** fastest time on unlimited processors (critical path length)
- **Parallelism:** T/T_∞ – average work per step along critical path
- **Speedup:** T/TP – factor faster on P processors
- **Linear speedup:** $T/TP = O(P)$; Perfect: $T/TP = P$
- **Parallel slackness:** $(T/T_\infty)/P$
- **Determinacy race:** two logically parallel instructions access same location, ≥ 1 modifies it
- **Strand:** chain of instructions with no parallel/procedural control
- **Fork-join parallelism:** model using spawn, sync, parallel for

Processes / Algorithms / Pathways

Work/Span Analysis

- **Goal:** Bound running time on P processors
- **Steps:** (1) Compute work T (serial projection). (2) Compute span T_∞ (longest path in trace DAG). (3) $TP \geq \max(T/P, T_\infty)$. (4) Greedy scheduler achieves $TP \leq T/P + T_\infty$.
- **Theorem 26.1:** Greedy scheduling: $TP \leq T/P + T_\infty$

Parallel Matrix-Vector Multiply (P-MAT-VEC)

- parallel for $i=1..n$ outer loop, inner serial loop sums row
- Work: $\Theta(n^2)$, Span: $\Theta(n)$, Parallelism: $\Theta(n)$

Comparisons & Trade-offs

Metric	Formula	Meaning
Work law	$TP \geq T/P$	At most P work per step
Span law	$TP \geq T_\infty$	Cannot beat critical path

Speedup $S = T/T_P$ in ideal case
Parallelism T/T_∞ Upper bound on S

Rules, Laws & Theorems

Corollary 26.2

- Greedy scheduler is within factor 2 of optimal

Corollary 26.3

- If $P \gg T/T_\infty$ (slackness $\gg 1$), then $T_P \approx T/P$ (near-perfect speedup)

Edge Cases & Common Pitfalls

- Determinacy races cause nondeterministic behavior (e.g., Therac-25, 2003 blackout)
- Span analysis: series composition adds, parallel composition takes $\max$
- High parallelism not always needed beyond $10\times$ slackness

Diagrams & Visuals

Trace DAG for P-FIB(4):
17 strands total, 8 on critical path
Work = 17, Span = 8, Parallelism = 2.125

End-of-Chapter Material

- **Key terms:** work, span, parallelism, speedup, slackness, determinacy race, strand, fork-join
- **Exercises:** 26.1-7 (more parallelism in mat-vec)
- **Chess lesson:** Optimization reduced work 2048→1024 but increased span 1→8; on 512 processors, time went from 5s to 10s

Cross-Chapter Links

- Ch 2-4 (divide-and-conquer recurrences)
- Ch 27 (online algorithms for scheduling)

Ch. 27 – Online Algorithms

Named Entities (Terms & Definitions)

- **Online algorithm:** input arrives progressively; decisions made without future knowledge
- **Offline algorithm:** knows entire input in advance
- **Competitive ratio:** $\max$ over all inputs I of $A(I)/F(I)$ where F is optimal offline
- **c-competitive:** competitive ratio = c
- **Inversion count:** $I(L, L') =$ number of pairs whose order differs between lists
- **Cache miss/hit:** block not in cache / in cache
- **Adversary:** oblivious (doesn't know random choices) vs nonoblivious (knows them)

Processes / Algorithms / Pathways

Move-to-Front (List Maintenance)

- **Goal:** Maintain search list to minimize total access + swap cost
- **Steps:** After searching for element x , swap x forward until it reaches front
- **Cost:** $2r - 1$ for element at position r
- **Competitive ratio:** 4 (Theorem 27.1)
- **Proof technique:** Potential function $\Phi = 2 \cdot I(\text{LM}, \text{LF})$

RANDOMIZED-MARKING (Caching)

- **Steps:** (1) On hit $\rightarrow$ mark block. (2) On miss: if all marked, unmark all. (3) Evict random unmarked block. (4) Load & mark new block.
- **Competitive ratio:** $O(\lg k)$ against oblivious adversary

LIFO/FIFO/LRU Caching Policies

- **LIFO:** evict newest block – competitive ratio $\Theta(n/k)$
- **LRU:** evict least recently used – competitive ratio $\Theta(k)$
- **Theorem 27.3:** LRU is $O(k)$ -competitive
- **Theorem 27.4:** Any deterministic online caching algorithm is $\Omega(k)$ -competitive

Comparisons & Trade-offs

Policy	Competitive Ratio	Strategy
LIFO	$\Theta(n/k)$	Evict newest
LRU	$\Theta(k)$	Evict least recently used
FIFO	$\Theta(k)$	Evict longest in cache
Randomized-Marking	$O(\lg k)$	Evict random unmarked

Formulas & Equations

Competitive Ratio (minimization)

$$\text{CR} = \max_{\{I, U\}} A(I)/F(I)$$

Elevator Hedging Strategy

- Wait k minutes, then take stairs $\rightarrow$ competitive ratio = 2 (independent of k, B)

Edge Cases & Common Pitfalls

- LIFO can have unbounded competitive ratio (depends on n)
- LRU and FIFO have $\Theta(k)$ but k is constant w.r.t. input size
- Randomized algorithms need oblivious adversary for theoretical guarantees

End-of-Chapter Material

- **Key terms:** competitive ratio, online/offline, inversion, adversary, cache eviction
- **Exercises:** 27.1-2 (ski rental 2-competitive alg), 27.2-4 (Move-to-Front 2-competitive with free moves), 27.3-1 (LRU epoch analysis)

Cross-Chapter Links

- Ch 35 (approximation algorithms, similar ratio analysis)
- Ch 15 (caching, offline furthest-in-future)
- Ch 16 (amortized analysis, potential functions)

Ch. 28 – Matrix Operations

Named Entities (Terms & Definitions)

- **LUP decomposition:** $PA = LU$ where L = unit lower-triangular, U = upper-triangular, P = permutation
- **LU decomposition:** $A = LU$ (no pivoting)
- **Schur complement:** $A' = A - vw^T/a$
- **Forward substitution:** solve $Ly = Pb$ for y
- **Back substitution:** solve $Ux = y$ for x
- **Symmetric positive-definite:** $A = A^T$ and $x^T A x > 0$ for all $x \neq 0$
- **Pivot:** diagonal entry used for elimination; pivoting = permuting rows for stability

Processes / Algorithms / Pathways

LUP-Solve

- **Goal:** Solve $Ax = b$ given LUP decomposition
- **Steps:** (1) Forward substitution: $Ly = Pb$. (2) Back substitution: $Ux = y$.
- **Complexity:** $\Theta(n^2)$

LU-Decomposition

- **Steps:** For $k=1..n$: (1) $u_{kk} = a_{kk}$. (2) $l_{ik} = a_{ik}/a_{kk}$ for $i>k$. (3) $u_{ki} = a_{ki}$ for $i>k$. (4) Update Schur complement: $a_{ij} = a_{ij} - l_{ik} \cdot u_{kj}$.
- **Complexity:** $\Theta(n^3)$

LUP-Decomposition (with pivoting)

- **Steps:** For $k=1..n$: (1) Find pivot row k' with $\max |a_{ik}|$ in column k . (2) Swap rows k and k' . (3) Update permutation π . (4) Compute multipliers and Schur complement.
- **Complexity:** $\Theta(n^3)$

Formulas & Equations

Forward substitution

$$y_i = b_{\pi[i]} - \sum_{j=1}^{i-1} l_{ij} \cdot y_j \quad \text{- Solves } Ly = Pb$$

Back substitution

$$x_i = (y_i - \sum_{j=i+1}^n u_{ij} \cdot x_j) / u_{ii} \quad \text{- Solves } Ux = y$$

Matrix inversion from LUP

- Solve $AX_i = e_i$ for each column $X_i \rightarrow O(n^3)$ total

Multiplication Inversion (Theorem 28.1-28.2)

- $M(n) = O(I(n))$ and $I(n) = O(M(n))$ equivalence

Rules, Laws & Theorems

Lemma 28.3

- Positive-definite nonsingular

Theorem 28.1

- Matrix inversion is no harder than multiplication (construct $3n \times 3n$ block matrix)

Theorem 28.2

- Matrix multiplication is no harder than inversion (for symmetric positive-definite via Schur complement recursion)

Edge Cases & Common Pitfalls

- LU fails when pivot = 0 $\rightarrow$ need pivoting
- Numerical stability: pivoting on max absolute value mitigates roundoff
- Symmetric positive-definite matrices require no pivoting

End-of-Chapter Material

- **Key terms:** LUP decomposition, forward/back substitution, Schur complement, pivoting, symmetric positive-definite
- **Exercises:** 28.1-2 (LU decomposition), 28.1-3 (LUP solve), 28.2-2 (LUP from mat-mul)

Cross-Chapter Links

- Ch 4 (Strassen's matrix multiplication, $O(n^{\lg 7})$)
- Ch 29 (linear programming)
- Appendix D (matrix basics)

Ch. 29 – Linear Programming

Named Entities (Terms & Definitions)

- **Linear program:** optimize linear function subject to linear constraints
- **Standard form:** $\max c^T x$ s.t. $Ax \leq b$, $x \geq 0$
- **Objective function:** $c^T x$ (linear function to optimize)
- **Feasible solution:** satisfies all constraints
- **Optimal solution:** feasible with max/min objective value
- **Feasible region:** set of all feasible solutions (convex polyhedron = simplex)
- **Dual:** minimization LP derived from primal max LP; optimal values equal
- **Simplex algorithm:** moves along edges of feasible region to reach optimum vertex
- **Integer linear program:** LP with x constraint – NP-hard

Processes / Algorithms / Pathways

Formulating an LP

- **Steps:** (1) Identify decision variables. (2) Specify linear constraints. (3) Define linear objective function. (4) Add nonnegativity constraints.
- **Example:** Political problem – minimize cost to win votes given effectiveness matrix

Simplex Algorithm (conceptual)

- **Steps:** (1) Start at feasible vertex. (2) Move along edge to neighbor with better objective. (3) Repeat until local (hence global) optimum reached.
- **Complexity:** Exponential worst-case, polynomial average-case

Ellipsoid & Interior-Point Methods

- Polynomial-time: ellipsoid (first polynomial LP algorithm), interior-point (practical)

Comparisons & Trade-offs

Algorithm	Type	Time	Notes
Simplex	Vertex-following	Exp. worst-case	Most used in practice
Ellipsoid	Polynomial	$O(nL^2)$	Theoretical, slow in practice
Interior-point	Polynomial	$O(n^3L)$	Moves through interior

Formulas & Equations

Standard Form LP

$$\max c^T x \text{ s.t. } Ax \leq b, x \geq 0 \quad - \quad x \in \mathbb{R}^n, A \in \mathbb{R}^{(m \times n)}, b \in \mathbb{R}^m, c \in \mathbb{R}^n$$

Dual LP (for max primal)

$$\min b^T y \text{ s.t. } A^T y \geq c, y \geq 0$$

Shortest Path as LP

$$\max d_t \text{ s.t. } d_v \leq d_u + w(u,v) \text{ for all } (u,v) \in E, d_s = 0$$

Rules, Laws & Theorems

Duality Theorem

- Optimal primal value = optimal dual value (if both feasible)
- **Weak duality:** $c^T x \leq b^T y$ for any feasible primal x , dual y
- **Strong duality:** equality at optimality

Edge Cases & Common Pitfalls

- Infeasible LP: no solution satisfying all constraints
- Unbounded LP: feasible but objective can increase without bound
- Integer LP is NP-hard; LP relaxation (drop integrality) gives bound

End-of-Chapter Material

- **Key terms:** objective function, constraints, feasible region, simplex, duality, primal/dual
- **Exercises:** 29.1-1 (feasible solutions), 29.2-1 (shortest path as LP)

Cross-Chapter Links

- Ch 22 (shortest paths as LP)
- Ch 24 (max flow as LP)
- Ch 34 (integer programming NP-complete)
- Ch 35 (LP for approximation algorithms)

Ch. 30 – Polynomials and the FFT

Named Entities (Terms & Definitions)

- **Polynomial:** $A(x) = \sum a_j x^j$, degree bound n
- **Coefficient representation:** vector $a = (a_0, a_1, \dots, a_{n-1})$
- **Point-value representation:** $\{(x, y), \dots, (x_{n-1}, y_{n-1})\}$ where $y_k = A(x_k)$
- **DFT (Discrete Fourier Transform):** $y_k = A(\omega_n^k)$ for $k=0..n-1$, ω_n principal n th root of unity
- **FFT (Fast Fourier Transform):** computes DFT in $\Theta(n \lg n)$ time
- **Convolution:** $c = a \cdot b$ where $c_j = \sum a_k \cdot b_{j-k}$ – polynomial multiplication
- **Interpolation:** convert point-value $\rightarrow$ coefficients (inverse DFT)
- **Vandermonde matrix:** $V_{jk} = \omega_n^{jk}$

Processes / Algorithms / Pathways

FFT (Cooley-Tukey)

- **Goal:** Compute polynomial evaluation at n th roots of unity in $\Theta(n \lg n)$
- **Steps:** (1) If $n=1$, return a . (2) Split into $A_{\text{even}}(x)$ (even indices) and $A_{\text{odd}}(x)$ (odd indices). (3) Recursively compute FFT on each (size $n/2$). (4) Combine: $y_k = y_{\text{even}_k} + \omega_n^k \cdot y_{\text{odd}_k}$; $y_{k+n/2} = y_{\text{even}_k} - \omega_n^k \cdot y_{\text{odd}_k}$.
- **Complexity:** $T(n) = 2T(n/2) + \Theta(n) = \Theta(n \lg n)$
- **Example:** Multiply $(6x^3+7x^2-10x+9)(-2x^3+4x-5)$ via FFT: (1) pad to degree 8, (2) FFT both, (3) pointwise multiply, (4) inverse FFT

Efficient Polynomial Multiplication

- **Steps:** (1) Double degree-bound (add n zeros). (2) Evaluate at $2n$ th roots of unity (FFT). (3) Pointwise multiply. (4) Interpolate (inverse FFT).
- **Complexity:** $\Theta(n \lg n)$

Formulas & Equations

Complex n th Roots of Unity

$\omega_n = e^{2\pi i/n}$ - ω_n^k for $k=0..n-1$ are all n th roots - **Cancellation lemma:**
 $\omega_n^{dn} = \omega_n^{dk}$ - **Halving lemma:** $(\omega_n^k)^2 = \omega_{n/2}^k$ for even n -
Summation lemma: $\sum_{j=0}^{n-1} (\omega_n^k)^j = 0$ if $n \nmid k$

DFT Matrix

$y = V_n^{-1} a$ where $V_n[jk] = \omega_n^{jk}$ - **Inverse:** $V_n^{-1}[jk] = \omega_n^{-jk}/n$

Lagrange Interpolation

$$A(x) = \sum y_k \cdot \prod_{j \neq k} (x - x_j) / (x_k - x_j)$$

Rules, Laws & Theorems

Theorem 30.1 (Uniqueness)

- n distinct point-value pairs determine unique degree-bound n polynomial

Theorem 30.2

- Polynomial multiplication in coefficient form in $\Theta(n \lg n)$ time

Convolution Theorem

- $a \cdot b = \text{IDFT}(\text{DFT}(a) \cdot \text{DFT}(b))$

Edge Cases & Common Pitfalls

- FFT requires n to be a power of 2 (pad with zeros otherwise)
- Inverse DFT: replace ω_n by ω_n^{-1} and divide by n
- For non-power-of-2, use Bluestein's algorithm or zero-padding

Diagrams & Visuals

FFT butterfly:

$$y_{\text{even}_k} \text{ — } y_k = y_{\text{even}_k} + \omega^k \cdot y_{\text{odd}_k}$$

$$\omega^k$$

$$y_{\text{odd}_k} \text{ — } y_{\{k+n/2\}} = y_{\text{even}_k} - \omega^k \cdot y_{\text{odd}_k}$$

End-of-Chapter Material

- **Key terms:** DFT, FFT, convolution, roots of unity, butterfly operation, twiddle factors
- **Exercises:** 30.1-1 (polynomial multiplication), 30.2-2 (DFT of $(0, 1, 2, 3)$)

Cross-Chapter Links

- Ch 28 (Vandermonde matrix)
- Ch 31 (modular arithmetic, number-theoretic transform analogue)

Ch. 31 – Number-Theoretic Algorithms

Named Entities (Terms & Definitions)

- **GCD:** greatest common divisor; $\text{gcd}(a, b) =$ smallest positive linear combination $ax + by$
- **Relatively prime:** $\text{gcd}(a, b) = 1$
- **Modular equivalence:** $a \equiv b \pmod{n}$ iff $n \mid (a - b)$
- **Group $(S, \cdot)$:** closed, associative, identity, inverses; abelian if commutative
- $+$: integers modulo n under addition; $*$: multiplicative group (units)

- mod n)
- **Euler's ϕ function:** $\phi(n) = |\ast| = n \cdot \prod_{p|n} (1 - 1/p)$
- **Primitive root (generator):** g such that $g = \ast$
- **Discrete logarithm:** $\text{ind}_{\{n,g\}}(a) = z$ where $g^z \equiv a \pmod{n}$

Processes / Algorithms / Pathways

Euclid's Algorithm (GCD)

- **Goal:** Compute $\text{gcd}(a,b)$ for $a \geq b \geq 0$
- **Steps:** (1) If $b=0$, return a . (2) Else return $\text{Euclid}(b, a \bmod b)$.
- **Complexity:** $O(\lg b)$ recursive calls; $O(\beta^2)$ bit ops for β -bit numbers
- **Example:** $\text{Euclid}(99,78) \rightarrow \text{Euclid}(78,21) \rightarrow \text{Euclid}(21,15) \rightarrow \text{Euclid}(15,6) \rightarrow \text{Euclid}(6,3) \rightarrow \text{Euclid}(3,0) \rightarrow 3$

Extended-Euclid

- **Goal:** Find x,y such that $d = \text{gcd}(a,b) = ax + by$
- **Steps:** (1) If $b=0$, return $(a,1,0)$. (2) Recursively get (d',x',y') for $(b, a \bmod b)$. (3) Return $(d', y', x' - a/b \cdot y')$.
- **Example:** $\text{Extended-Euclid}(99,78) \rightarrow (3, -11, 14)$ since $3 = 99 \cdot (-11) + 78 \cdot 14$

Modular Linear Equation Solver

- **Goal:** Solve $ax \equiv b \pmod{n}$
- **Steps:** (1) Compute $d = \text{gcd}(a,n)$, x' from Extended-Euclid. (2) If $db \rightarrow$ no solution. (3) $x = x' \cdot (b/d) \bmod n$. (4) Solutions: $x_i = x + i \cdot (n/d)$ for $i=0..d-1$.

Repeated Squaring (Modular Exponentiation)

- **Goal:** Compute $a^b \bmod n$ efficiently
- **Steps:** (1) If $b=0$ return 1. (2) If b even: $(a^{\{b/2\}})^2 \bmod n$. (3) If b odd: $a \cdot (a^{\{b-1\}}) \bmod n$.
- **Complexity:** $O(\lg b)$ multiplications

Formulas & Equations

GCD Recursion Theorem

$$\text{gcd}(a,b) = \text{gcd}(b, a \bmod b)$$

Euler's Theorem

$$a^{\{\phi(n)\}} \equiv 1 \pmod{n} \text{ for } \text{gcd}(a,n)=1$$

Fermat's Theorem

$$a^{\{p-1\}} \equiv 1 \pmod{p} \text{ for prime } p, a \text{ not divisible by } p$$

Chinese Remainder Theorem

- $n = n_1 \dots n_k$, pairwise coprime
- $a \mapsto (a \bmod n_1, \dots, a \bmod n_k)$ is bijection
- $a = \sum a_i \cdot c_i \pmod{n}$, where $c_i = m_i \cdot (m_i^{-1} \bmod n_i)$, $m_i = n/n_i$

Rules, Laws & Theorems

Lamé's Theorem

- $\text{Euclid}(a,b)$ makes $< k$ recursive calls if $b < F_{k+1}$ (Fibonacci)
- $O(\lg b)$ calls

Lagrange's Theorem

- $|\text{subgroup}|$ divides $|\text{group}|$

Unique Prime Factorization (Theorem 31.8)

- Every integer has unique prime factorization

Edge Cases & Common Pitfalls

- Modular inverse exists iff $\gcd(a,n)=1$
- If n has nontrivial square roots of 1 $\rightarrow n$ is composite (Corollary 31.35)
- Working modulo prime vs composite: $*$ is cyclic, $*$ may not be

Diagrams & Visuals

Extended-Euclid recursion (99,78):

a	b	a/b	d	x	y
99	78	1	3	-11	14
78	21	3	3	3	-11
21	15	1	3	-2	3
15	6	2	3	1	-2
6	3	2	3	0	1
3	0	-	3	1	0

End-of-Chapter Material

- **Key terms:** gcd, modular arithmetic, extended Euclid, CRT, $\phi(n)$, discrete log, repeated squaring
- **Exercises:** 31.1-2 (infinitely many primes), 31.2-2 (Extended-Euclid(899,493)), 31.4-1 ($35x \equiv 10 \pmod{50}$)

Cross-Chapter Links

- Ch 30 (roots of unity in complex vs. finite fields)
- Ch 33 (machine learning, randomness)
- RSA cryptosystem (Sec 31.7)

Ch. 32 – String Matching

Named Entities (Terms & Definitions)

- **String matching:** find all occurrences of pattern $P[1:m]$ in text $T[1:n]$
- **Valid shift:** s where $T[s+1:s+m] = P[1:m]$
- **Prefix:** $w \preceq x$ if $x = wy$; **Suffix:** $w \preceq x$ if $x = yw$
- **Suffix function:** $\sigma(x) =$ length of longest prefix of P that is suffix of x
- **Prefix function π :** $\pi[q] = \max\{k < q : P[1:k] = P[q-k+1:q]\}$
- **Finite automaton:** 5-tuple $(Q, q, A, \Sigma, \delta)$ for string matching

Processes / Algorithms / Pathways

Naive String Matching

- **Steps:** For each shift $s=0..n-m$, check $P[1:m] == T[s+1:s+m]$
- **Complexity:** $O((n-m+1) \cdot m)$ worst-case
- **Example:** $P=aaab, T=aaaaaaab \rightarrow O((n-m+1)m)$

Rabin-Karp (Rolling Hash)

- **Goal:** Average-case $O(n+m)$ string matching
- **Steps:** (1) Compute $p = P \bmod q$. (2) Compute $t = T[1:m] \bmod q$. (3) For each shift: if $t_s = p$ (hit), verify equality. (4) Compute $t_{s+1} = (d \cdot (t_s - T[s+1] \cdot h) + T[s+m+1]) \bmod q$.
- **Complexity:** $O(m)$ preprocessing, $O((n-m+1) \cdot m)$ worst-case, $O(n+m)$ expected
- **Example:** $P=31415, q=13, T=...$ (Fig 32.4). Window values mod 13; spurious hit at position 13

Finite Automaton Matcher

- **Steps:** (1) Precompute $\delta(q,a) = \sigma(P[:q]a)$. (2) For each $T[i]$: $q = \delta(q, T[i])$; if $q=m \rightarrow$ match.
- **Complexity:** $O(m|\Sigma|)$ preprocessing, $O(n)$ matching

Knuth-Morris-Pratt (KMP)

- **Goal:** $O(n)$ matching with $O(m)$ preprocessing (no Σ factor)
- **Steps:** (1) Compute prefix function $\pi[1:m]$. (2) Scan T : while $q>0$ and $P[q+1] \neq T[i]$, $q=\pi[q]$; if match, $q++$. If $q=m$, found.
- **Complexity:** $O(m) + O(n)$
- **Key insight:** $\pi[q]$ gives longest proper prefix of $P[:q]$ that is also a suffix

Comparisons & Trade-offs

Algorithm	Preprocessing	Matching	Space
Naive	O	$O((n-m+1)m)$	$O(1)$
Rabin-Karp	$O(m)$	$O((n-m+1)m)$ / expected $O(n+m)$	$O(1)$
Finite Automaton	$O(m)$	Σ	$)$
KMP	$O(m)$	$O(n)$	$O(m)$
Suffix Array	$O(n \lg n)$	$O(m \lg n + k \cdot m)$	$O(n)$

Formulas & Equations

Rabin-Karp Rolling Hash

$t_{s+1} = (d \cdot (t_s - T[s+1] \cdot h) + T[s+m+1]) \bmod q$ - $h = d^{m-1} \bmod q$ - $d = \text{radix}$ (e.g., 10 for decimal, 256 for ASCII)

Prefix Function

$\pi[q] = \max\{k < q : P[:k] = P[q-k+1:q]\}$

Rules, Laws & Theorems

Lemma 32.5 (Prefix iteration)

- $\pi^*[q] = \{k < q : P[:k] = P[q-k+1:q]\}$; iterating π generates all proper prefix-suffix matches

Edge Cases & Common Pitfalls

- Rabin-Karp: spurious hits when hash matches but strings differ (choose large prime q to minimize)
- Naive is $O(n^2)$ worst-case (e.g., $P = a^m$, $T = a^n$)
- KMP: after finding match at $q=m$, set $q=\pi[m]$ to continue scanning

Diagrams & Visuals

KMP prefix function π for $P=ababaca$:

```
q:  1 2 3 4 5 6 7
π:  0 0 1 2 3 0 1
```

Finite automaton states:

$0 \rightarrow a \rightarrow 1 \rightarrow b \rightarrow 2 \rightarrow a \rightarrow 3 \rightarrow b \rightarrow 4 \rightarrow a \rightarrow 5 \rightarrow c \rightarrow 6 \rightarrow a \rightarrow 7$ (accept)

Back edges from each state on mismatch use prefix function

End-of-Chapter Material

- **Key terms:** valid shift, prefix/suffix, automaton, prefix function, rolling hash
- **Exercises:** 32.1-1 (naive comparisons), 32.2-1 (Rabin-Karp spurious hits), 32.4-1 (compute π)

Cross-Chapter Links

- Ch 31 (modular arithmetic for Rabin-Karp)
- Ch 12 (Trie, suffix tree)

Ch. 33 – Machine-Learning Algorithms

Named Entities (Terms & Definitions)

- **Supervised learning:** training data has labels $\rightarrow$ learn hypothesis for prediction
- **Unsupervised learning:** no labels (e.g., clustering)
- **k-means clustering:** partition n points into k clusters minimizing sum of squared distances to centroids
- **Centroid (mean):** d -dimensional point where each coordinate = mean of cluster values
- **Lloyd's procedure:** iterates assign-to-nearest-center + recompute-centroids
- **Dissimilarity:** squared Euclidean distance $\Delta(x,y) = \sum (x_a - y_a)^2$
- **Multiplicative weights:** maintain weights per expert, decrease on mistakes
- **Weighted majority:** predict based on weighted vote of experts
- **Gradient descent:** iterative optimization following negative gradient
- **Convex function:** $f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y)$

Processes / Algorithms / Pathways

Lloyd's Procedure (k-means)

- **Goal:** Find locally optimal k -clustering
- **Steps:** (1) Choose k initial centers randomly. (2) Assign each point to nearest center. (3) If no change $\rightarrow$ stop. (4) Recompute centers as centroids. (5) Go to step 2.
- **Complexity:** $O(T \cdot d \cdot k \cdot n)$ where T = iterations
- **Example:** 49 US capitals $\rightarrow k=4$ clusters after 11 iterations

Weighted-Majority (Multiplicative Weights)

- **Goal:** Limit mistakes vs best expert
- **Steps:** (1) Initialize all weights = 1. (2) For each event: (a) Sum weights predicting 1 vs 0. (b) Predict majority. (c) Multiply incorrect experts' weights by $(1-\gamma)$.
- **Theorem 33.4:** $m \leq 2(1+\gamma) \cdot m^* + (2 \ln n)/\gamma$
- **Corollary 33.5:** $m \leq 2 \cdot m^* + O(\log n)$ when $\gamma = 1/2$

Gradient Descent

- **Goal:** Minimize convex function $f: \mathbb{R}^n \rightarrow \mathbb{R}$
- **Steps:** (1) Start at $x(0)$. (2) For $t=0..T-1$: $x(t+1) = x(t) - \gamma \cdot \nabla f(x(t))$. (3) Return avg of points.
- **Theorem 33.8:** $f(x_{\text{avg}}) - f(x^*) \leq O(LR/T)$ where L bounds gradient, R bounds distance from start

Comparisons & Trade-offs

Algorithm	Type	Guarantee	Complexity
Lloyd's k-means	Clustering	Local minimum	$O(T \cdot d \cdot k \cdot n)$
Weighted-Majority	Online learning	$2 \cdot m^* + O(\log n)$ mistakes	$O(n \cdot T)$
Gradient Descent	Optimization	$f(x) - f(x^*) \leq \epsilon$	$O(L^2 R^2 / \epsilon^2)$

Formulas & Equations

k-means Objective

$f(S, C) = \sum_{\ell=1}^k \sum_{x \in S(\ell)} \Delta(x, c(\ell))$ = sum of squared distances to cluster centroids

Weighted-Majority Mistake Bound

$m \leq 2(1+\gamma) \cdot m^* + (2 \ln n)/\gamma$ - m = algorithm mistakes, m^* = best expert mistakes, $\gamma \in (0, \frac{1}{2}]$

Edge Cases & Common Pitfalls

- k-means: NP-hard to find global optimum (local minimum only)
- Gradient descent may converge to local minimum (not global) for non-convex functions
- Step size too large $\rightarrow$ overshoot; too small $\rightarrow$ slow convergence
- Weighted majority: weights may approach 0 for perpetually wrong experts

End-of-Chapter Material

- **Key terms:** clustering, centroid, Lloyd's procedure, multiplicative weights, gradient descent, convex function
- **Exercises:** 33.1-1 (alternative k-means objective), 33.2-2 ($\ln(1-x)$ bounds), 33.3-1 (convex function property)

Cross-Chapter Links

- Ch 27 (online algorithms, competitive analysis)
 - Ch 29 (linear programming)
 - Ch 34 (NP-hardness of k-means)
-

Ch. 34 – NP-Completeness

Named Entities (Terms & Definitions)

- **P**: problems solvable in polynomial time
- **NP**: problems verifiable in polynomial time (certificate)
- **NPC (NP-complete)**: hardest problems in NP; all NP problems reduce to them
- **NP-hard**: at least as hard as all NP problems (may not be in NP)
- **Reduction**: polynomial-time transformation from problem A to problem B ($A \leq_P B$)
- **Decision problem**: yes/no answer (vs optimization problem)
- **Circuit-SAT**: given boolean circuit, is there satisfying assignment?
- **SAT**: boolean formula satisfiability
- **3-CNF-SAT**: SAT where formula is AND of ORs of 3 literals each
- **CLIQUE**: does graph contain complete subgraph of size k ?
- **VERTEX-COVER**: does graph have vertex cover of size $\leq k$?
- **HAM-CYCLE**: does graph have Hamiltonian cycle?
- **TSP**: traveling salesperson tour of cost $\leq k$?
- **SUBSET-SUM**: does subset sum to target t ?

Processes / Algorithms / Pathways

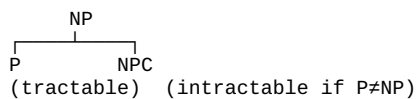
Reduction Methodology

- **Goal**: Prove problem B is NP-complete
- **Steps**: (1) Show $B \in \text{NP}$ (certificate verifiable in polynomial time). (2) Reduce known NP-complete A to B. (3) Show reduction is polynomial time. (4) Show $x \in A \iff f(x) \in B$.

Cook-Levin Theorem (Circuit-SAT is NP-complete)

- Any NP problem can be reduced to CIRCUIT-SAT by encoding the verification algorithm as a circuit
- Key insight: configuration of Turing machine computation $\rightarrow$ boolean circuit

Classifications & Hierarchies



Comparisons & Trade-offs

Problem	Known Time	Status
Shortest path (unweighted)	$O(V+E)$	P
Longest simple path	?	NP-complete
Euler tour	$O(E)$	P
Hamiltonian cycle	?	NP-complete
2-CNF SAT	$O(m+n)$	P
3-CNF SAT	?	NP-complete
Linear programming	polynomial	P
Integer linear programming?		NP-hard

Formulas & Equations

Reduction for NP-Completeness Proof

$L \leq_P L'$: polynomial-time f s.t. $x \in L \iff f(x) \in L'$ - If $L' \in P$ then $L \in P$ - If $L' \in \text{NPC}$ then $L \in \text{NPC}$ (when $L \notin \text{NP}$)

Rules, Laws & Theorems

Theorem 34.4

- If any NP-complete problem is polynomial-time solvable, then $P = \text{NP}$

Lemma 34.8

- If $L' \in \text{NPC}$ and $L' \leq_P L$, then L is NP-hard; if $L \in \text{NP}$, then $L \in \text{NPC}$

Cook-Levin Theorem (34.7)

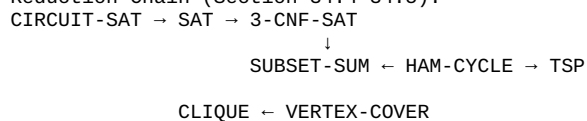
- CIRCUIT-SAT is NP-complete

Edge Cases & Common Pitfalls

- Optimization vs decision: show decision version hard $\rightarrow$ optimization version hard
- Encoding matters (unary vs binary can change complexity class)
- Reduction must be polynomial in instance size

Diagrams & Visuals

Reduction Chain (Section 34.4-34.5):



End-of-Chapter Material

- **Key terms:** P, NP, NPC, reduction, certificate, Cook-Levin, 3-CNF-SAT, HAM-CYCLE, TSP, CLIQUE, VERTEX-COVER, SUBSET-SUM
- **Exercises:** 34.2-1 (graph isomorphism $\notin \text{NP}$), 34.5-1 (CLIQUE reduction)

Cross-Chapter Links

- Ch 35 (approximation algorithms for NP-complete problems)
- Ch 29 (integer programming NP-hard)
- Ch 22 (shortest path $\in P$ vs longest path NP-complete)

Ch. 35 – Approximation Algorithms

Named Entities (Terms & Definitions)

- **Approximation ratio $\rho(n)$:** $C/C^* \leq \rho(n)$ for minimization; $C^*/C \leq \rho(n)$ for maximization
- **$\rho(n)$ -approximation algorithm:** achieves ratio $\rho(n)$
- **Approximation scheme:** $(1+\epsilon)$ -approximation for any $\epsilon > 0$
- **PTAS:** polynomial-time approximation scheme (time polynomial in n for fixed ϵ)
- **FPTAS:** fully polynomial-time (time polynomial in n and $1/\epsilon$)
- **Maximal matching:** matching where no more edges can be added

Processes / Algorithms / Pathways

APPROX-VERTEX-COVER (2-approximation)

- **Goal:** Find vertex cover within factor 2 of optimal
- **Steps:** (1) $C = \emptyset$, $E' = E$. (2) While $E' \neq \emptyset$: pick any $(u,v) \in E'$, add u,v to C , remove all edges incident on u or v . (3) Return C .
- **Complexity:** $O(V+E)$
- **Proof:** Let A = edges picked. $|C| \geq |A|$ (each edge needs distinct vertex). $|C| = 2|A| \leq 2|C|$.

APPROX-TSP-TOUR (2-approx with triangle inequality)

- **Goal:** TSP tour within factor 2 of optimal (when triangle inequality holds)
- **Steps:** (1) Compute MST T . (2) Preorder walk of T gives tour H .
- **Complexity:** $O(V^2)$
- **Proof:** $c(T) \leq c(H)$; full walk W costs $2 \cdot c(T)$; triangle inequality gives $c(H) \leq c(W) \leq 2 \cdot c(T)$.

Greedy-Set-Cover ($O(\lg n)$ -approximation)

- **Goal:** Find subfamily C covering all elements, near-minimum size
- **Steps:** (1) $U = X$, $C = \emptyset$. (2) While $U \neq \emptyset$: pick $S \in C$ maximizing $|S \cap U|$; $U_{i+1} = U_i - S$; $C = C \cup \{S\}$.
- **Complexity:** $O(|X| \cdot \sum_{S \in C} |S|)$
- **Theorem 35.4:** $|C| \leq |C^*| \cdot \ln |X|$

APPROX-MIN-WEIGHT-VC (LP-rounding, 2-approximation)

- **Steps:** (1) Form LP relaxation: $\min \sum w(v) \cdot x(v)$ s.t. $x(u)+x(v) \geq 1$, $0 \leq x(v) \leq 1$. (2) Solve LP. (3) Include v iff $x(v) \geq \frac{1}{2}$.

Randomized MAX-3-CNF-SAT ($8/7$ -approximation)

- **Steps:** Set each variable to 1 with prob $\frac{1}{2}$, 0 with prob $\frac{1}{2}$.
- **Guarantee:** Expected clauses satisfied = $7m/8$, ratio $\leq 8/7$.

APPROX-SUBSET-SUM (FPTAS)

- **Goal:** Subset sum within $(1+\epsilon)$ of optimal, time polynomial in n and $1/\epsilon$
- **Steps:** (1) Maintain sorted list of subset sums $\leq t$. (2) After each merge, trim list $\delta = \epsilon/(2n)$. (3) Return max in L_n .
- **Complexity:** $O(n \cdot \log t / \delta) = \text{poly}(n, 1/\epsilon)$

Comparisons & Trade-offs

Problem	Algorithm	Ratio	Time	Type
Vertex Cover	APPROX-VC	2	$O(V+E)$	Deterministic
Weighted VC	LP-rounding	2	$\text{poly}(n)$	LP-based
TSP (triangle)	MST+preorder	2	$O(V^2)$	Deterministic
TSP (general)	—	Any $p \geq 1$	—	Imposs. unless $P=NP$
Set Cover	Greedy	$O(\ln n)$	$O(\sum S_i)$	X
Subset Sum	APPROX-SS	$1+\epsilon$	$\text{poly}(n, 1/\epsilon)$	FPTAS
MAX-3-CNF-SAT	Random	$8/7$	$O(m)$	Randomized

Formulas & Equations

Approximation Ratio

$$\max(C/C^*, C^*/C) \leq \rho(n)$$

TSP lower bound via MST

$$c(T) \leq c(H^*) \text{ (deleting edge from tour gives spanning tree)}$$

Rules, Laws & Theorems

Theorem 35.3 (TSP inapproximability)

- Unless $P=NP$, no constant-factor approximation for general TSP

Theorem 35.4 (Set cover)

- Greedy Set-Cover is $O(\ln |X|)$ -approximation

Theorem 35.6 (Weighted VC)

- LP-rounding gives 2-approximation

Theorem 35.7 (Subset Sum)

- APPROX-SUBSET-SUM is FPTAS

Edge Cases & Common Pitfalls

- General TSP has NO constant-factor approximation unless $P=NP$
- Vertex cover: 2-approximation, but vertex cover is NP-complete
- Set cover lower bound: cannot be approximated better than $\Omega(\ln n)$ unless $P=NP$
- FPTAS exists only for some NP-complete problems (subset-sum yes, TSP no)

End-of-Chapter Material

- **Key terms:** approximation ratio, PTAS, FPTAS, vertex cover, TSP, set cover, subset sum, LP rounding
- **Exercises:** 35.1-1 (suboptimal vertex cover example), 35.2-3 (closest-point TSP heuristic), 35.3-1 (set cover tie-breaking)

Cross-Chapter Links

- Ch 34 (NP-completeness: all problems here are NP-hard)
- Ch 29 (linear programming for approximation)
- Ch 27 (competitive analysis analogous to approximation ratio)

Appendix A – Summations

Named Entities

- **Arithmetic series:** $\sum k = n(n+1)/2$
- **Geometric series:** $\sum x^k = (x^{n+1}-1)/(x-1)$
- **Harmonic series:** $H_n = \sum 1/k = \ln n + \gamma + O(1/n)$, $\gamma \approx 0.57721$

- **Telescoping series:** $\sum (a_k - a_{k-1}) = a_n - a_0$

Formulas & Equations

Key Summations

$\sum_{k=1}^n k = n(n+1)/2 = \Theta(n^2)$ $\sum_{k=1}^n k^2 = n(n+1)(2n+1)/6$ $\sum_{k=0}^n x^k = (x^{n+1}-1)/(x-1)$ for $x \neq 1$ $\sum_{k=0}^\infty x^k = 1/(1-x)$ for $|x| < 1$ $\sum_{k=0}^\infty k \cdot x^k = x/(1-x)^2$ for $|x| < 1$ $\sum_{k=1}^n 1/k = \ln n + \gamma + O(1/n)$ $\sum_{k=1}^n k^3 = n^2(n+1)^2/4$ $\sum_{k=1}^n k \cdot 2^k = (n-1)2^{n+1} + 2$ $\sum_{k=0}^n k^2 \cdot x^k = x(1+x - (n+1)^2 x^{n+1} + (2n^2+2n-1)x^{n+1} - n^2 x^{n+2})/(1-x)^3$ for $x \neq 1$

Bounding by integrals

For monotonic f : $\int_n^{\infty} f(x)dx \leq \sum_{k=1}^{\infty} f(k) \leq \int_1^{n+1} f(x)dx$

Harmonic Series Bound Proof

- $\sum_{k=1}^n 1/k = \int_1^{n+1} 1/x dx$
- Using integral bounds: $\ln(n+1) \leq H_n \leq 1 + \ln n$
- The Euler-Mascheroni constant $\gamma = \lim_{n \rightarrow \infty} (H_n - \ln n) \approx 0.57721$

Splitting Technique (Geometric Series)

- Split a sum at a midpoint to bound a geometrically growing sequence:
 - $\sum_{k=0}^n k \cdot a^k \leq (a^{n+1}-1)/(a-1) \cdot n$ for $a > 1$
 - More generally: bound early terms directly, later terms by geometric series

Ratio Test for Convergence

- For $\sum a_k$ with $a_k > 0$: if $\lim_{k \rightarrow \infty} a_{k+1}/a_k = r$, then
 - $r < 1$: series converges
 - $r > 1$: series diverges
 - $r = 1$: inconclusive

Comparison Test

- If $0 \leq a_k \leq b_k$ for all k and $\sum b_k$ converges, then $\sum a_k$ converges
- If $0 \leq b_k \leq a_k$ for all k and $\sum b_k$ diverges, then $\sum a_k$ diverges

Telescoping Series Examples

- $\sum_{k=1}^n 1/(k(k+1)) = \sum_{k=1}^n (1/k - 1/(k+1)) = 1 - 1/(n+1)$
- $\sum_{k=1}^n k \cdot (k+1)! = \sum_{k=1}^n ((k+1)! - k!) = (n+1)! - 1$
- $\sum_{k=1}^n 2/((2k-1)(2k+1)) = \sum_{k=1}^n (1/(2k-1) - 1/(2k+1)) = 1 - 1/(2n+1)$
- $\sum_{k=1}^n (1/\sqrt{k} - 1/\sqrt{k+1}) = 1 - 1/\sqrt{n+1}$ (telescoping with square roots)
- $\sum_{k=1}^n k \cdot k! = \sum_{k=1}^n ((k+1)! - k!) = (n+1)! - 1$ (same as $k \cdot (k+1)!$ but reformulated)

Splitting Sums to Bound Recurrences

- Merge sort: $T(n) = 2T(n/2) + n$. Expanding the sum produces a geometric series: $\sum_{k=0}^{\lg n} 2^k \cdot (n/2^k) = n \lg n$
- Quickselect: $T(n) \leq T(3n/4) + cn$. Expanding: $cn \cdot \sum_{k=0}^\infty (3/4)^k = O(n)$ (geometric series with $r < 1$)

Product-to-Sum for Bounding Factorials (Stirling)

- $n! = \prod_{k=1}^n k \rightarrow \ln(n!) = \sum_{k=1}^n \ln k$
- Using integral bound: $\int_1^n \ln x \, dx \leq \sum_{k=2}^n \ln k \leq \int_1^{n+1} \ln x \, dx$
- Result: $n \cdot \ln n - n + 1 \leq \ln(n!) \leq (n+1) \cdot \ln(n+1) - n$
- Stronger: $n! = \sqrt{2\pi n} \cdot (n/e)^n \cdot (1 + \theta(1/n))$ (Stirling's approximation)

Cross-Chapter Links

- Used throughout for algorithm analysis (insertion sort, merge sort, quicksort, hashing)

Appendix B – Sets, Etc.

Named Entities

- **Power set (2^S):** set of all subsets; $|2^S| = 2^{|S|}$
- **Cartesian product:** $A \times B = \{(a,b) : a \in A, b \in B\}$; $|A \times B| = |A| \cdot |B|$
- **Equivalence relation:** reflexive, symmetric, transitive
- **Equivalence class:** $[a] = \{x \in S : x \sim a\}$; partition S into disjoint classes
- **Partial order:** reflexive, antisymmetric, transitive
- **Total order:** partial order where every pair is comparable
- **Hasse diagram:** graphical representation of a partial order (edges implied upward, transitively reduced)
- **Function types:** injection (1-to-1), surjection (onto), bijection (both)

Laws of Set Algebra

DeMorgan's Laws

$A \cap B = A \cap B$ and $A \cup B = A \cup B$ ##### Distributive Laws $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
 $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

Formulas & Equations

Inclusion-Exclusion

$$|AB| = |A| + |B| - |A \cap B| \quad |ABC| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

Handshaking Lemma (Proof)

$\sum_{v \in V} \deg(v) = 2|E|$ - **Proof:** Each edge contributes 1 to degree of each of its two endpoints. Summing degrees counts each edge exactly twice.

Free Tree Properties (Theorem B.2)

Any two of the following imply the third: connected, acyclic, $|E| = |V| - 1$. Equivalently: a free tree (undirected) is a connected, acyclic graph with exactly $|V| - 1$ edges and a unique simple path between any two vertices.

Complete k-ary Tree Formulas

- **Full k-ary tree (each internal node has exactly k children):**
 - Internal nodes I , leaves L : $L = (k-1)I + 1$
 - Total nodes N : $N = kI + 1$

- Height h (root at 0): $N \leq (k^{h+1}-1)/(k-1)$

Kraft Inequality

For any prefix-free binary code with codeword lengths ℓ_i : $\sum 2^{-\ell_i} \leq 1$. Conversely, given lengths satisfying Kraft, there exists a prefix-free code with those lengths. - **Proof intuition**: In a full binary tree of depth L , each codeword at depth ℓ_i uses $2^{L-\ell_i}$ leaves; total leaves used cannot exceed 2^L . - **Used in**: Huffman coding optimality proof (Ch 15).

Graph Isomorphism Definition

- Graphs $G = (V, E)$ and $G' = (V', E')$ are isomorphic if there exists a bijection $f: V \rightarrow V'$ such that $(u, v) \in E$ iff $(f(u), f(v)) \in E'$
- Isomorphism is an equivalence relation on graphs.

Common Set Identities

- $A \cup A = A$, $A \cap A = A$ (idempotence)
- $A \cup U = U$, $A \cap \emptyset = \emptyset$ (domination)
- $A \cup \emptyset = A$, $A \cap U = A$ (identity)
- $A \cup (A \cap B) = A$, $A \cap (A \cup B) = A$ (absorption)
- $A = \overline{\overline{A}}$ (double complement)

Edge Cases & Common Pitfalls

- **Empty set vs. $\{\}$** : $\emptyset$ has 0 elements; $\{\}$ has 1 element (the empty set itself)
- **Relation properties**: A relation can be none of reflexive/symmetric/transitive (e.g., "is parent of" on people)
- **Partial order vs. total order**: In a partial order, some elements may be incomparable (e.g., set inclusion on subsets)
- **Power set size**: $|2^S| = 2^{|S|}$ even when S is empty ($2^\emptyset = \{\emptyset\}$)

Cross-Chapter Links

- Foundational for all graph algorithms (Ch 20-24), tree structures (Ch 10-13)

Appendix C – Counting and Probability

Named Entities

- **Permutation**: ordered sequence; $P(n, k) = n! / (n - k)!$
- **Combination**: k -subset; $C(n, k) = n! / (k!(n - k)!)$
- **Binomial coefficient**: $C(n, k)$
- **Conditional probability**: $\Pr\{A|B\} = \Pr\{A \cap B\} / \Pr\{B\}$
- **Independent events**: $\Pr\{A \cap B\} = \Pr\{A\} \cdot \Pr\{B\}$
- **Pairwise vs. mutual independence**: pairwise independence does NOT imply mutual independence
- **Random variable**: function from sample space to real numbers
- **Bernoulli trial**: success (p) or failure ($q = 1 - p$)
- **Geometric distribution**: #trials until first success
- **Binomial distribution**: #successes in n trials
- **Entropy function $H(\lambda)$** : $H(\lambda) = -\lambda \lg \lambda - (1 - \lambda) \lg(1 - \lambda)$ for $0 < \lambda < 1$

Formulas & Equations

Binomial Theorem

$$(x+y)^n = \sum_{k=0}^n C(n,k) \cdot x^k \cdot y^{n-k}$$

Pascal's Identity

$$C(n,k) = C(n-1,k-1) + C(n-1,k)$$

Binomial bounds

$C(n,k) \geq (n/k)^k$ (lower), $C(n,k) \leq (ne/k)^k$ (upper, Stirling) $C(n,\lambda n) \leq 2^{n \cdot H(\lambda)}$ for $0 < \lambda < 1$ (entropy bound) $\sum_{k=0}^{\lambda n} C(n,k) \leq 2^{n \cdot H(\lambda)}$ for $\lambda \leq 1/2$ (binomial tail bound)

Bayes' Theorem

$$\Pr\{A|B\} = \Pr\{A\} \cdot \Pr\{B|A\} / \Pr\{B\} \quad \text{Extended form: } \Pr\{A_i|B\} = \Pr\{A_i\} \cdot \Pr\{B|A_i\} / \sum_j \Pr\{A_j\} \cdot \Pr\{B|A_j\}$$

Linearity of Expectation

$$E[X+Y] = E[X] + E[Y] \quad (\text{always, even for dependent variables})$$

Variance

$$\text{Var}[X] = E[X^2] - E^2[X] \quad \text{Var}[aX + b] = a^2 \cdot \text{Var}[X] \quad \text{For independent } X, Y: \text{Var}[X+Y] = \text{Var}[X] + \text{Var}[Y]$$

Geometric Distribution

$$\Pr\{X=k\} = q^{k-1}p \quad \text{for } k \geq 1 \quad E[X] = 1/p, \quad \text{Var}[X] = q/p^2$$

Binomial Distribution

$$b(k;n,p) = C(n,k) \cdot p^k \cdot q^{n-k} \quad E[X] = np, \quad \text{Var}[X] = npq$$

Markov's Inequality

$$\Pr\{X \geq t\} \leq E[X]/t \quad \text{for nonnegative } X, \quad t > 0$$

Chernoff Bound (Theorem C.8)

For $X \sim \text{Bin}(n,p)$, $\mu = E[X]$, $\delta > 0$: - $\Pr\{X \geq (1+\delta)\mu\} \leq (e^\delta / (1+\delta)^{1+\delta})^\mu$ - $\Pr\{X \leq (1-\delta)\mu\} \leq e^{-\mu\delta^2/2}$ - For $r > \mu$: $\Pr\{X \geq r\} \leq (\mu/r)^r \cdot e^{r-\mu}$

Jensen's Inequality

For convex function ϕ : $\phi(E[X]) \leq E[\phi(X)]$ For concave function ϕ : $E[\phi(X)] \leq \phi(E[X])$ Used to bound expectations and in proof of the EM algorithm.

Law of Total Probability

$$\Pr\{B\} = \sum_i \Pr\{B|A_i\} \cdot \Pr\{A_i\} \quad \text{for partition } A_i$$

Law of Total Expectation

$$E[X] = \sum_i E[X|A_i] \cdot \Pr\{A_i\} \quad \text{for partition } A_i$$

Conditional Expectation

$E[Y|X=x] = \sum_y y \cdot \Pr\{Y=y|X=x\}$ $E[Y] = E[E[Y|X]]$ (law of iterated expectation)

Union Bound (Boole's Inequality)

$\Pr\{\bigcup_{i=1}^n A_i\} \leq \sum_{i=1}^n \Pr\{A_i\}$

Covariance

$\text{Cov}[X, Y] = E[XY] - E[X]E[Y]$ For independent X, Y : $\text{Cov}[X, Y] = 0$ (converse not always true)

Moment-Generating Function

$M_X(t) = E[e^{tX}]$ Used to derive Chernoff bounds: $\Pr\{X \geq (1+\delta)\mu\} \leq \min_t \frac{E[e^{tX}]}{e^{t(1+\delta)\mu}}$

Edge Cases & Common Pitfalls

- **Not all random variables have expectation:** Cauchy distribution has no finite mean
- **Conditional probability with $\Pr\{B\}=0$:** $\Pr\{A|B\}$ is undefined, not zero
- **Correlation vs. causation:** $\text{Cov}[X, Y] = 0$ does not imply independence (e.g., $Y = X^2$ for symmetric distribution)
- **Bayes' theorem with rare events:** Even with a highly accurate test, $\Pr\{\text{Disease}|\text{Positive}\}$ can be low if the disease is very rare (base rate fallacy)

Cross-Chapter Links

- Hashing (Ch 11), randomized algorithms (Ch 5, 7, 9), balls-and-bins (Ch 5, 19)

Appendix D – Matrices

Named Entities

- **Matrix:** rectangular array; $m \times n$ has m rows, n columns
- **Transpose A^T :** exchange rows and columns
- **Identity I_n :** diagonal matrix with 1s; multiplicative identity
- **Inverse A^{-1} :** $AA^{-1}=I$ if exists
- **Determinant $\det(A)$:** scalar; $\det(A)=0$ iff A is singular
- **Rank:** size of largest set of linearly independent rows ($\equiv$ columns)
- **Positive-definite:** $x^T A x > 0$ for all $x \neq 0$
- **Cofactor:** $c_{ij} = (-1)^{i+j} \cdot \det(A_{-ij})$ where A_{-ij} is A with row i , col j removed
- **Adjugate:** $\text{adj}(A) = C^T$ where C is the cofactor matrix

Formulas & Equations

Matrix Multiplication

$C = AB$: $c_{ij} = \sum_k a_{ik} \cdot b_{kj}$ for $A(p \times q) \cdot B(q \times r)$ Standard: $\Theta(n^3)$, Strassen: $\Theta(n^{\lg 7})$

Transpose Properties

- $(A^T)^T = A$
- $(A + B)^T = A^T + B^T$

- $(AB)^T = B^T A^T$
- $(A^{-1})^T = (A^T)^{-1}$

Determinant (recursive via cofactors)

$\det(A) = \sum_j a_{ij} \cdot c_{ij} = \sum_j a_{ij} \cdot (-1)^{i+j} \cdot \det(A_{-i, -j})$ $A^{-1} = \text{adj}(A) / \det(A)$
(when $\det(A) \neq 0$)

Determinant Properties (Theorem D.4)

- $\det(0 \text{ row}) = 0$; scaling row by λ multiplies $\det$ by λ
- Adding row multiple to another: $\det$ unchanged
- $\det(A^T) = \det(A)$; swapping rows flips sign
- $\det(AB) = \det(A) \cdot \det(B)$
- $\det(cA) = c^n \cdot \det(A)$ for $n \times n$ A
- $\det(A^{-1}) = 1/\det(A)$

Positive-Definite Properties

- If A is positive-definite, then all eigenvalues are positive
- Leading principal minors are all positive
- A has a unique Cholesky decomposition: $A = LL^T$ where L is lower-triangular with positive diagonal entries
- $x^T A x > 0$ defines an ellipsoid in $\mathbb{R}^n$

LUP Decomposition Steps ($A = LUP$)

1. Permute rows of A for numerical stability (partial pivoting)
2. For each column k : find pivot $|a_{ik}| \geq |a_{jk}|$ for $j \geq k$
3. Swap row i with row k (record in permutation matrix P)
4. For each row below pivot: compute multiplier $\ell_{ik} = a_{ik}/a_{kk}$
5. Subtract $\ell_{ik} \times$ pivot row from row i (update U)
6. Store ℓ_{ik} in L (lower-triangular, 1s on diagonal)
7. Solve: forward substitution $Ly = Pb$, then back substitution $Ux = y$

- Complexity: $O(n^3)$

Matrix inversion from LUP

- Forward substitution ($Ly=Pb$) $\rightarrow$ back substitution ($Ux=y$)
- Complexity: $O(n^3)$
- $M(n)=O(I(n))$ and $I(n)=O(M(n))$ – multiplication and inversion are equivalent in difficulty

Matrix Multiplication Properties

- $A(BC) = (AB)C$ (associative, but not commutative)
- $(A+B)C = AC + BC$ (distributive)
- $C(A+B) = CA + CB$ (distributive)
- $AB \neq BA$ in general

Special Matrix Types

- **Diagonal:** $d_{ij} = 0$ for $i \neq j$; inverse has $1/d_{ii}$ on diagonal
- **Symmetric:** $A^T = A$ (e.g., adjacency matrix of undirected graph)
- **Orthogonal:** $A^T A = AA^T = I$; $\det(A) = \pm 1$
- **Triangular:** lower or upper; determinant = product of diagonal entries
- **Idempotent:** $A^2 = A$ (projection matrices)
- **Nilpotent:** $A^k = 0$ for some k

Eigenvalues and Eigenvectors

- $Av = \lambda v$: scalar λ is eigenvalue, v is eigenvector
- $\det(A - \lambda I) = 0$: characteristic polynomial, degree n
- $\text{Trace}(A) = \sum \lambda_i = \sum a_{ii}$
- $\det(A) = \prod \lambda_i$
- For symmetric A : all eigenvalues are real; eigenvectors are orthogonal

Matrix Norms

- **Frobenius norm**: $\|A\|_F = \sqrt{\sum_i \sum_j a_{ij}^2}$
- **Spectral norm**: $\|A\|_2 = \max\{\|x\|=1\} \|Ax\| = \sigma_{\max}(A)$ (largest singular value)
- **Max norm**: $\|A\|_{\infty} = \max_i \sum_j |a_{ij}|$

Edge Cases & Common Pitfalls

- **Singular matrix**: $\det(A) = 0$ means no inverse; this can happen even for non-zero matrices with linearly dependent rows/columns
- **Ill-conditioned matrix**: Near-singular matrices cause numerical instability; condition number $\kappa(A) = \|A\| \cdot \|A^{-1}\|$ measures sensitivity
- **Strassen vs. standard**: Strassen is only faster for large n (typically $n > 1000$) due to constant factors and numerical precision loss
- **LU without pivoting fails**: If $A_{kk} = 0$ at any step, pivoting (LUP) is required

Cross-Chapter Links

- Ch.4 (Strassen), Ch.28 (LUP decomposition), Ch.30 (Vandermonde in FFT)

Cross-Cutting Topics

Design Paradigms & Meta-Methods

Paradigm	Where Used
Divide-and-Conquer	Merge sort (Ch 2), Strassen (Ch 4), Quicksort (Ch 7), FFT (Ch 30), Parallel algorithms (Ch 26), LUP (Ch 28)
Dynamic Programming	Rod cutting (Ch 14), Matrix-chain (Ch 14), LCS (Ch 14), OBST (Ch 14), Floyd-Warshall (Ch 23), Subset-sum (Ch 35)
Greedy	Activity selection (Ch 15), Huffman codes (Ch 15), MST Kruskal/Prim (Ch 21), Dijkstra (Ch 22), Set cover approx (Ch 35)
Randomization	Randomized quicksort (Ch 7), Randomized selection (Ch 9), Universal hashing (Ch 11), Randomized marking (Ch 27)
Reduction	Max flow $\rightarrow$ max bipartite matching (Ch 24), NP-completeness proofs (Ch 34), Linear programming duality (Ch 34)

Amortization	29) Stack/queue analysis (Ch 16), Dynamic tables (Ch 16), Disjoint-set forests (Ch 19), Move-to-Front (Ch 27)
Online / Competitive	Elevator problem (Ch 27), Move-to- Front (Ch 27), LRU/FIFO caching (Ch 27)
LP-Rounding	Weighted vertex cover approx (Ch 35)
Fork-Join Parallelism	Spawn/sync model (Ch 26)

Proof & Argument Patterns

Pattern	Where Used
Loop invariants	Insertion sort (Ch 2), PARTITION (Ch 7), BUILD-MAX-HEAP (Ch 6), HEAPSORT (Ch 6), BFS (Ch 20), Generic-MST (Ch 21), Dijkstra (Ch 22), RB-INSERT- FIXUP (Ch 13), KMP (Ch 32)
Induction	Merge sort correctness (Ch 2), Radix sort (Ch 8), Bellman-Ford (Ch 22), Euclid's algorithm (Ch 31), Red- black height bound (Ch 13)
Contradiction	Dijkstra correctness (Ch 22), Gale- Shapley (Ch 25)
Cut-and-paste	MST cut property (Ch 21), Optimal substructure (Ch 14)
Exchange argument	Activity selection greedy (Ch 15), Huffman optimality (Ch 15), Augmenting paths (Ch 24, 25)
Decision tree	Comparison sort lower bound (Ch 8)
Potential function	Amortized analysis – stack, counter, dynamic table (Ch 16), Disjoint-set (Ch 19), Move-to-Front (Ch 27), Gradient descent (Ch 33)
Probabilistic method	Indicator variables for hire/hash/quicksort analysis (Ch 5, 7, 11)
Reduction proof	NP-completeness (Ch 34), Problem equivalence (Ch 28: inversion $\rightarrow$ multiplication)

Probability & Statistics Foundation

- **Indicator random variables:** Ch 5, Ch 7, Ch 11, Ch 9
- **Linearity of expectation:** Ch 5, Ch 7, Ch 9, Ch 11
- **Harmonic series:** $H_n \approx \ln n + \gamma$ – appears in hiring (Ch 5), quicksort (Ch 7), hashing (Ch 11)
- **Geometric distribution:** $E[t] = 1/p$ – used in RANDOMIZED-SELECT analysis (Ch 9)
- **Binomial distribution:** bucket sort (Ch 8), hashing (Ch 11)
- **Chernoff/Markov bounds:** randomized algorithms (Ch 5, 13)
- **Birthday paradox:** hashing (Ch 11), indicator variables (Ch 5)
- **Balls-and-bins:** hashing (Ch 11), load balancing (Ch 5)
- **Bayes' theorem:** conditional probability (App C)

Mnemonics & Memory Aids

1. **Big-O order:** " $O(1) < O(\lg n) < O(n) < O(n \lg n) < O(n^2) < O(2^n) < O(n!)$ "
2. **Heap index:** $\text{parent}=i/2$, $\text{left}=2i$, $\text{right}=2i+1$
3. **RB tree properties:** "Root Black, Red children Black, same Black-height on all paths"
4. **BFS vs DFS:** "BFS with Queue (breadth), DFS with Stack (depth)"
5. **Kruskal:** "K-sort and union" – sort edges, union components
6. **Prim:** "P-priority queue, one tree" – priority queue, single tree
7. **Bellman-Ford:** "V-1 passes, then check" – V-1 relaxations, then cycle check
8. **Floyd-Warshall:** "For k, For i, For j" – triple loop order
9. **Rabin-Karp:** rolling hash = $(\text{old_val} - \text{digit} \cdot h) \cdot d + \text{new_digit} \bmod q$
10. **KMP:** $\pi[q]$ = longest proper prefix also suffix
11. **NP-complete reduction chain:** CIRCUIT-SAT $\rightarrow$ SAT $\rightarrow$ 3-SAT $\rightarrow$ CLIQUE $\rightarrow$ VERTEX-COVER $\rightarrow$ HAM-CYCLE $\rightarrow$ TSP, and 3-SAT $\rightarrow$ SUBSET-SUM
12. **FFT pattern:** "FFT = pad $\rightarrow$ Transform $\rightarrow$ multiply $\rightarrow$ Inverse Transform"
13. **RSA:** "Repeated squaring + Extended Euclid + $\phi(n)$ + CRT"
14. **B-tree degree:** "t = minimum children; node has t-1 to 2t-1 keys"
15. **Master theorem:** compare $f(n)$ vs $n^{\log_b a}$ – 3 cases: leaf-dominated (1), equal (2), root-dominated (3)
16. **DP vs Greedy:** "DP: solve subproblems first, then choose; Greedy: choose first, then solve subproblem"

People & Dates

Person	Contribution	Chapter	Year
J.W.J. Williams	Heapsort, priority queue	6	1964
R.W. Floyd	BUILD-MAX-HEAP	6	–
C.A.R. Hoare	Quicksort, RANDOMIZED-SELECT	7, 9	~1960/1961
H.H. Seward	Counting sort	8	1954
Blum, Floyd, Pratt, Rivest, Tarjan	Worst-case linear selection	9	1973
A.M. Turing	Stacks for subroutine linkage	10	1947
H.P. Luhn	Hash tables, chaining	11	1953
Carter & Wegman	Universal hashing	11	1979
Adel'son-Vel'skiĭ & Landis	AVL trees	13	1962
Bayer	Red-black trees	13	1972
Guibas & Sedgwick	Red/black color convention	13	–
R. Bellman	Dynamic Programming	14	1950s
D. Huffman	Huffman codes	15	1952
R.E. Tarjan	Amortized analysis, disjoint-set forests	16, 19	1970s
Bayer & McCreight	B-trees	18	1972
E.W. Dijkstra	Dijkstra's algorithm	22	1959
R. Bellman	Bellman-Ford	22	1958
R.W. Floyd	Floyd-Warshall	23	1962
L.R. Ford Jr., D.R. Fulkerson	Max flow, min-cut	24	1956
J. Edmonds, R.M. Karp	Edmonds-Karp	24	1972
D. Gale, L.S. Shapley	Stable marriage	25	1962
J. Hopcroft, R.M. Karp	Maximum bipartite matching	25	1973
J. Cooley, J. Tukey	FFT	30	1965

Euclid	GCD algorithm	31	~300 BCE
R. Rivest, A. Shamir, L. Adleman	RSA cryptosystem	31	1977
R.M. Karp, M.O. Rabin	Rabin-Karp string matching	32	1987
D.E. Knuth, J.H. Morris, V.R. Pratt	KMP string matching	32	1977
S. Cook	NP-completeness, Cook-Levin theorem	34	1971
R.M. Karp	21 NP-complete problems	34	1972

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[PASS]	All chapters from the book are covered (35 chapters + 4 appendices)
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[PASS]	Every algorithm/formula/process has ≥ 1 worked example with numeric values
[PASS]	Every formula has all variables defined
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